Corpus Approaches to Evaluation

Phraseology and Evaluative Language

Susan Hunston

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Corpus Approaches to Evaluation

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Phraseology and Evaluative Language
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Page 187. Figure 9.1 'Statement types' (From chapter 'Evaluation and the planes of discourse' by Susan Hunston) from "Evaluation in Text: authorial stance and the construction of discourse" edited by Susan Hunston and Geoff Thompson (2000 hb). Reproduced by permission of Oxford University Press.

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For Linda and for John

Contents

List of Figures ix
List of Tables xi

1 Evaluative Language, Phraseology and Corpus Linguistics 1

2 Appraisal, Stance, Evaluation 10

3 Status in Written Texts and Multi-Modal Texts 25

4 Evaluation, Quantity and Meaning 50

5 Modal-Like Expressions 66

6 Corpus Approaches to Investigating Status 92

7 Grammar Patterns, Local Grammars and Evaluation 119

8 Phraseology, Intensity and Density 151

9 Conclusion 166

Appendix to Chapter 5 173
Notes 183
Bibliography 187
Index 195

Figures

2.1	Concordance lines for *trust*.	15
3.1	From Hunston (2000a: 187).	32
4.1	Examples of *terrible*.	57
4.2	*CAUSE* and *BRING ABOUT*	59
4.3	*a shred of*.	60
4.4	PAY . . . *price*.	61
5.1	Concordance lines for *decide wh* (from Hunston 2003b).	70
5.2	Concordance lines for *distinguish between*.	71
5.3	Concordance lines for *distinguishing between*.	72
6.1	Concordance lines for *assumption that*.	94
6.2	Phraseologies associated with the negative evaluation of assumptions.	100
6.3	adjective *to assume that*.	101
6.4	Writer's assumptions.	102
6.5	*we assume that*.	103
6.6	*we assume that*.	104
6.7	Concordance lines for *this discovery*.	107
6.8	Examples of **Vn *as* n.**	111
7.1(a)	Co-text: *RECOVER* is followed by a noun phrase (**V n**); also included here is the passive equivalent (***be* V-ed**).	124

7.1(b)	Co-text: *RECOVER* is followed by a prepositional phrase beginning with *from* (**V *from* n**).	124
7.1(c)	Co-text: *RECOVER* is followed by a noun phrase and a prepositional phrase beginning with *from* (**V n *from* n**).	124
7.1(d)	Co-text: *RECOVER* occurs at the end of a clause or is followed by a non-dependent prepositional phrase or adverb (**V**).	125
7.2(a)	Adjectives indicating emotion.	129
7.2(b)	Adjectives indicating human qualities.	129
7.2(c)	Adjectives indicating qualities of things.	129
7.2(d)	Adjectives indicating attitudes.	130
7.3	Examples of **ADJ *about* n.**	140
8.1	Intensifying sequences.	158

Tables

3.1	Status in Example 3.5	29
3.2	Status in Example 3.6	31
3.3	A Classification of Source Types	35
3.4	'Engagement' Account of Example 3.16	45
3.5	'Status' Account of Example 3.16	45
3.6	'Clause Relations' Account of Example 3.16	46
3.7	'Engagement' Account of Example 3.17	48
3.8	'Status' Account of Example 3.17	48
5.1	Frequencies of Left Collocates before *distinguish* (Rounded Percentages)	74
5.2	Sequences, Patterns and Meanings with *to distinguish*	75
5.3	Frequencies of Left Collocates before *distinguishing between* (Rounded Percentages)	76
5.4	Modal-Like Elements Identified by Selecting ***of* + -ing form**	78
5.5	Rounded Percentage of Each Wordform in Four Lemmas	82
5.6	Left Collocates of Four Base Forms (200 Lines Each)	82
5.7	Relative Frequency of Wordforms for Four Lemmas	83
5.8	Modal Meaning and *whether*	84
5.9	Sentences with *for fear of* and Their Paraphrases	87
5.10	Summary of Uses of 10 Selected Verb Sequences	89

6.1 Frequencies of *fact* in the BNC 109

6.2 Frequencies of *hypothesis* in the BNC 109

7.1 Parsing the **V n n** Pattern 121

7.2 Parsing the **V n *from* n** Pattern 121

7.3 Adjectives in Two Patterns 132

7.4 Adjectives in Appraisal Types 137

7.5 Parsing the **V n *as* adj, V n n** and **V n adj** Patterns 141

7.6 Parsing Definitions 143

7.7 ***It* v-link ADJ –ing** 145

7.8 ***It* v-link ADJ *for* n to-inf** 145

7.9 **V *it* ADJ to-inf** (Example from FrameNet) 145

7.10 **V *for* n to-inf** 145

7.11 **V *for* n to-inf** 146

7.12 Using 'Evaluation Limiter' 146

7.13 Using 'Evaluation Limiter' 146

7.14 Parsing **v-link ADJ to-inf** 148

7.15 Parsing **v-link ADJ to-inf** 148

7.16 Parsing **v-link ADJ to-inf** 149

7.17 Parsing **v-link ADJ to-inf** 149

8.1 Verbs and Nouns with *months of* 163

8.2 Instances of Intensifying Phrases 164

1 Evaluative Language, Phraseology and Corpus Linguistics

This book examines the role of corpus linguistics in the study of evaluative language, prioritising that approach to corpus linguistics which focuses on phraseology.

There are clearly three terms in this opening sentence—evaluative language, corpus linguistics and phraseology—that require explanation. This is the purpose of this chapter.

1.1 EVALUATIVE LANGUAGE

Evaluative language is that language which indexes the act of evaluation or the act of stance-taking (Du Bois 2007). It expresses an attitude towards a person, situation or other entity and is both subjective and located within a societal value-system (Hunston 1994: 210). As a brief illustration of these points, here is an example of an evaluative act, taken from a newspaper review of a book which represents a new genre, 'monster mash-ups'. These are books which combine characters and language from classical literature with plots taken from modern horror literature, for comic effect. The reviewer writes:

Example 1.1

> But the other obvious problem with monster mash-ups is that the joke very quickly grows old. *Pride and Prejudice and Zombies* is often very funny, but by the third or fourth chapter you've well and truly got the idea; by the time you come to *Sense and Sensibility and Sea Monsters*, the novelty has thoroughly faded. (Merritt 2009)

This paragraph performs an act of evaluation, fulfilling the purpose of the book review. It uses particular resources of English to indicate this function: the individual words *problem* and *funny*, and the word combinations *joke grows old* and *novelty has faded*. The evaluation is strengthened by the intensifiers *very*, *well and truly* and *thoroughly*.

A particularly interesting phrase is *you've well and truly got the idea*. This might be interpreted in a positive light—true understanding has been

achieved. However, following as it does the comment *the joke very quickly grows old*, it implies a negative evaluation: the reader achieves an understanding that is too complete too early to convey maximum enjoyment of the book. In this way the paragraph appeals to a shared value between writer and reader: that jokes are funny when they are novel and that once a joke is familiar it is no longer as funny.

The paragraph gives information to the reader, but this is information that is entirely subjective. Another reviewer might offer a completely different assessment of the books concerned. Nonetheless, there is in this particular review no indication that the opinion held is purely personal, no overt concession to other points of view. A dialogue is enacted, between *Pride and Prejudice and Zombies is often very funny* and *by the third or fourth chapter you've well and truly got the idea*, but that dialogue does not admit of alternative viewpoints. To interpret the paragraph as 'this is what this writer thinks' rather than as 'this is what is objectively verifiable' depends upon understanding the nature of evaluation as a subjective act.

The paragraph does, however, engage the reader in addressing him or her as *you*. This underlines the paragraph as an act of persuasion; the expectation is that the reader will agree with the opinions expressed.

There are clearly a number of ways in which the brief explication of this paragraph could be extended, each of which corresponds to a tradition in the study of evaluative language. Perhaps the most obvious would be to study in more detail the particular language resources that are used to convey evaluation: the words, collocations and phrases. This is an endeavour that was recommended by Stubbs (1986) and that has been carried out in numerous contexts, for example, the extensive literature on stance, engagement and metadiscourse (e.g. Biber 2006a; Hyland 2009; Hyland and Tse 2004). Once such features have been identified, one set of texts can be measured against another to compare the amount and type of evaluative language in each (Biber 2006a; Hyland 2009; Charles 2006a, 2006b). A second approach might be to use identified linguistic resources to assess how a book such as *Pride and Prejudice and Zombies* (in Example 1.1) has been greeted by a large number of critics. This is the approach taken by Sentiment Analysis (e.g. Taboada and Grieve 2004), where the aim is to identify positive and negative evaluation in a very large collection of texts. The means to achieving that aim involves the automatic identification of evaluative language.

Most researchers find it advantageous and indeed instructive to break down the mass of language often referred to as 'evaluative' into different types. For example, G. Thompson (2001) distinguishes between 'interactive' and 'interactional', a distinction taken up by Hyland and Tse (2004). In other publications Hyland distinguishes between 'stance' and 'engagement'. Hunston (1994) identifies three functions of evaluation and suggests that they identify the epistemic object being evaluated (status), the value given to the object (value) and the relevance of parts of the text (relevance). Martin and White (2005) offer what is probably the most theory-grounded study of the functions and

forms of evaluative meaning in English, based on systemic-functional linguistics. They would point out that Example 1.1 can be discussed in a number of ways: as an example of a unilateral statement as opposed to a debate ('engagement'), as an intensified act of evaluation ('intensification') and as evaluation construed by assessing both the quality of the book—*funny*—and the reader's reaction to it—*the joke very quickly grows old* ('appraisal').

The study of evaluative language is important for a number of reasons. Indicating an attitude towards something is important in socially significant speech acts such as persuasion and argumentation. Taking a stance towards something and negotiating alignment or non-alignment is a crucial aspect of interaction between individuals. As linguistic study moves away from truth-value and towards a focus on the interactive, the importance of this aspect has become more apparent.

In spite of its importance, evaluative language can pose difficulties for linguistic theories. In some theories, language is classified into broad functions (or metafunctions). These may include an interpersonal function: building and expressing relationships. Evaluative language most obviously belongs to this function. On the other hand, Halliday and Hasan (1985) and Sinclair (1987) suggest that only some evaluation has an interpersonal, or interactive, function, while other acts of evaluation are located within the ideational function.

Evaluative language presents difficulties in analysis because there is no set of language forms, either grammatical or lexical, that encompass the range of expressions of evaluation. It is true that adjectives and adverbs frequently express evaluative meaning (e.g. Turney 2002; Conrad and Biber 2000), and, as shall be argued in Chapter 7, some patterns of use are associated with such meaning, but this does not mean that every adjective and adverb marks evaluation or that all evaluation can be identified in this way.

In fact, evaluation is frequently expressed cumulatively and implicitly. Example 1.2 is an example of cumulative evaluation (in which the names of the professor and the university have been replaced by pseudonyms):

Example 1.2

> As I write this, Professor Smith, now a distinguished scholar, has her job under threat from the ghastly, grey accountants who run the University of Biggin-on-Sea. We are now in an epoch of production-line universities with celebrities paid fortunes to teach eight hours a week and genuine scholars dumped in the bin. (Ali 2008)

Evaluation of the University of Biggin-on-Sea, and of British universities in general, is cumulative: *ghastly*, *grey accountants*, *production-line universities*, *genuine scholars dumped in the bin*. This accumulation of evidence allows us to interpret *celebrities paid fortunes to teach eight hours a week* in a negative light.

Example 1.3 is an example of implicit evaluation:

Example 1.3

> To determine whether any branched species [of alga] was differentially important to the establishment of surfgrass, I followed seedlings for 7 months. During this time 90% of the seedlings died . . . but the seeds attached to each algal species died in about the same proportion . . . These results suggest that all the branched species facilitate surfgrass and that the magnitude of the facilitation is proportional to the number of seeds originally attached to the alga. (Turner 1983)

In this paragraph, a question is asked—'Does the species of alga affect how well surfgrass becomes established?'—and answered in the negative—'Any species of alga helps surfgrass become established; it is the number of seeds that makes the difference'. Hunston (1993a) argues that evaluation here consists of setting a goal ('To determine whether . . .') and then noting that the goal has been achieved ('The results suggest that . . .'), thereby evaluating the experiment as successful. Arguably, then, an act of evaluation has taken place that is marked by lexical repetition (*any branched species—all the branched species*; *important to—facilitate*; *differentially—proportional to*) but not by any recognisable instance of evaluative language.

These features of cumulation and implicitness make it appear that evaluative language is more suited to text-based than to corpus-based enquiry. The detail involved in explicating the evaluative language in a given text suggests that very close reading is required. It would seem, then, that the broader sweep adopted by corpus linguistic methods would be suitable only for adding quantitative detail to the more interesting small-scale studies. It is one of the tasks of this book to argue that this is not the case.

1.2 CORPUS LINGUISTICS

Corpus linguistics is, by now, a term covering a wide range of activities and approaches (e.g. Baker 2009; Sampson and McCarthy 2005; Teubert 2005). At its most basic, shared concerns include collecting quantities of text in electronic form so that they are open to data-manipulation techniques. Such techniques range from finding a search term and observing its immediate environments (key-word-in-context or concordance lines); to calculations of relative frequency (as in, for example, collocation studies); to annotation for such categories as word class, grammatical function or semantic class; and frequency calculations based on such categories. Frequencies of various kinds can be compared in different corpora, leading to observations about different registers or different languages and about the development of a language over time.

Corpus linguistics is more than a simple set of techniques, but it is a field where technological advancement and theoretical development go hand in hand. For example, one of the most basic corpus linguistics

practices—isolating a node word and its immediate co-text in each instance of occurrence, and manipulating that output to highlight similarity in co-text (i.e. sorting concordance lines)—was developed because words were considered to be more important than grammatical categories, and the immediate co-text of a word, including its significant collocations, was considered to offer the most important information about it. In turn, the prevalence of concordancing software has facilitated the observation of language patterning and so influenced the development of theories based on that phenomenon.

Corpus linguistics, then, is more than a way of investigating existing models of language. The methodologies it uses can be used both to change and to complement our understanding of those models. For example, from different perspectives, each of Biber et al. (1999), Carter and McCarthy (2006), Holmes and Nesi (2009) and Matthiessen (2006) focus on differentials between language in different registers or modes. The concept of register pre-dates large-scale corpus research of the kind used today. Corpus research has assisted, however, in quantifying the occurrence of various features in different registers (see Biber et al. 1999; Matthiessen 2006, though each is working with a very different concept of 'register'), and in identifying features which are unique to a specific mode (Carter and McCarthy 2006).

On the other hand, research at the lexis-grammar interface (e.g. Sinclair 1991) offers a revision to previous models of language by suggesting that a sharp divide between syntax and semantics may not be advisable. More specifically, corpus linguistics has identified what in this book I shall call 'phraseology' as a key concept in the way that language works.

1.3 PHRASEOLOGY

'Phraseology' is a very general term used to describe the tendency of words, and groups of words, to occur more frequently in some environments than in others. As used in this book, the term takes its inspiration from Sinclair's (1991) summary of a number of corpus studies that together indicate that (a) more language occurs in 'fixed phrases' than might otherwise be thought and, furthermore, that (b) 'fixed phrases' are more varied than might otherwise be thought. This observation leads on the one hand to a concern for multi-word units (MWUs), that is, stretches of language consisting of two, three or four (or more) words which occur frequently in a given corpus and which can often be seen to play a particular role in a given register. A large number of studies today focus on MWUs. On the other hand, it also leads to the identification of what Sinclair called 'units of meaning', that is, flexible word sequences which display some consistency in aspects of form but more so in aspects of meaning. Added to this is Hoey's work on 'lexical priming' (2005). Hoey argues for the priming connected with individual words as a key psycholinguistic mechanism by which language is acquired,

used and interpreted. Alongside the psycholinguistic arguments lie a set of observations which indicate that differential frequency relating to individual words is much more patterned than had previously been thought. Not only do words generally co-occur with specific others (collocation) and more frequently in one set of grammatical environments than others (colligation), but many words occur differentially in different parts of a text, such as in paragraph or text initial position. Some of these differential frequencies are register-independent and others are register-specific.

Under the very broad heading of phraseology, therefore, comes a range of diverse phenomena of which the following are examples:

- The noun *cold* co-occurs with the adjectives *bitter*, *icy* and *extreme*. The noun *heat* collocates with *intense*, *sweltering* and *extreme*.
- The verb *afford* mainly occurs with modal verbs such as *can* and *could*. On average, over half of occurrences have negative forms such as *cannot* or phrases such as *can ill afford*. This is remarkable when it is borne in mind that the ratio of positive and negative clauses in English is 9:1 (Halliday 1993).
- The noun *consequence* is much more likely to occur as the subject of a verb than as its object (Hoey 2005: 46–47).
- In US university discourse there are a number of MWUs or 'lexical bundles' that are frequently used by teachers to focus attention on a new topic (Biber 2006b: 142–143). These include *want to talk about* and *what I want to do is*.
- The two-word phrase *true feelings* is used in fairly restricted contexts, expressing difficulty or reluctance to express genuine emotion. The contexts are varied in form but all convey, often implicitly, this sense. Explicit examples include: *she hid her true feelings*; *when I'm able to reveal my true feelings*; *we lose the ability to express our true feelings*; *only her close female friends . . . had any idea of her true feelings* (Sinclair 2003).
- An MWU may be infrequent overall but may still have a quality of 'phrase-ness' because each element is frequent relative to the other elements. An example is *after a few moments of*, which occurs relatively rarely but in which *few* is a significant collocate of *moments of* and *after* is a significant collocate of *a few moments of* (Hunston 2002).

Although phraseology as a topic of research has gained considerably from the input of corpus linguistics, Gries (2008a) notes that it is also a central tenet in cognitive linguistics and construction grammar, and that it is not ignored either by generative grammarians. Commenting on the lack of agreement, or even explicitness, as to what might constitute the object of attention in phraseological studies (or 'phraseologism'), Gries (ibid.: 4) identifies six features which provide a useful yardstick, not in the sense that all phraseological studies adopt the same attitude towards them, but in the

sense that such studies might legitimately be compared in terms of their stance towards them. The six features (quoted verbatim from Gries with comments from myself in parentheses) are:

i. the *nature* of the elements involved in a phraseologism (studies may be divided into those that restrict themselves to wordforms or lemmas and those which adopt a wider brief, with elements that may be words, or grammatical forms, or broadly-defined elements of meaning);
ii. the *number* of elements involved in a phraseologism (studies of collocation typically restrict themselves to two elements, not necessarily contiguous, while units of meaning as identified by Sinclair are not restricted in the number of words they may contain);
iii. the *number of times* an expression may be observed before it counts as a phraseologism (Gries adopts the criterion that to be counted as a phraseologism, an expression must be observed more frequently than expected);
iv. the permissible *distance* between the elements involved in a phraseologism (studies of contiguous collocations allow no distance whereas many studies specify discontinuous elements; Sinclair and Renouf (1991) is an example of the latter; Gries might have added 'whether or not flexibility in the order of the elements is taken into account');
v. the degree of *lexical and syntactic flexibility* of the elements involved (as noted earlier, evidence suggests that most phrases admit considerable variation, and indeed much linguistic creativity depends on taking an apparently fixed phrase and treating it as flexible);
vi. the role that *semantic unity* and *semantic con-compositionality/non-predictability* play in the definition (idioms are items that are both unified semantically—an idiom 'means something'—and non-compositional—their meaning cannot be predicted from the meaning of each individual word; most studies of phraseology, however, apply the first but not the second of these restrictions, and work such as Hoey's applies neither).

Looking at evaluative language from the perspective of phraseology is not new. Biber et al. (1999), for example, identify lexical bundles that express stance, such as *no doubt*, *in fact* and *according to*. (See also Biber 2006b for a more detailed discussion.) In addition, one of the insights owing to Sinclair's 'unit of meaning' theory is that units of meaning express attitude, often implicitly. In this book it will be argued that phraseology as broadly defined plays a number of roles in the study of evaluative language.

1.4 THE ORGANISATION OF THIS BOOK

The second and third chapters of this book focus on evaluative language as a discoursal, rather than as a corpus, phenomenon. Chapter 2 expands the

introduction to the ways that evaluative language has been studied without corpora. Chapter 3 provides an account of the concept of epistemic status, as an introduction to the work in Chapter 6.

The rest of the book focuses on corpus studies of evaluative language. Chapter 4 reviews the literature in this area. In Chapter 5, the role of phraseology in the expression of modal meaning is investigated. Evidence is put forward to support the argument that some verbs co-occur with modal meaning more frequently than others do, and that modal meaning can be expressed by a large number of grammatically varied phrases as well as by traditional modal verbs. In Chapter 6, it is argued that status nouns, which evaluate the epistemic status of a proposition, can themselves be observed to co-occur differentially with phrases that evaluate them in one respect or another. Chapter 7 considers the possibility of devising a local grammar of evaluation, based on the co-occurrence of words and grammatical units. Finally, in Chapter 8, the role of particular phrases in supporting and intensifying evaluative meaning is studied. Chapter 9 provides a conclusion to the book.

1.5 A NOTE ON CORPORA AND TERMINOLOGY

For the most part, the corpus information in this book is drawn from the Bank of English corpus held at the University of Birmingham. The corpus itself is jointly owned by the University of Birmingham and HarperCollins publishers. The version used in this book comprises 450 million tokens and consists of a number of sub-corpora consisting of, among others, news journalism from Britain, US and Australia, fiction and non-fiction books published in Britain and the US, spontaneous spoken English from Britain and the US and issues of a large number of special interest magazines. The study in Chapter 6 is based on the sub-corpus containing issues of *New Scientist* magazine. Otherwise, unless otherwise stated, information about collocation, frequency and usage is based on the whole Bank of English corpus, and examples and concordance lines are taken from there.

The software used to search the Bank of English determines some definitions of the terminology used in this book. A **random** sample of concordance lines takes each nth instance of the node word from across the corpus or the selected sub-corpora. For example, if there are 1,000 instances of a given word, and the search request specifies 100 examples, the software will take display every tenth occurrence. For calculations of **collocation**, the software specifies a span of four words each side of the node. For the most part, and unless otherwise indicated, I use raw frequency to place collocates in order, rather than measures of significance such as z-score, t-score or Mutual Information. In some discussions, reference is made to collocates that are in the L1 or R1 positions relative to the node (that is,

immediately preceding or following the node word). Where these are placed in order, again it is raw frequency that is used to establish the order.

Throughout this book, italics are used to indicate wordforms and capitals are used to indicate lemmas. A lemma is defined as the abstraction from a set of wordforms belonging to the same word class. Thus, *WALK* indicates *walk*, *walks*, *walking* and *walked* (but not the noun *walk*).

2 Appraisal, Stance, Evaluation

2.1 INTRODUCTION

This chapter is one of two that consider various approaches to the concept of 'evaluation', focusing on points of agreement and difference. This chapter prioritises researchers whose work takes individual texts (written or spoken) as the raw material, and uses methodologies appropriate to those data, although it also makes some mention of those whose work is corpus-oriented. Corpus approaches will, however, be taken up again in Chapter 4 (including the phenomenon known as 'semantic prosody').

One aim of this chapter is to problematise the notion of evaluative language and to ask how it is possible for such a slippery and context-dependent aspect of language to be amenable to investigation via corpus techniques. Another is to tease out the conceptual differences that lie behind the various terminologies and approaches used.

It is clear that a broad range of terminology is used to refer to similar areas of language use: appraisal, stance, sentiment, evaluative, attitudinal or affective language, metadiscourse and evaluation, to name only the most obvious. When considering common ground I shall treat these as largely equivalent, but I shall also discuss the differences in approach that lead to the divergence in terminology.

What lies behind some of the distinctions in terminology is a variance in what kind of phenomenon 'evaluation' is taken to be. Some possibilities are:

- 'Evaluation' is an action—something which a person does. The action may be purely private and unexpressed (in which case it lies outside a study of language) or it may be performed in language. Englebretson (2007: 3) notes that the term 'stance-taking' implies that 'stances are something that people actively engage in'. This in turn draws on a tradition (largely conversation analysis) which regards interactions as the sites where people accomplish social activities (e.g. Hutchby and Wooffitt 2008: 12).
- 'Evaluation', or its near synonyms, is the set of words and phrases which express evaluative meaning. Hyland and Tse (2004: 157,

following Hyland 2000) define metadiscourse as 'the *linguistic resources* used to organize a discourse or the writer's stance towards either its content or the reader' (italics mine). Conrad and Biber (2000: 57) define stance as 'a cover term for the *expression* of personal feelings and assessments' (italics mine). The tradition used here is that of corpus linguistics, which takes the recurrence of linguistic items (words, phrases, grammatical categories) as its object of study.

- 'Evaluation' is a set of meanings which might be expressed in a given text using a wide variety of language resources. Introducing their concept of appraisal, Martin and White (2005: 42) refer to a 'system of meanings' to which a speaker or writer has access and which that speaker/writer may use to 'approve and disapprove, enthuse and abhor, applaud and criticise, and . . . position their reader/listeners to do likewise' (ibid: 1). Martin and White seem to share with Du Bois a concern with actions performed in discourse, in particular the tripartite actions of expressing attitude, construing a subjective position and aligning self with audience. However, their concern for identifying and systematising semantic resources comes from the Systemic Functional Linguistics (SFL) tradition with which they identify themselves.
- 'Evaluation' is a function performed by a text, or part of a text. This treats the text as an agentive entity, independent of its interactants. Hoey (2001), for example, talks of sentences providing elements of text patterns. Hunston (1989) offers a model for considering what functions evaluation performs in a set of academic texts. Thompson and Hunston (2000) discuss more generally the functions of evaluation. It should perhaps be noted that researchers who adopt metalanguage of this type are not ignorant of the role of language in construing (inter)subjective positions (Hoey's book, for example, is titled *Textual Interaction*), but treat texts as amenable to analysis without explicit recourse to their producers.

To illustrate further what this means, consider the sentence from an article written by Randolph Quirk and analysed in detail by Sinclair (1993; reprinted in Sinclair 2004: 98–100), shown here as Example 2.1.

Example 2.1

The implications are daunting.

This sentence creates a stance towards an object (the previous sentence which has 'implications') which is shared between writer and reader. It includes a word, *daunting*, which expresses evaluative meaning. The sentence draws on a set of meanings which speakers of English have at their

disposal to evaluate 'facts' about the world. In this case, arguably, the facts are evaluated in terms that construe Reaction: quality (Martin and White 2005: 56), that is, in terms of the effect they have on the writer and/or reader. The sentence performs a function in 'encapsulating' (Sinclair 2004: 83) and evaluating a previous one, thereby not only offering opinion but indicating the organisation of the text: the information in the previous sentence is no longer active but has been 'wrapped up' by this evaluation. The sentence therefore illustrates a writer performing an action, and in doing so interacting with the reader, the use of a subjective word (*daunting*) which has negative polarity, and the text-organising function of evaluation.

2.2 ITEMISING CONSENSUS

The differences summarised here suggest ways in which various approaches to investigating language in use have taken account of the phenomenon (or collection of phenomena) termed here 'evaluation'. These approaches will be described in more detail in the following, but in this section I wish to identify points of consensus, of which six are discussed here.

The first point of agreement is that evaluation is both subjective and intersubjective. Evaluative utterances express a personal opinion. This is made very clear in Sentiment research, where the aim is to distinguish between 'information' and 'opinion' as an aid to Information Extraction (Wiebe et al. 2005: 166). Thompson and Hunston (2000: 1) talk about 'positive or negative opinions' and Martin and White (2005: 42) gloss attitude as 'ways of feeling'. Gallie (1956) discusses 'essentially contested concepts', that is, statements which by definition cannot be discovered to be true or false in any objective way. This might be taken as a partial definition of an evaluative utterance. Evaluation is personal, private, subjective; evaluative statements are endorsed only by their speaker.

The concept of intersubjectivity is equally important. It is acknowledged that evaluation has the function of interacting with a social other. Martin and White (2005) locate appraisal systems within the interpersonal metafunction; the stance triangle proposed by Du Bois (2007) indicates that all acts of evaluation determine an alignment between speaker and hearer; Thompson and Hunston (2000: 8) identify 'building and maintaining relations' as one of the functions of evaluation. Many writers distinguish between evaluation and interaction, with evaluation (or the expression of personal opinion) being a contributing factor in a rather larger phenomenon (e.g. G. Thompson 2001 on writer–reader interaction; Hyland and Tse 2004 on metadiscourse), but all see evaluation as an important method by which interaction is achieved and interaction as a key function of evaluation.

Secondly, evaluation construes an ideology that is shared by writer and reader (or speaker and hearer). This is the other side of the 'subjectivity' coin. Evaluation is personal and subjective but it also takes place within a

social and ideological framework. The two interactants in the act of evaluation are located within a value system (even when they are construed as disagreeing). Hunston (1993a) argues that some evaluation in academic discourse can be interpreted as such only in light of ideological positioning related to 'how knowledge is made' or 'what counts as good research'; in these cases there may be no obvious expressions of evaluation or stance. Similarly, Martin and White (2005: 67) distinguish utterances which 'afford' (in their terms) the reader the opportunity to infer evaluation even when no explicit (or 'inscribed') evaluation is present. The likely inferences are based on shared ideological assumptions. The relationship between evaluation and ideology is multifaceted. One the one hand, implicit evaluation works in a text because writer and reader share assumptions. On the other hand, the ideological position that lies behind a text can be inferred by the analyst by examining the evaluation in it. See, for example, Benwell and Stokoe (2006: 174–175), where the writers infer cultural assumptions about health, appearance and lifestyle from advertisements for items such as cosmetics and cosmetic surgery, although these writers do not use the term 'evaluation'. Finally, of course, speakers and writers tend to use certain evaluations because of the ideological positions that they hold and that they believe their interactants hold.

The third point of consensus is the broad range of lexical and other indicators of evaluative meaning. Although researchers differ in what they include under the heading 'evaluation' (for example, whether indicators of modal meaning are included or excluded), restricting the term to something like 'desirable or undesirable' leads to a number of areas of agreement. One is that typical evaluative items include adjectives such as *terrible* and *wonderful* (or even, simply, *good* and *bad*) and that ordinary readers would often identify these as evaluative even out of context. (In White's terms [personal communication 2009] these items are 'stable'.) Another, however, is that evaluation is heavily dependent on context (see the next point) so that taking items out of context is potentially an unreliable indicator of evaluative meaning. Finally, there is consensus that evaluation is indicated by such a large range of lexical and other items that it would be pointless to try and list them. Riloff and Wiebe (2003), for example, note that 'subjective language can be exhibited by a staggering variety of words and phrases'. Lexical items with a strong likelihood of indicating evaluation in context include nouns (e.g. *success*), verbs (e.g. *fail*), adjectives (e.g. *excellent*) and adverbs (e.g. *unfortunately*). Lexical items do not need to be single words; as Sinclair has noted, phrases such as *my cup of tea* are evaluative even though the individual words *cup* and *tea* are unlikely to be so. Some grammatical frames normally have an evaluative function, such as '*it was* adjective *of* person *to* do something', where frequently occurring adjectives include: *nice*, *good*, *typical*, *wrong*, *kind*, *brave*, *stupid*, *naïve*, *unfair*, *foolish*, *sweet*, *clever*, *unprofessional*, *lovely*, *cruel* and *selfish* (see Chapter 7). Most importantly, however, evaluation may be implied rather

than stated explicitly. Martin and White (2005: 67) go into this in detail, distinguishing four levels of explicitness from 'Inscribed' and 'Afforded'. Citing a speech by the former prime minister of Australia, Paul Keating, which evaluates negatively the actions of European settlers against the indigenous people of Australia, they give the utterance *it was our ignorance and our prejudice* ('that made us perform these actions') as an example of Inscribed, or explicit, appraisal and the utterance *we brought the diseases* ('that infected and killed the indigenous people') as an example of Afforded appraisal, where the reader is given evidence for an appraisal rather than the appraisal itself. Distinguishing, and agreeing on, levels of explicitness is never easy, but there is agreement that they do exist.

As suggested earlier, a fourth point of consensus is that evaluation is both contextual and cumulative. At its simplest, this means that a word not encountered in naturally occurring discourse cannot reliably be identified as evaluative or not, or if it is assumed to be evaluative, as positive or negative. This is, of course, not something that is necessarily restricted to evaluative meaning. Teubert (e.g. 2003: 9) argues that no word has a meaning except when it is encountered in context. To put this another way, the meaning of any word cannot be identified reliably if the word is encountered in isolation. What happens, presumably, when a speaker of, say, English, is presented with an isolated word (such as *horse*) and asked to say what it means or what its word class is, the speaker imagines a context in which he or she might regularly encounter that word (e.g. 'In the field a horse was quietly eating grass') and so assigns it a meaning based on that context—animal used for riding—and a word class—noun. There are, of course, many other contexts for *horse*, each of which would suggest meanings and word classes at variance with these, such as 'rocking horse', 'on his high horse' or 'horsing around'.

The same is true of evaluative meanings. A speaker may consider that the most likely context for *electric* is 'electric fire' or 'electric storm' and so consider that *electric* is a non-evaluative word. On the other hand, in a context such as 'her performance was electric', the meaning is evaluative and positive. There is widespread corroboration of this. Groom (2004), for example, notes that in a corpus of book reviews in academic journals, *scholarly* has a positive evaluative meaning in some instances (e.g. *this is a fine, scholarly . . . book*) and a neutral meaning in others (e.g. *provoking scholarly debate*). Wilson et al. (2005) distinguish between 'prior polarity', which they define as 'out of context . . . [evoking] something positive or something negative' and 'contextual polarity', which is the meaning in a given phrase, which 'may be different from the word's prior polarity'. The contexts which deny prior polarity are of two types. One type is exemplified by the word *trust* which, according to Wilson et al (2005), has a positive prior polarity but which in the phrase *National Environment Trust* is neutral. (It could equally well be argued that this is in fact two words, *trust* and *Trust*, that happen to share the same spelling and pronunciation.) In

the other type, polarity is reversed in context by negation, as happens to the word *reasonable* in the clause *There is no reason to believe that polluters are suddenly going to become reasonable* (ibid.).

In the preceding examples, 'context' is taken to be the words occurring immediately before and after the word (or phrase) under consideration. It is assumed that whether a word is, in a given instance, neutral, positive or negative can be ascertained by looking at a context no longer than a concordance line of something between 80 and 500 characters long. Twelve sample concordance lines for *trust* illustrate this (Figure 2.1).

Figure 2.1 Concordance lines for *trust*.

1 me. I, in turn, was affirmed in my trust in the process of feminist therapy
2 She's very blonde. How much can we trust her, you see? Because she's very
3 RHODES, Croydon </subh> People don't trust the consortium, because no one
4 n a cold sweat. `Everything is OK, I trust?" Mrs Parsons smiled a little
5 HIV charity, the Terrence Higgins Trust, numb with shock" said director
6 ne outcross from a good friend who I trust and who helps me if I ever I need
7 problem is to transfer into a unit trust portfolio service. Henderson
8 their own weapons, but not ready to trust them in Indian hands. The West and
9 was organized as a Delaware Business Trust on October 24, 1997, is an open-en
10 ch higher returns. If we assume the trust produces 10% a year many have done
11 meningitis, the National Meningitis Trust runs a 24-hour helpline on 0845–60
12 rswick. <p> After retiring from the trust in 1976 he continued to act as a

An evaluative meaning ('have confidence in someone or something') can be identified in lines 1, 2, 3, 6, 8 and a non-evaluative one ('an organisation') in lines 5, 7, 9, 10, 11, 12. Line 4, being a clausal adverbial, is different from either.

Talking about evaluation as contextual and cumulative, however, can imply rather more than this interpretation of words in their immediate phraseology. Hoey, for example, quotes a paragraph (cited here as Example 2.2) which he labels 'evaluation' even though it contains no obviously subjective or attitudinal language:

Example 2.2

Trials have been carried out with freight-dropping at rates from 19 feet to 42 feet per second. The charge weighed about one and half tons, but the system can handle up to eight tons. At low altitudes freight can be dropped without a parachute. (Hoey 1983: 68)

It may be argued that the paragraph does contain indications of capability such as *can handle* and *can be dropped*, but the function of the paragraph as evaluation really depends on its position as the final element in an article

about the development of an air-cushion system to solve the problem of freight being damaged when it is dropped by parachute from a helicopter. Similarly Hunston (1993a: 59) quotes the following paragraph (which was discussed in more detail in Chapter 1 and is repeated here as Example 2.3):

Example 2.3

To determine whether any branched species [of alga] was differentially important to the establishment of surfgrass, I followed seedlings for 7 months. During this time 90% of the seedlings died . . . but the seeds attached to each algal species died in about the same proportion . . . These results suggest that all the branched species facilitate surfgrass and that the magnitude of the facilitation is proportional to the number of seeds originally attached to the alga.

In this example, the three clauses *the seeds attached to each algal species died in about the same proportion*; *all the branched species facilitate surfgrass*; and *the magnitude of the facilitation is proportional to the number of seeds originally attached to the alga* contradict and therefore evaluate negatively the hypothesis 'some branched species are more important to the establishment of surfgrass than others'. Apart from the contrastive *but* in the second sentence, which alerts the reader to the presence of evaluation, there is no attitudinal language, and indeed no expression of opinion, in the paragraph. The evaluative function is based purely on the contrast between *differentially* in the first sentence and *the same proportion* and *proportional* in the second and third sentences. No automatic search of texts or scrutiny of concordance lines would identify this contextually determined evaluation.

As well as being contextual, evaluation is also cumulative. In one sense, this is simply another way of saying that evaluative meanings tend to cluster together. In a book review, for example, the assessment of the book could be said to be the accumulation of all the different things said about it. Where evaluation is highly implicit, however, it could be said that it is only the accumulation of evaluation that makes it noticeable. In the *surfgrass* example, for instance, the hypothesis is evaluated three times, and it could be said that because the evaluation is implicit the repetition is required to make the point.

Context and accumulation can work together. The following example is taken from Jullian (2008), and is an extract from a *Time* article. Example 2.4 cites the opening lines of the article:

Example 2.4

The new museum of Earth history that opened last week in Eureka Springs, Arkansas, isn't nearly as big as more famous natural history museums in Chicago, Washington and New York City; in fact the whole thing would probably fit

> neatly inside one of their exhibition halls. And its nine replicas of dinosaur skeletons and skulls don't quite measure up to the rich fossil collections on display elsewhere. But it's got something the others don't: an account of Earth's history that hews to the most literal version of biblical creationism. (Barnes 2005)

A reader may feel that the museum is being criticised, even sneered at, without being entirely certain what language cues indicate this. In Martin and White's (2005: 67) terms, judgement is being 'flagged', such that the apparently positive *it's got something the others don't* is actually interpreted as negative. The flagging cues are, arguably, *most literal version* and *creationism*. (A survey of Creation Science websites suggests that neither is a term adherents to these views apply to themselves.) It could equally well be argued, however, that the evaluation of the article is so far indeterminate. A Creation Scientist or a sceptic could both interpret the article as being on their side. The next sentence of the article, however, is determinate (Example 2.5):

> Example 2.5
>
> Nestled close to the 20-m-tall Christ of the Ozarks statue, the museum is the latest addition to a theological theme park established almost four decades ago by the late Gerald L. K. Smith, *a right-wing zealot and notorious anti-Semite*. (emphasis mine) (Barnes 2005)

The judgement of Smith, to follow Martin and White again, is inscribed, and this in turn resolves any doubt surrounding the quoted paragraph. In other words, it is possible that some evaluation in some texts remains only potential until it is actualised by the introduction of a more explicit statement or by the sheer accumulation of evidence. Martin and White might argue that in this case judgement is afforded but the reader is not clearly positioned until the inscribed appraisal is encountered.

The fifth point of consensus is that evaluation involves a target, or object, and a source. This is essential to the 'stance triangle' proposed by Du Bois (2007) and to every other model of evaluation. Essentially, a person evaluates an object. There is also consensus that the nature of the evaluation will depend on the nature of the object. This is perhaps stated most explicitly by Hunston (2000a: 199): 'the status of something constrains the criteria or grounds on which it can be given value', but it is implied also in Martin and White's distinction between Judgement (the value of actions and abilities) and Appreciation (the value of things). The source of evaluation is apparently simple (it is the person speaking or writing) but in practice complicated by the process of attribution. This is discussed by Hunston (2000a, 2004a) and by Wiebe et al. (2005), among others, and is noted also by Martin (2000; Martin and White 2005). The complication is of at least two types. Firstly, evaluations themselves might be attributed to speakers other than the author

of the text under discussion. This leads to a layering of evaluation. Wiebe et al. (2005: 173–174) cite an example of so-called reported speech (Example 2.6) which comprises not one but three statements:

Example 2.6

"The US fears a spill-over," said Xirao-Nima . . .

Simplifying the argument somewhat: the writer is the source of an objective speech event (Xirao-Nima uttered some words); Xirao-Nima, mediated through the writer, is the source of a judgement about the state of mind of the US;[1] the US (mediated through Xirao-Nima and the writer) is the source of a subjective speech event about the state of affairs described as *a spill-over*. If an algorithm seeks to identify 'the spill-over is to be feared' as the evaluation, it must also identify the source and the 'layering' of evaluation. Hunston (2000a: 180–181) agrees that the layering can be complex. She quotes an example cited here as Example 2.7:

Example 2.7

Sir James may have caused apoplexy in the Tory party with his tirades against Eurodoctrine, but he has not so far moved enough voters for them to register on an opinion poll.

Here there is an object ('Eurodoctrine') which is evaluated negatively by 'Sir James'. His comments in turn are negatively evaluated by 'the Tory party'. These two evaluations are reported by the writer but also evaluated negatively by him/her in the selection of *tirades* to describe Sir James's utterances and *apoplexy* to describe the feelings of the Tory party. In addition, the writer enacts a dialogue between two voices, one of whom argues that Sir James has caused apoplexy (and so is a force to be reckoned with) and one who argues that Sir James has not attracted popular support (and so is insignificant). This might be represented as:

The writer reports [a second voice who reports that [the Tory party has responded to [Sir James' negative comments about [Eurodoctrine]]]]

suggesting four levels of embedding.

Secondly, the process of attribution may itself evaluate the attributed proposition. This has been extensively researched, from Stubbs (1986) to Hunston (1989, 2000a) and Thompson and Ye (1991). Stubbs (1986: 2–3), for example, cites a BBC news report where a statement about the cause of an explosion ('methane gas caused the explosion') is consistently attributed to sources other than the BBC itself. Example 2.8 is one example (Stubbs 1986: 2):

Example 2.8

A spokesman from the Water Board refused to speculate on whether methane gas could have caused the explosion.

Stubbs comments: 'It is difficult to imagine a more guarded statement'. In this case the source of the proposition is not even mentioned and the BBC is doubly removed from commitment to it. The effect in this case is to mark the proposition ('methane gas caused the explosion') as in no way endorsed by the BBC. As Stubbs (1986: 3) notes, however, attribution can have the opposite effect if the source is identified as authoritative, as in Example 2.9:

Example 2.9

The noted educationalist A. H. Halsey has claimed that . . .

In this case, as Stubbs suggests, the writer's commitment to the proposition is strengthened rather than weakened by the attribution. Hunston (1989) suggests that the choice of attributing verb (*point out* as opposed to *argue*, for example) contributes to the effect of the attribution. Along with others, she argues that whereas responsibility for the proposition is delegated by attribution it can be reclaimed by the choice of an appropriate verb or by an attributive adverbial clause with *as* (*as x says*).

The sixth point, which is less frequently stated openly, is that once researchers set themselves the task of identifying evaluation it becomes difficult reliably to identify anything that is not evaluative. Indeed, it may be said that subjectivity and ideological value permeate even the most objective of discourses. It can reasonably be argued that every text and every utterance is evaluative, so that the phenomenon itself disappears, to be replaced simply by 'language'. In the end, then, researchers have to either insist on an evaluation–non-evaluation distinction (while acknowledging the difficulties therein) or deal with evaluation as a self-contained system, ignoring what might be considered to lie outside it.

2.3 A BRIEF ACCOUNT OF A NUMBER OF APPROACHES

In this section, four of the most relevant approaches to evaluation are briefly outlined, as background information to discussion in the rest of this book.

2.3.1 Appraisal theory

Appraisal theory is the approach developed by Martin and White who work within the tradition of SFL. SFL regards language as performing three major functions: construing a world of experience (ideational); construing

relationships between people (interpersonal); organising each instance of discourse (textual). It also regards language as a system of choices: meaning is construed by making one choice from out of the possibilities. The choices are arranged in a system: at each point in the system the choices are limited, but each choice opens up a new set of possibilities. It is recognised that each of the functions works rather differently (Halliday 1982). Construing experience (the ideational metafunction) is atomistic—the language consists of 'particles' such as processes and participants, such that the parts add up to the whole. Organising the discourse (the textual metafunction) is wave-like—there is no sharp distinction between elements of the clause or the discourse, but there might be a gradual movement from, say, old to new information. Construing relationships (the interpersonal metafunction) is prosodic—evaluative meaning in particular is scattered throughout a clause or a text and can be difficult to pin down. Dealing with evaluative meaning is a relatively recent development in SFL. In defining appraisal as 'one of three major discourse semantic resources construing interpersonal meaning (alongside involvement and negotiation)', Martin and White (2005: 34–35) define their object of attention as 'semantic resources'. That is, they are interested in which meanings are made, rather than in specific linguistic features. Although they may use specific lexical items in exemplifying categories, the category does not consist of the example words themselves. They make it clear (e.g. ibid.: 42) that the systems they envisage are systems of meaning as construed by English. Appraisal itself is divided into three 'interacting domains—attitude, engagement and graduation' (ibid.: 35). These are defined both as systems of meaning and as topics of investigation: '**Attitude** is concerned with our feelings, including emotional reactions, judgements of behaviour and evaluation of things. **Engagement** deals with sourcing attitudes and the play of voices around opinions in discourse. **Graduation** attends to grading phenomena whereby feelings are amplified and categories blurred' (ibid.: 35).

I shall discuss engagement more fully in Chapter 5 and graduation in Chapter 8. Here I focus on the Attitude systems, which are divided into Affect, Judgement and Appreciation. How these work as choices might be illustrated by invented examples each designed to express an opinion about a film. A reviewer has a choice between these meaning types:

- 'I loved the film' (Affect—expressing attitude by construing an emotional response).
- 'The film is carefully balanced' (Appreciation—expressing attitude by construing an aesthetic view of the film as an object).
- 'This director really knows what she is doing' (Judgement—expressing attitude by construing a view of the behaviour of the director as a social being).

Each of the invented utterances in a sense implies all the others but a speaker can choose to express his or her evaluation of the film prioritising

one kind of meaning or another. Although Martin and White present the three kinds of meaning as equidistant from each other in terms of system choice, it seems to me that Affect is different in kind from the other two. If evaluation is both 'expression of a subjective reaction to something' and 'expression of the social value of something', as argued in the previous section, then Affect relates to the first of these and both Judgement and Appreciation to the second. The difference between Affect on the one hand and Appreciation/Judgement on the other is important to the discussion in Chapter 7.

Martin and White sub-divide each of Affect, Judgement and Appreciation into a number of different areas of meaning. For example, Affect is divided into (un)happiness, (in)security and (dis)satisfaction. Each of these is further sub-divided. Judgement falls into two major types: social esteem and social sanction, each again sub-divided. In the sample analyses given by Martin and White (2005: 74–75), each word or phrase construing some kind of attitude is coded with the relevant category. There is much value to this exercise. The predominant meanings in a given text can be identified, for example, or (as in Martin 2000) the resources used by different people expressing attitude in a single text can be separated out. Inevitably, though, there is considerable subjectivity involved in allocating a given expression to one or other category, and there might be some dispute over whether a particular phrase construes 'unhappiness' or 'dissatisfaction', for example. Martin (2000) makes it clear that what he offers is a 'reading' (open to debate) rather than an 'analysis' (a definitive account). What the exhaustive taxonomy does provide, however, is a sense of the very wide expanse of meaning covered by the term 'attitude'. Even if the various distinctions cannot always be nailed down to everyone's satisfaction, the attempt to do so gives a sense of the very wide scope covered by the notion of evaluative language.

2.3.2 Status, Value, Relevance

Hunston's work on evaluation in written academic prose (more specifically, in research articles reporting experimental work) pre-dates that of Martin and White and to some extent has been superseded by it. Working in an area where evaluation is usually highly implicit, it is perhaps not surprising that Hunston, having identified evaluation, found it impossible to distinguish this from non-evaluation, and so focused instead on the types and functions of evaluation identified. Her proposals can be interpreted as suggesting a three-move evaluative act: identification and classification of an object to be evaluated, ascribing a value to that object and identifying the significance of the information. Each move is different in kind from the others, however. The 'objects' are mainly discoursal and epistemic, in the form of propositions giving information, raising questions, drawing conclusions and so on. As each proposition is a discoursal object in this sense, these classified objects occur throughout the text. The value given

to the objects is context-dependent, often implicit, and cumulative. Text construing value may encompass anything from a single clause to several paragraphs. Significance, on the other hand, is indicated only at relatively infrequent intervals in the text and is identified through items of encapsulation among other features. Hunston identifies these 'moves' as the three functions of evaluation: Status (makes an object out of propositions); Value (gives a value to both objects external to the text and to propositions in the text); Relevance (occasionally marks the relevance to the discussion of stretches of text, typically paragraphs). The work on Value has largely been overtaken by Martin and White's model, while Relevance, though arguably of interest, has been largely overlooked. The concept of Status is still used and will be discussed further in Chapter 3.

Although the details of Hunston's model may be said to have been superseded by that proposed by Martin and White, it is perhaps worth drawing attention to one point of difference between the two. Hunston's work, like Martin and White, is bottom-up and based on the investigation of individual texts. Martin and White extrapolated from these individual instances to a general taxonomy of attitudinal meaning. Hunston did not do this, though she did propose a more limited taxonomy of the value meanings construed by empirical research articles. It is difficult to find these meanings accounted for in Martin and White's model. Furthermore, Hunston noted that what counts as evaluative meaning depends very much on what is being evaluated—in other words, the status of the appraised item constrains the interpretation of the appraisal. This is a point that might still be worth making, particularly in instances where the appraisal is highly implicit.

2.3.3 Stance

Stance is a term that is used in two distinct ways in the literature. A number of writers use 'stance' in a way that corresponds to some aspects of evaluation. Conrad and Biber (2000), for example, consider stance adverbials in three different corpora. They sub-divide the adverbials into groups according to two criteria: grammatical form (single word, prepositional phrase and clause) and kind of stance expressed (style, attitude and epistemic). They are then able to quantify the groups in each of the corpora. This enables them to show, for example, that conversation has more adverbial markers of stance than either academic prose or news journalism. In all the corpora the single adverb adverbial is more frequent than other grammatical forms, but the prepositional phrase is proportionally more frequent in academic prose than in the other corpora. In all the corpora, markers of epistemic stance are the most frequent. Style adverbials are more frequent than attitude adverbials in conversation, but these proportions are reversed in academic prose.

It is clear that what Conrad and Biber are investigating are markers or expressions of stance rather than stance as a more abstract concept itself.

A further restriction is that they limit their investigation to one kind of grammatical marker, as opposed to lexical indications of subjectivity. In other words, they investigate 'the use of a grammatical device to provide a personal framing for some other proposition' (Conrad and Biber 2000: 59). In later work, however, Biber (2006a) extends the features quantified to include, for example, attributive markers that also indicate stance (see also Chapter 4 for a discussion of Biber's work on stance).

The work by Biber and others on stance, therefore, takes a quantitative approach to a large amount of data. Much more fine-grained is work in conversation analysis on stance-taking (e.g. Englebretson 2007). The term 'stance-taking' emphasises that these researchers are interpreting stance as an activity rather than as a set of markers or expressions. This precludes quantitative work but allows detailed exploration of individual instances. Du Bois (2007) proposes the 'stance triangle' to represent the act of stance-taking in spoken dialogue. That act comprises all three of: evaluation, positioning and alignment. Du Bois (2007: 163) notes that '[i]n taking a stance, the stancetaker (1) evaluates an object; (2) positions a subject (usually the self); and (3) aligns with other subjects'. Evaluation, then, is the aspect of the action that comprises giving value to an object. The necessary corollaries of being subjective and being intersubjective are separated conceptually from evaluation itself. Perhaps paradoxically, this has the effect of emphasising the interdependence of the three concepts. Evaluating something necessarily indicates 'where you and I stand in relation to the object' and also necessarily indicates 'where you and I stand in relation to each other'. These two aspects relate to the ideological aspect of evaluation ('where you and I stand') and the interactional aspect ('how you and I relate to each other').

2.3.4 Metadiscourse

Researchers in metadiscourse prioritise the interaction between writer and reader and invoke a distinction between 'primary', or informative, discourse and 'secondary', or interactional, discourse. For example, Crismore and Farnsworth (1990: 119) define metadiscourse as 'an author's overt or nonovert presence in the discourse in order to direct rather than to inform readers: the linguistic material given to readers so they will understand what is said and what is meant in the primary discourse'. They follow Vande Kopple's (1985) categorisation of metadiscourse, which includes 'attitude markers' as one of the items expressing the interpersonal function of metadiscourse. Hyland and Tse (2004: 157) offer this definition: 'Metadiscourse is . . . the linguistic resources used to organise a discourse or the writer's stance towards either its content or the reader'. Drawing on G. Thompson's (2001) distinction between 'interactive' and 'interactional', Hyland and Tse (2004: 168–169) propose a two-dimensional model of metadiscourse that incorporates both 'interactive' resources that 'set out an argument to explicitly establish the writer's preferred interpretations' and 'interactional'

resources that 'involve readers in the argument by alerting them to the author's perspective towards both propositional information and readers themselves'. Hyland (2005: 39) aligns his view of metadiscourse with that of Sinclair's (2004: 51–66) concept of the 'interactive plane' that is distinguishable from the 'autonomous plane' (the record of experience) and that negotiates with the reader the structure and meaning of the text itself.[2] Linguistic features listed under 'interactive resources' include indicators of semantic relations between clauses and text sections (*however*, *therefore*, *in conclusion*, etc.) and evidentials (e.g. *X says*, *according to X*) while those under 'interactional resources' include hedges, boosters, attitude markers (*unfortunately*, *I agree*, etc.) and explicit mentions of either the writer (*I*, *we*) or the reader (*you*) (Hyland and Tse 2004: 169).

For Hyland at least, it appears that metadiscourse is subsumed entirely under the concept of interaction or engagement between writer and reader. On the other hand, the distinction between 'interactive' and 'interactional' recalls the 'positioning' and 'alignment' acts proposed by Du Bois. Whereas the more obvious aspects of evaluative language are limited to the 'attitude markers', almost all the features listed in his model have been discussed under the heading of 'evaluation' by one writer or another.

2.4 CONCLUSION: THE PROBLEM OF EVALUATION IN CORPUS STUDIES

This chapter has discussed the consensus on evaluation and has outlined very briefly a number of different approaches to the topic. (More approaches are discussed in Chapter 4.) Priority in this chapter has been given to text-based approaches that stress that evaluation is an action performed in discourse, that it is a set of meanings and that it is always contextually determined. These nuanced, highly interpretative approaches pose a challenge for corpus work. Obviously a corpus approach can perform a function in quantifying expressions and in providing the basis, with substantial evidence, for comparisons between discourse types. What it cannot do is replicate the kinds of analysis performed by Du Bois or by Martin and White. As will be seen in Chapter 4, corpus studies effectively answer a different set of questions. The purpose of this book is to identify what kind of questions they are and how the answers might be arrived at.

3 Status in Written Texts and Multi-Modal Texts

3.1 INTRODUCTION

This chapter revisits the concept of epistemic status, as discussed in Hunston (1989) and subsequent publications (e.g. Hunston 1993a, 1993b, 1994, 2000a). In Hunston (1989) it was presented as one function of evaluation, alongside 'value' and 'relevance'. As noted in Chapter 2, of the three suggested concepts, status is the one that may still have some currency. This chapter reviews the concept of status and the linguistic resources associated with it (section 3.2), including an exploration of the relationship between status and attribution (section 3.3). The next section of the chapter (3.4) takes us beyond written text into the world of documentary films, and considers how 'status' might be applied to visual texts as well as verbal ones. There follows a discussion of how status is related to other concepts, especially 'Engagement', that apparently account for roughly similar phenomena (section 3.5). The chapter includes numerous examples in which a proposition is identified along with markers of status. The convention used in the examples in the chapter is that propositions are underlined and status markers are put into bold.

It is argued in this chapter that one of the functions of evaluation is to reify texts and propositions by assigning them an epistemic status. The status so assigned aligns a text or a proposition to a construed world. As will be argued in the following, the alignment is not inevitable but is averred by the text writer. Examples of the averred alignments of texts include assertions that a book is fiction or non-fiction, or that a segment of film is news reportage or reconstruction. Instances of the averred alignments of propositions include their labelling as hypotheses, possibilities or facts.

Before going further, three important points need to be made concerning status and its evaluative nature. Firstly, like all evaluation, the assignment of status to a text or proposition is an exercise in subjective judgement by the writer or speaker of the text in which the assignment is made. Following Teubert's (2005: 3) stipulation that '[t]here is no direct link between the discourse and the "real world"', the analyst's task is to identify the nature of that judgement, not to agree or disagree with it. For instance, in Example

3.1, the proposition 'the position and momentum of the electron are defined all along' is evaluated by the writer of the text as an 'interpretation'. This evaluation by the writer makes an assertion about the relationship between the proposition and the physical world. For the analyst, however, identifying the status as an 'interpretation' makes an observation about the text itself, not about the behaviour of electrons.

Example 3.1

There is an alternative **interpretation** . . . the position and momentum of the electron are defined all along.

Secondly, as with all evaluation, the assignment of status is a matter of social significance, as evidenced by the focus on status in reviews of books and films. Section 3.4 discusses examples of television documentaries whose topic is status itself. The film *Truth and Lies* asks whether the book *Fragments* is autobiography or fiction (Hunston 2006b). The documentary *The Real Da Vinci Code* questions whether claims in the book *Holy Blood, Holy Grail* are based in fact or speculation (Hunston 2008b). The case of *Fragments* is particularly interesting: the book recounts the life of a child in a concentration camp during the Second World War but was denounced by critics who claimed, with subsequent substantiation, that the writer had not experienced these events himself. Even though the book was acknowledged to reflect events of the kind that did happen, and might have been very successful as a novel, the outcry when the 'fraud' was discovered was such that the book was withdrawn from publication.

So-called 'reality TV' seems to be a site of frequent debate concerning status. In June 2007 controversy over a Dutch TV show where three contestants apparently competed to receive a dying woman's kidneys was resolved when the programme was labelled a 'hoax' by its makers.[1] The makers of the UK's *Big Brother* programme, responding to the judgement by (the UK media regulator) Ofcom about racist abuse broadcast in one transmission, said, 'We are grateful to Ofcom for recognising that the events of this series were in no way engineered or manufactured'. This suggests that the status of the offending segments as 'real' was perceived as a defence against accusations of poor management of the issue.[2]

Finally, status is intrinsically linked with evaluations of 'good' and 'bad', especially in texts which are part of a knowledge-building agenda, or texts which seek to influence actions using rationality as a means of persuasion. Example 3.2 is an extract from a website maintained by the British National Health Service and designed both to give parents information about the combined Measles-Mumps-Rubella (MMR) vaccine and to persuade them to allow this vaccine to be given to their children. Uptake of the vaccine in the UK declined in the 1990s and 2000s because of reports that it was linked to an increase in numbers of children with autism.

Example 3.2[3]

It has been **suggested** that there is a new condition called Regressive Autism (where a child seems to be developing normally but starts to show signs of autism between 1 and 3 years of age)—where children lose skills like speech and actions, and that this occurred after MMR.

This has been studied and it has been **proved** that the proportion of autistic children with regression is the same whether they have had MMR or not, and that there has been no increase in the proportion with regression after MMR was introduced.

This means that MMR has not caused their regression.
. . . .
The **scientific evidence** is that MMR does not cause autism.

In this example, a proposition 'there is a new condition . . . this occurred after MMR' is evaluated as a 'suggestion' whereas other, contrary, propositions like 'there has been no increase in the proportion . . . MMR has not caused their regression . . . MMR does not cause autism' are evaluated as 'proven' and the outcome of 'scientific evidence'. This implies that the statement 'MMR does not cause autism' is more trustworthy than its opposite, and also that this evaluation is the outcome of objective, rational thinking rather than of a subjective appraisal. The rhetorical devices of setting a 'suggestion' against 'proof', and of maintaining an apparently objective stance, are key in the persuasive strategy of the website.

Evaluation of status appears to be particularly important to academic registers (Hunston 1993a, 1993b, 1994). In academic writing in the empirical sciences in particular, the status of propositions comprises an essential aspect of the discipline concerned. Evaluation of status essentially reifies propositions into the objects of which the discipline is comprised: hypotheses, results, conclusions, assumptions, implications and so on.

3.2 EVALUATION OF STATUS: THE LINGUISTIC RESOURCES

One of the basic tenets of the concept of status is that all propositions in texts are evaluated in terms of how they are aligned with the world. In other words, evaluation of status is an obligatory element throughout all texts. Status may be marked or unmarked, as demonstrated in Examples 3.3 and 3.4 (both from a website hosted by the UK Foreign & Commonwealth Office dealing with Avian Flu).[4] Where the status is unmarked, a default interpretation is made based on the status of the text as a whole.[5]

Example 3.3

Experts began monitoring a form of avian influenza . . . eight years ago.

Example 3.4

Almost all [human victims of avian flu] **are thought to** have caught the disease directly from infected poultry.

In each sentence, a proposition has been underlined. In Example 3.3 the proposition is presented without modification. The default interpretation is that this proposition is directly aligned with the world; the reader is asked to believe that a group of people called 'experts' undertook an activity labelled as 'monitoring' and that this activity began eight years before the text was written. In Example 3.4 the proposition is modified by the phrase *are thought to*, which alters the represented alignment: that infected poultry is the sole source of the disease is represented not as 'what is' but as 'what might be' or 'what probably is' or 'what people think is'.

Typical resources for status modification include:

- Verbs, nouns and adjectives governing that-clauses e.g. *It has been suggested that . . . The discovery that . . . It is probable that . . .*
- Those same verbs, nouns and adjectives in other grammatical contexts e.g. *As was suggested . . . This discovery . . . the probable cause . . .*
- Adverbs and adverbials such as *probably*, *supposedly*, *in fact*.
- Modal auxiliaries e.g. *may*, *might*, *could*.
- Other forms of attribution or identification of the source of a proposition (often called 'evidentials') e.g. *According to Smith . . . There is a general consensus that . . . As is well known . . .*

In many texts whose declared purpose is to give information, the evaluation of status is relatively straightforward. Most propositions are unmodified and express 'what is'; others are modified by adverbs, modal verbs or clauses governing that-clauses. Table 3.1 shows a commentary on each of six propositions from Example 3.5 (from the same website as Examples 3.3 and 3.4).

Example 3.5[6]

[1]A severe form of avian influenza or 'bird flu'—called H5N1—has affected poultry flocks and other birds in several countries since 2003.[2]France has been the most recent country to announce a case of H5N1 in a wild bird.[3]As of 23 February 2006, 170 people have also caught the infection, as a result of close and direct contact with infected birds. [4]Ninety-two of these have subsequently died. [5]**There is no firm evidence that** H5N1 has acquired the ability to pass easily from person to person. [6]However, **concern remains that** the virus **might** develop this ability, or that it **might** mix with human flu viruses to create a new virus.

Table 3.1 Status in Example 3.5

Sentence	*Proposition*	*Rough paraphrase of Status*	*Indicated by . . .*
1	a severe form of avian influenza or 'bird flu' has affected poultry flocks and other birds in several countries since 2003	'what is'	No modification
2	France has been the most recent country to announce a case of H5N1 in a wild bird	'what is'	No modification
3	as of 23 February 2006, 170 people have also caught the infection	'what is'	No modification
3	[this has come about] as a result of close and direct contact with infected birds	'what is'	No modification
4	ninety-two of these have subsequently died	'what is'	No modification
5	H5N1 has acquired the ability to pass easily from person to person	'what is unlikely to be'	Modified by clause governing that-clause: *There is no firm evidence that*
6	the virus will develop this ability	'what might be'	Modified by clause governing that-clause: *concern remains that* and by modal *might*
6	it will mix with human flu viruses to create a new virus	'what might be'	Modified by clause governing that-clause: *concern remains that* and by modal *might*

As Example 3.5 illustrates, evaluation of status may be the most important type of evaluation in an apparently 'objective' text. The more overtly subjective or emotive the text, however, the more complex the evaluations of status are likely to be. Example 3.6 is from a web-based publication entitled 'What Doctors Don't Tell You' and it both gives and assesses information:

Example 3.6[7]

[1]We are all in the grip of avian-flu hysteria. [2]**The World Health Organisation (WHO) estimates that** the virus **could** claim up to 7.4 million lives around the

world if it successfully mutates and becomes transmissible among humans,
[3]while **the UK's chief medical officer Sir Liam Donaldson says** it's a matter of when and not if.
[4]When it happens, it **could** claim the lives of up to 750,000 Britons, **he estimates**
[5]while, in America, it **may** kill up to two million people, **warns President Bush**.
. . .
[6]**Why should scientists suddenly fear that** the H5N1 virus is likely to mutate soon, and become transmissible among humans, **when it has been around for at least 50 years?**
[7]There **appears to be** no scientific basis for ***the dramatic public warnings*** that are being issued. . .

Sentences 2–5 each contain a proposition that is modified by a clause governing a that-clause—*The World Health Organisation estimates*; *the UK's chief medical officer says*; *he estimates*; *warns President Bush*—as well as by modal verbs (*could*, *may*). Each of these propositions, then, is 'what might be' or 'what is said to be' rather than 'what is'. Sentence 1 is different, however. It is unmodified and so represents a direct alignment between the proposition and the world. Some of the lexis, however, is marked for subjectivity: *in the grip of* and *hysteria*. This suggests an 'opinion' rather than a 'fact': 'what we think is' rather than 'what is'. Sentence 6, an interrogative, also presents problems in terms of identifying propositions and their status. The proposition 'the H5N1 virus will mutate soon and become transmissible among humans' is modified by a clause governing a that-clause (*scientists fear that*) but there is the added modification of *why should . . . suddenly . . . when it has been around . . .* These modifications turn a 'what might be' proposition into a 'what is unlikely to be' one. On the other hand, two further propositions: 'Scientists fear x' and 'it has been around for at least 50 years' are 'what is'. (Further complications concerning sentence 6 will be discussed in Section 3.3.) Sentence 7 contains a proposition 'There is no scientific basis for the dramatic public warnings that are being issued', modified by *appears to be* and paraphrasable as 'what is likely'. Because *the dramatic public warnings* is anaphoric, however, and summarises or encapsulates (Francis 1986; Sinclair 1995, 2004) the reported speech in sentences 2–5, sentence 7 also re-evaluates the status of the propositions in those sentences as 'what is unlikely to be'. Table 3.2 summarises this commentary on Example 3.6.

A question that has not so far been addressed is whether the status of propositions in a text can be the subject of textual analysis. I take 'analysis' to mean the identification, on named criteria, of categories of objects, and the labelling of each object according to the category it belongs to. Status labels, for example, might include FACT, HYPOTHESIS, SUPPOSITION,

Table 3.2 Status in Example 3.6

Sentence	*Proposition*	*Rough paraphrase of Status*	*Indicated by . . .*
1	we are all in the grip of avian-flu hysteria	'what we think is'	*in the grip of*; *hysteria*
2	the virus will claim up to 7.4 million lives around the world if it successfully mutates and becomes transmissible among humans	'what is said to be' → 'what is unlikely to be'	Modified by clause governing that-clause: *The World Health Organisation (WHO) estimates* and by modal *could* Modified by *there is no basis for*
3	it's a matter of when and not if	'what is said to be' → 'what is unlikely to be'	Modified by clause governing that-clause: *the UK's chief medical officer Sir Liam Donaldson says* Modified by *there is no basis for*
4	it will claim the lives of up to 750,000 Britons	'what is said to be' → 'what is unlikely to be'	Modified by clause governing that-clause: *he estimates* and by modal *could* Modified by *there is no basis for*
5	in America, it will kill up to two million people	'what is said to be' → 'what is unlikely to be'	Modified by clause governing that-clause: *warns President Bush* and by modal *may* Modified by *there is no basis for*
6	scientists suddenly fear that the H5N1 virus is likely to mutate soon	'what is'	Presupposed
6	it has been around for at least 50 years	'what is'	Presupposed
6	the H5N1 virus will mutate soon, and become transmissible among humans	'what is unlikely to be' → 'what is unlikely to be'	Interrogative with *why should* and clause *when . . . years* Modified by *there is no basis for*
7	there is no scientific basis for the dramatic public warnings that are being issued	'what is likely to be'	Modified by *appears to be*

ESTIMATE and so on. It will be noted, however, that in Tables 3.1 and 3.2 such categorisation was avoided in favour of far less rigorous 'rough paraphrases'. For this reason, the tables represent commentaries on, rather than analyses of, the examples they relate to. It would accord better with linguistic practice to offer a set of distinct categories that relate to each other in specified ways (cf. Sinclair and Coulthard 1975: 15–17). I attempted to do this (Hunston 2000a: 187), suggesting a system network of 'statement types' (see Figure 3.1).

The network makes, I believe, a number of useful distinctions. The most basic distinction is between sentences which are metalinguistic and which refer to the text itself (e.g. *Table 3.1 provides a commentary on example 3*) and others (termed 'informing'). The metalinguistic propositions have a focusing function and are not aligned in any way with the world outside the text. Therefore, evaluation of status does not apply to them. Also of interest is the basic division with 'informing' statements between 'world-reflecting' and 'world-creating' propositions. This distinction is similar to Searle's notion of 'direction of fit' (1985: 18): world-reflecting propositions fit the world, whereas world-creating propositions make a world that fits them. For example, *the virus could claim up to 7.4 million lives* is a statement that fits the world (however tentatively), whereas *Let's assume a worst-case scenario in which avian flu has taken hold* creates a (hypothetical) world that fits the assumption. World-reflecting propositions invite further discourses in which the alignment of the proposition to the world is debated; world-creating propositions do not. World-reflecting propositions are much more frequent than world-creating ones are; Examples 3.1–3.4 in this chapter consist exclusively of world-reflecting propositions. In essence, then, the network serves to exclude from consideration a few, relatively infrequent

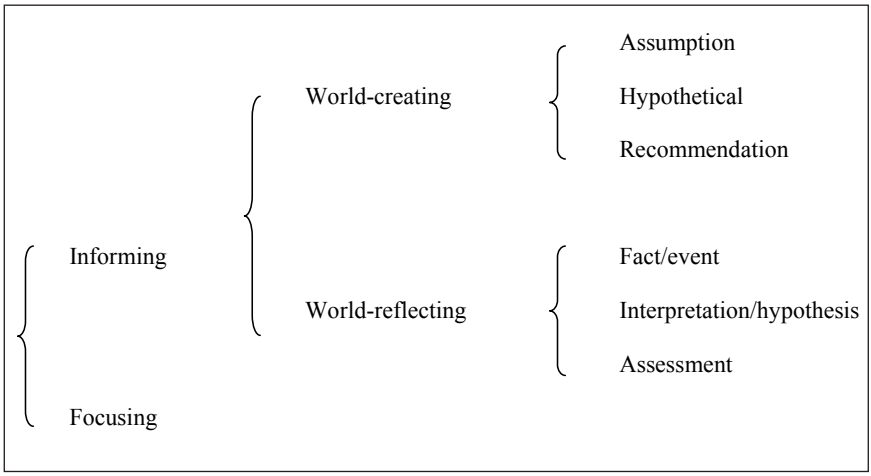

Figure 3.1 From Hunston (2000a: 187).

statement types such as *Figure 1 shows . . .*, *I assume . . .*, *I recommend . . .* to allow the analyst to focus on the more frequent, world-reflecting propositions. Attempts to create more delicate categories are not wholly successful, however. In Hunston (2000a) I suggested three categories of world-reflecting propositions: 'facts' or 'events', where the proposition is expressed as being aligned with the world ('what is'); 'interpretations' and 'hypotheses', where the proposition is aligned with a possible world ('what might be' or 'what is said to be'); and 'assessments', where the proposition is aligned with a subjective world ('what we think is'). Such a broad classification works to a certain extent, but it masks more delicate distinctions and the possibility of modification within categories. It also masks more interesting issues, such as the complex dialogue between propositions that will be explored in the next section. On the whole, then, I feel that a taxonomy of status is probably at least premature and possibly ultimately not a useful exercise.

3.3 STATUS AND ATTRIBUTION

As noted earlier, one of the key modifiers of the status of a proposition is its attribution by the writer to another speaker. Attribution is contrasted by Sinclair (1987) with averral. All propositions, according to Sinclair, are either averred (construed as spoken by the writer) or attributed (construed as spoken by someone else). This section deals with the interaction between status and attribution.

There are clear resonances here with research into related topics, including citation and reporting practices (e.g. Swales 1986; Thompson and Ye 1991; Hyland 2000; P. Thompson 2001; Charles 2006a, among many others), speech and thought representation (e.g. Semino and Short 2004) and evidentiality (e.g. Chafe and Nichols 1986).

Explicit modification of a proposition indicating its status often takes the form of a clause governing a that-clause. Examples from Examples 3.3 and 3.4 are:

i. There is no firm evidence that . . .
ii. Concern remains that . . .
iii. The World Health Organisation estimates . . .
iv. The UK's chief medical officer says . . .
v. President Bush warns . . .

In these cases, what is grammatically the main clause in the sentence acts as a modification of the that-clause, modifying the status of the proposition therein. Halliday and Matthiessen (2004: 613) treat clauses such as (i.) and (ii.) as metaphors of modality. The clause *there is no firm evidence that* indicates that the following proposition is possible but

unlikely to be the case, whereas *concern remains that* indicates that the following proposition is possible but not certain. Attributing a proposition to a specific other speaker can, arguably, perform a similar function. Because the author takes no responsibility for the attributed proposition, its certainty is modified. On the other hand, as is well known, attributing a proposition to a respected source can confirm rather than question veracity, and the confirmation can be strengthened if the attributing mechanism functions to reaffirm author commitment. Examples include the attribution verb *point out* and the prepositional phrase *as x says/ argues/ claims/ etc*. Thus, whereas attribution always modifies status, it can do so in a number of ways, depending on the verb or other mechanism of attribution and on the attributed source. There have been a number of proposals for taxonomies of source types (e.g. Stubbs 1996: 150; Hunston 2000a: 190; Wickens 2000), each of which highlights distinctions such as between the author and other, between named people and unspecified groups and between sources that are human and those that are textual in nature. Taxonomies tend to be complicated by the fact that they often try to deal not only with the question of who is the Speaker of the proposition, but also with issues such as how that Speaker is represented and whether the author delegates or reclaims responsibility for the proposition, as each of these affects the status of the proposition. To illustrate this point, consider the following invented sentences:

i. I am afraid that bird flu will spread.
ii. It is likely that bird flu will spread.
iii. Scientists point out that bird flu will spread.
iv. Concern remains that bird flu will spread.

In each case, the proposition is 'bird flu will spread'. The source for this proposition in (i.) and (ii.) is the author; in (iii.) and (iv.) it is others, probably 'scientists' in both cases. In (i.) and (iii.) the source is named, though more explicitly in (i.) than in (iii.); in (ii.) and (iv.) it is not. The author has or reclaims responsibility for the proposition in each of (i.)–(iii.) but delegates responsibility in (iv.). Representing these different types of distinction is more readily done in a matrix than in a network: Table 3.3 illustrates how this might be done.

So far in this chapter, status has been treated as a monologic entity. That is, it has been assumed that a proposition such as 'the virus could claim up to 7.4 million lives around the world' (from Example 3.4) has a single status based on the fact that it is attributed, that the source is the WHO and that the verb of attribution is *estimates*. Inevitably, however, because intertextuality is an issue here, the status is dialogic. In other words, the proposition has a status that is attributed to the WHO as

Table 3.3 A Classification of Source Types

Example	*Source?*			*Specified?*		*Responsibility?*	
	Self	*Other*		*Specified*	*Unspecified*	*Author has respon-sibility*	*Author delegates respon-sibility*
		Group	*Indiv*				
I am afraid that	√	x	x	√	x	√	x
It is likely that	√	x	x	x	√	√	x
Scientists point out that	x	√	x	√	x	√	x
Concern remains that	x	√	x	x	√	x	√

well as the status from the point of view of the author of this text. (It is important to recognise, as with all attribution, that we as readers of Example 3.4 do not know whether the WHO ever produced such a figure or whether they reported it as an estimate.) In this case, the attributed status is an estimate: 'a rough approximation of what might be', and the averred status is an estimate-for-which-no-responsibility-is-taken.

The importance of the dialogue between statuses is clearer if we consider sentence 6 from Example 3.6, repeated here as Example 3.7:

Example 3.7

Why should scientists suddenly fear that <u>the H5N1 virus is likely to mutate soon</u>, and become transmissible among humans, when it has been around for at least 50 years?

As pointed out in Table 3.2, there are a number of propositions here, but we will look only at the attributed one, underlined in Example 3.7. The sentence construes a difference of opinion between the unnamed 'scientists' to whom the proposition is attributed, who regard it as closely aligned with the (future) world, and the author, who regards the same proposition as less closely aligned to the world. According to the author, the scientists regard the mutation of the virus as 'possible' whereas the author regards it as 'unlikely'. In other words, the sentence construes a debate in which what

is at issue is the status of the underlined proposition. A representation such as Table 3.3, which treats status as monologic, misses this intertextuality.

A final point to be made about status, attribution and sources is that how a source is regarded depends very much on the wider context of the text concerned. In Example 3.6, the WHO and the UK chief medical advisor are effectively treated as unreliable sources because their utterances are referred to as *hysteria* and *dramatic public warnings* for which there is no evidence. Example 3.8 is from the same Foreign & Commonwealth Office website as Example 3.5.

Example 3.8[8]

> The World Health Organisation (WHO) has warned that recent outbreaks of bird flu, also known as avian influenza, could trigger a future human flu pandemic. This factsheet explains why there is international concern about this. It describes sensible precautions for travellers and residents in affected areas, both in respect of avian influenza, and in respect of a possible future human flu pandemic.

In this example, the WHO is treated as an expert institution whose utterances are to be taken seriously. There is nothing in the rest of the text to suggest otherwise. Thus, whereas the proposition attributed to the WHO in Example 3.6 might be glossed as 'WHO thinks this is possible but we think it is unlikely', the equivalent proposition in Example 3.8 might be glossed as 'WHO and we both think this is possible'. Example 3.6 construes conflict, but Example 3.8 does not, and this has an effect on the evaluation of status of each proposition.

Status, then, may be dialogic, and may be particularly complex when different points of view are construed. This is possible because the writer has a voice even in attributed statements (Hunston 2004a). Example 3.9 contains a number of propositions that are attributed by the writer to another speaker, Rajagopalan. (In this example, LI is an abbreviation of 'Linguistic Imperialism'. It might be glossed in this example as 'people who ascribe to views informed by linguistic imperialism'.)

Example 3.9

> He [Rajagopalan] states that LI's fundamental premises are '(a) that in a monolingual setting communication is always perfect and (b). . .'. He thus characterises LI as motivated by 'romantic imagination' and 'religious credos'. He goes on to indirectly accuse them of holding the naïve belief that power relationships can be eradicated altogether. . .. (Canagarajah 1999: 208)

These propositions are:

i. 'LI's fundamental premises are that in a monolingual setting communication is always perfect . . .'
ii. 'LI is motivated by romantic imagination and religious credos'.
iii. 'LI [researchers] hold the naïve belief that power relationships can be eradicated altogether'.

In each case the writer avers that something fitting the writer's paraphrase is attributable to Rajagopalan. In other words, whereas the proposition 'LI is motivated by romantic imagination and religious credo' is attributed to Rajagopalan, the proposition 'He thus characterises LI as motivated by romantic imagination and religious credo' is averred by the writer.

Of course, the situation in Example 3.9 is further complicated by the fact that the writer also avers that Rajagopalan attributes propositions to other speakers, namely to the body of people characterised as 'LI'. These propositions are:

iv. 'in a monolingual setting communication is always perfect'
v. 'power relationships can be eradicated altogether'

What we read in Example 3.9, then, is the writer's interpretation of Rajagopalan's interpretation of the words and thoughts of a non-specifically identified group of people. In the terms introduced by Leech and Short (1981) and further developed by Semino and Short (2004), this is the presentation of speech and thought, not the reporting of speech and thought. In addition to the interpretation demanded in the process of selection and paraphrase, the writer evaluates each proposition in terms of its status. The grammatical resource used to attribute is in each case: a verb followed by a that-clause (*states that*); a verb followed by a noun phrase indicating the topic and a prepositional phrase with *as* (*characterises LI as*); a verb followed by a noun phrase indicating the topic and a prepositional phrase with *of* (*accuse them of*); a noun followed by a that-clause (*belief that*); and a noun and that-clause separated by a copula verb (*premise is that*). The noun or verb selected (*state*, *characterise*, *accuse*, *belief*, *premise*) modifies the assessment of the proposition. The assessment can be complex. In the final sentence of Example 3.9, for instance, repeated here as Example 3.10:

Example 3.10

He goes on to indirectly accuse them of holding the naïve belief that power relationships can be eradicated altogether.

there is a complex interaction between the writer's own assessments and Rajagopalan's, in each case as construed by the text. The proposition 'LI researchers hold the naïve belief . . .' is construed as having alignment

of word and world for Rajagopalan, while the writer's own assessment of its status is indeterminate. (To oversimplify the matter, Rajagopalan thinks the proposition is true but the writer does not divulge an opinion as to its truth or otherwise.) However, the writer also evaluates the mental action of 'holding the naïve belief' as undesirable though the choice of *accuse . . . of*. In addition, the proposition 'power relationships can be eradicated altogether' is evaluated as 'possibly true' by the word *belief* but as 'unlikely to be true' by the word *naïve*. It is open to debate whose evaluation this is. Whereas it is clear that Rajagopalan is construed as evaluating the proposition as *naïve belief*, it is not clear whether the writer concurs with that evaluation. Also, 'power relationships can be eradicated altogether' is construed as something that is evaluated as aligned with the world by the L1 researchers, though it is not certain whether the writer holds responsibility for that construal or whether this is delegated to Rajagopalan.

In sum, attribution is one of the ways in which status can be modified. There is a wide variety of linguistic resources used in attribution, and these impinge on the effect of attribution on status. Firstly, there are a variety of grammatical structures (verb + that-clause, verb with *as* and so on) that are used in attribution. If we assume that attribution involves both the writer's voice and that of the attributee, we might say that the writer's voice is stronger where the attribution is not expressed through a that-clause. In other words, it is argued that in

> *Rajagopalan states that LI's fundamental premises. . .*

the writer's own voice is subordinated to that of Rajagopalan and might be overlooked completely. However, in

> *He thus characterises LI as motivated by romantic imagination and religious credos*

the writer's voice as interpreter of Rajagopalan's views is much stronger. Secondly, the specific noun or verb chosen in the attribution affects status. Although attribution itself devolves responsibility for the proposition (and thus makes the averred alignment between word and world less exact), the choice of verb or noun allows the writer either to reclaim that responsibility or to distance him/herself still further. For example, in

> *Rajagopalan* ***states*** *that* <u>*LI's fundamental premises*</u> *. . .*

the status of the underlined proposition can be glossed as 'what is said by others to be', or 'what might be', whereas in

Rajagopalan ***points out*** *that* *LI's fundamental premises* . . .

the status is glossed as 'what I and others believe' or 'what is probably the case', and in

Rajagopalan ***naively claims*** *that* *LI's fundamental premises* . . .

the status is 'what others wrongly say' or 'what is unlikely to be the case'. Thirdly, the construed source of each proposition—whether this is the writer, another speaker, 'common sense' and so on—affects its construed status.

It has also been argued in this section that attribution is essentially a dialogic feature of text (in agreement with Martin and White 2005) and that this leads to complexity when status and attribution come into play, as writer and attributee may be construed as being in conflict in terms of the status of a proposition. Both dialogue and conflict are important to the presentation of an argument; the complexities demonstrated earlier arguably make it more difficult for a reader to isolate a single voice for (dis)agreement.

3.4 STATUS IN MULTI-MODAL TEXTS

Thanks largely to the work of Kress and van Leeuwen (2006), it is now widely acknowledged that concepts originally developed to describe meaning in language can and should be extended to account for other semiotic systems. The notion of status has been applied to filmed images as well as to language (Hunston 2006b, 2008b), and this work will be summarised briefly here. It is based on discussions of two documentary films shown on British television:

Truth and Lies (BBC 2000)
The Real Da Vinci Code (Channel 4 2005)

What these documentaries have in common is that both assess the status of other, written texts. In *Truth and Lies* two views are presented of a book (entitled *Fragments*) about a child growing up in a concentration camp during the Second World War. Either the book is a factual account of the writer's childhood as a concentration camp victim, or it is an imaginary account of such a childhood written by a man who was born and grew up in Switzerland. In *The Real Da Vinci Code*, texts that assert the truth of the basic tenets of the novel *The Da Vinci Code* are assessed, not only for

their veracity, but also for the extent to which they are based on research or are pure speculation.

In their seminal book *Reading Images* (2006), Kress and van Leeuwen apply concepts from systemic-functional semiotics to visual images as well as to verbal language. The semantics of modality is one of the areas they apply to images, referring to aspects such as colour and amount of detail that make the image appear more or less 'real' (2006: 160–163). They stress that to talk of 'reality' in this context is not to make an actual judgement about the relationship between the image and the world, but only to analyse the representation (ibid.: 154). Just as the semiotic concept of modality can be applied to images, so can the evaluative concept of status.

In two previous publications (Hunston 2006b, 2008b) I have discussed how evaluation of status may operate in documentary film texts. It is argued that documentaries are compiled from a sometimes very large number of component texts. These include texts consisting of words only, typically the unseen presenter's voice-over, texts consisting of images only, such as photographs or filmed sequences without speech, and texts consisting of both image and speech, such as filmed interviews and the like. The concept of status can be applied both to the spoken language and to the images. Both speech and image demonstrate an evaluation of their alignment with the world. In speech this is done using the same resources of language as are used in writing: a proposition may be unmodified, or modified through modal verbs, adverbs and so on, or it may be attributed. Example 3.11 comes from the documentary *Truth and Lies* and is part of the narrative spoken by the film's unseen presenter. In this passage, the presenter suggests that Wilkomirski (the writer of *Fragments*) may not be giving a true account of his childhood, arguing that all the information Wilkomirski claims to have through memory might actually be the product of research. The passage is heavily modified throughout (status indicators are shown in bold) and construes a conflict between Wilkomirski on the one hand and his detractors (*a Jewish historian*, *the official historian at Majdanek*) on the other.

Example 3.11

In Riga, which **he [Wilkomirski] says** he left when he was two or three, could he really have remembered his house? . . . Bernstein and Verena returned with him to Riga, which he **seemed** to know like the back of his hand. . . . But **a Jewish historian says** his account of escaping with his mother and brothers across the river is **highly unlikely**. **The fact is that** almost everything about the Nazi occupation of Riga, and the murder of the Latvian Jews, is well known. . . . **Wilkomirski admits** he studied the holocaust from college onward. Before this trip to Majdanek in 1993 his passports disclose several previous trips to Poland. On any one he **could** have visited Majdanek. He **could** have stitched his story together from what he'd read. **The official historian at Majdanek says** Wilkomirski could only have survived if the SS hadn't

> known he was a Jew. Since **he says** he's circumcised, that **seems unlikely**. . . . **He claims** he was . . . transferred to Auschwitz. 38 women and children were transported from Majdanek to Auschwitz in April 1944 but **as far as is known** they were killed. . . . [transcribed from broadcast]

Similar resources are used in the film *The Real Da Vinci Code*. A typical example (3.12) comes from an interview in which the interviewee comments on manuscripts claimed to be alternative accounts of Jesus's life and teaching (that is, alternative gospels).

> Example 3.12
>
> These were remarkable texts that- One of the first of them opens with the words 'These are the secret sayings which the living Jesus spoke'.

Here, as in the written texts discussed earlier, there is complexity in the assignment of status due to layering of attribution. The film-makers give a representation of the interview; the interviewee represents the gospel writers as writing: 'These are the secret sayings which the living Jesus spoke'. According to the interviewee's representation, the gospel writers present their statement as aligned with the world. It is not clear whether the interviewee agrees with alignment, although one interpretation of 'remarkable' suggests that she regards it as at least highly possible. The film-maker's evaluation remains obscure in that there are no verbal clues to it, neither are there obvious visual clues. The interviewee is represented as speaking for herself (the interviewer is not in shot), and the setting and camera angles give her dignity: she is a person whose views might be respected. This does not amount, however, to an assertion that her views are necessarily correct.

The averred alignment of image and world has been extensively discussed by Kress and van Leeuwen (2006) under the heading of 'modality'. Kress and van Leeuwen describe the choices that image-makers have in 'expressing meanings of truth and falsehood, fact and fiction, certainty and doubt, credibility and unreliability' (2006: 154). For example, in discussing colour saturation in an image, they note that:

> We judge an image real when, for instance, its colours are approximately as saturated as those in 35mm photographs. When they are more saturated, we judge them exaggerated, 'more than real', excessive. When they are less saturated we judge them 'less than real', 'ethereal', for instance, or 'ghostly'. (Kress and van Leeuwen 2006: 159)

As well as colour (differentiation and modulation as well as saturation), Kress and van Leeuwen identify degree of detail of context, degree of abstraction or representation, level of illumination and brightness as other

contributors to modality. For films we might add speed of movement, as 'unreal' segments may appear to be slowed down.[9]

In both the documentaries mentioned earlier there are segments that are visually 'real', such as the interviews. What are more interesting, though, are the segments in each film that represent 'what might have been' rather than 'what was'. In *Truth and Lies*, events from Wilkomirski's childhood—the death of his father, his journey to the concentration camp and so on—are shown with low modality, indicated by low colour saturation, low colour differentiation and lack of contextualisation. Clearly, these segments are reconstructions rather than 'the real thing'; it is understood that the boy is not Wilkomirski himself, the train is not the one he travelled on and so on. This would be the case whether the book *Fragments* were considered to be factually accurate or not. I argue that the status of the reconstructions is ambiguous or indeterminate. These segments of film could represent reconstruction of actual events—'This is like what was in all important aspects'—or could represent a deceptive fiction—'This is what Wilkomirski would like us to believe'. The ambiguity is not resolved until the second half of the film, when evidence is offered for the book *Fragments* being invention rather than record. At that point the viewer is in a position to re-evaluate the reconstructions in terms of their status.

Very different visual resources are used in *The Real Da Vinci Code* to a similar end. In that film, scenes from the hypothesised life of Mary—showing her preaching, attracting jealousy from Jesus's disciples, kissing Jesus and holding their baby—are depicted in a series of cartoon-like drawings. These are abstract, low in contextualisation and of higher than natural colour saturation. Like the reconstructed scenes in *Truth an Lies*, these are clearly not intended to be 'real' in any sense (it would be unreasonable to expect contemporary drawings of Mary to exist), but the highly unrealistic representations seem designed to draw attention to their indeterminate status.

In both cases, then, reported propositions ('my father was killed by Nazis in Riga', 'Mary was the lover of Jesus') are given visual representation in reconstructions, the status of which is indeterminate. It could be 'this is how we think things were' or 'this is how other people, wrongly, say things were'. The film-makers are able, in both cases, to exploit the indeterminacy to create suspense in terms of what the viewer is to believe.

3.5 STATUS, MODALITY AND ENGAGEMENT

Before ending this summary of the notion of status, I shall focus on two concepts from SFL: modality (Halliday and Matthiessen 2004) and engagement (Martin and White 2005). Each of these cover some of the

same ground as what has been discussed here under the heading of 'status', and might be regarded as alternatives to the approach I am suggesting here.

Halliday and Matthiessen (following Halliday in earlier editions) treat epistemic modality ('modalization') as a semantic category realised most congruently by the central modal verbs (*may*, *might*, *must*, etc.) and a small group of adverbs (*possibly*, *probably*, etc.) and metaphorically by clauses such as *I think*, *it's possible* and so on (Halliday and Matthiessen 2004: 620). The function of modalization is to place an utterance on a scale between 'it is' and 'it isn't', indicating the degree of probability or usuality associated with a clause (ibid.: 619). There are obvious similarities between the notion of modality and that of status. Given the clause shown in Example 3.13, for example, both approaches would note the modification by *might*.

Example 3.13

The virus might develop the ability to pass from person to person.

A modality approach might note that the orientation of the modalization is subjective and internal and that the value is low (the development is 'possible' rather than 'probable'). A status approach might comment that only a possible alignment with the world is indicated. Put more briefly, both might gloss the proposition in the clause as 'possible'.

It may seem, then, that 'modality' and 'status' are the same thing. Indeed, Kress and van Leeuwen (2006) extend the notion of modality away from a strictly clause-centred, grammatical notion to a more general semantic one in a way that makes it overlap more clearly with notions of status. I believe, however, that it is preferable to keep the two concepts separate. Although evaluation of status can be paraphrased in terms of degree of certainty, the concept of alignment between proposition and world is rather more than that. Furthermore, although evaluation of status may involve elements of clause grammar such as adverbs or modal verbs, it does not necessarily do so.[10] Three examples will be considered to illustrate this. Firstly, although a hypothesis and a deduction from an experiment may both in modality terms lie in the 'possible' range, their roles in the epistemology of a discipline are very different, and their alignment with the world is different too. Secondly, although it can be argued that modality may be expressed metaphorically both by a marker of opinion such as *it is possible that* and by an attribution such as *concern remains that*, the difference between 'I think this is possible' and 'I report that someone else thinks this is possible' is a distinction in status. Finally, it is worth considering clauses that are grammatically unmarked but which have a particular status because of their textual context. For example, a sentence such as Example 3.14 is unmarked for

modality but the status is determined by the fact that it comes from a work of fiction, a fairy story.

Example 3.14[11]

Once there was a little girl who, after the death of her parents, went to live near a great wood with her old uncle and his two grandsons.

Although grammatically it is not dissimilar from Example 3.15 the status of each is very different:

Example 3.15[12]

Mr Donaldson moved out of his Belfast home last December, and had been living in the run-down cottage which had neither electricity nor running water.

The language resources involved in status include modal auxiliaries, nouns and adverbs, *it-* structures and structures of attribution. Many of these are accounted for by Martin and White (2005) under the heading of Engagement.[13] Comparing status and engagement is useful, partly because it shows how looking at texts of a different kind can lead to different priorities and also because it throws into relief the importance of theories of discourse. To develop this argument, I shall here consider two short extracts from texts of different types, from the points of view of Engagement (using Martin and White 2005) and of Status (using Hunston 2000a). The first text (Example 3.16) is from an opinion article in the *Guardian* newspaper, written shortly after the murder of the Russian Alexander Litvinenko in London in 2006.

Example 3.16

[1]Russia has killed people abroad, it is true, and recently. [2]In 2004 two military-intelligence agents blew up the Chechen separatist leader Zelimkhan Yandarbiyev in Qatar (the Americans helped). [3]That was wrong, but Yandarbiyev was a real threat, channelling funding to militants on Russian soil. [4]Litvinenko, however, was a spent force as a critic of Putin. [5]Even in purely pragmatic terms, the notion that the president would order his murder in Britain on the eve of an EU-Russia summit seems unlikely. (Parfitt 2006)

This text enacts a dialogue between the writer and the imagined reader. The reader is positioned to take the view that the Russian government was responsible for Litvinenko's death. The writer counters that view but makes two concessions to the imagined position: that the Russian government has been known to kill people (e.g. Yandarbiyev), and that the killing of Yandarbiyev was wrong. An account of the extract using Martin and White's terminology is shown in Table 3.4.

Table 3.4 'Engagement' Account of Example 3.16

Sentence	*Text*	*Choice from Engagement system*
1	*Russia has killed people abroad, it is true, and recently.*	Heteroglossic: contract: proclaim: concur: concede
2	*In 2004 two military-intelligence agents blew up the Chechen separatist leader Zelimkhan Yandarbiyev in Qatar*	Monoglossic
2	*the Americans helped*	Monoglossic
3	*That was wrong*	Heteroglossic: contract: proclaim: concur: concede
3	*but Yandarbiyev was a real threat, channelling funding to militants on Russian soil.*	Heteroglossic: contract: disclaim: counter
4	*Litvinenko, however, was a spent force as a critic of Putin.*	Heteroglossic: contract: disclaim: counter
5	*Even in purely pragmatic terms, the notion that the president would order his murder in Britain on the eve of an EU-Russia summit seems unlikely.*	Monoglossic

This account stresses that the extract alternates between sentences that are dialogic (heteroglossic) and those that merely assert (monoglossic). It also argues that the writer 'contracts' rather than 'expands' the dialogic space, allowing no room for disagreement. What matters for this account is the relationship between two construed individuals—writer and reader—and the voices of each in the text. What does not matter is the relationship between the statements in the text and the outside world.

An account of the same extract using status terminology is shown in Table 3.5.

Table 3.5 'Status' Account of Example 3.16

Sentence	*Propositions*	*Status paraphrase*
1	*Russia has killed people abroad and recently.*	'What is agreed to be'
2	*In 2004 two military-intelligence agents blew up the Chechen separatist leader Zelimkhan Yandarbiyev in Qatar*	'What is'
2	*the Americans helped*	'What is'
3	*That was wrong*	'What is agreed to be'
3	*Yandarbiyev was a real threat, channelling funding to militants on Russian soil.*	'What I think is'
4	*Litvinenko was a spent force as a critic of Putin.*	'What I think is'
5	*the president ordered his murder in Britain on the eve of an EU-Russia summit*	'What is unlikely to be'

This analysis makes no mention of the opening or closing of dialogic space but does distinguish between an averred record of events (e.g. sentence 2), an expression of opinion (e.g. sentence 3), and an assessment of something as unlikely (sentence 5). The reader is construed not as a participant in the text but as an interpreter of it, and is positioned to attach more credence to *two military-intelligence agents blew up . . . Yandarbiyev in Qatar* (evaluated in the text as 'what is') than to *Litvinenko, however, was a spent force as a critic of Putin* (evaluated as 'what I think') or to *the president would order his murder on the eve of an EU-Russia summit* (evaluated as 'what is unlikely to be').

It is perhaps worth adding that both accounts make implicit reference to an understanding that the extract is built around two examples of the concession—counter-assertion clause relation (Hoey 1983: 191; Martin and White 2005: 124–126):

Russia has killed people before (CONCESSION)

But Litvinenko was a spent force (COUNTER-ASSERTION)

and:

Killing Yandarbiyev was wrong (CONCESSION)

But he was a real threat (COUNTER-ASSERTION)

A more detailed analysis of the clause relations in the extract is shown in Table 3.6.

Table 3.6 'Clause Relations' Account of Example 3.16

Sentence	*Text*	*Analysis*	
1	*Russia has killed people abroad, it is true, and recently.*	claim=	
2	*In 2004 two military-intelligence agents blew up the Chechen separatist leader Zelimkhan Yandarbiyev in Qatar (the Americans helped).*	=evidence	conces-sion=
3	*That was wrong*	concession=	
3	*but Yandarbiyev was a real threat, channelling funding to militants on Russian soil.*	=counter-assertion	
4	*Litvinenko, however, was a spent force as a critic of Putin*	claim=	=counter-assertion
5	*Even in purely pragmatic terms, the notion that the president would order his murder in Britain on the eve of an EU-Russia summit seems unlikely.*	=consequence	

Returning to the engagement–status debate, there are arguments in favour of both positions, but my own view is that Martin and White's model of Engagement is here more successful in accounting for how a persuasive argument is put together. The text construes a world in which the writer is both deferential ('You are correct to argue that Russia has killed abroad before') and assertive ('Believe me—Litvinenko was a spent force'). The precise alignment of each proposition to the world, in particular the tricky and ultimately unworkable distinction between 'facts' and 'opinions', is not an issue of importance. How the world of politics and espionage is construed matters much less than how the argument is.

When the text is a report of academic research, however, I would argue that the positions are reversed. Example 3.17 is a short extract from a Sociolinguistics research paper that was used also in Hunston (1994). The research question under discussion is at what age boys and girls learn to use politeness formulae such as *thank you*. The extract comes from the discussion section of the paper.

Example 3.17

[1]Greif and Gleason (1980) . . . found that preschool-aged boys and girls were equally likely to say thank you spontaneously. [2]The present results indicate that preschool-aged girls say thank you spontaneously more frequently than boys. [3]This finding is in accord with other research which supports sex differences in language development (Clarke-Stewart 1973; Maccoby & Jacklin 1974; Nelson 1973). [4]Moss (1974) has suggested that girls receive more social training from adults than do boys and this may result in faster acquisition of the more social aspects of language. [5]Sex differences in the spontaneous use of thank you by preschoolers may also have a cultural basis, with Western culture placing more emphasis on the use of politeness routines such as thank you by females. (Becker and Smenner 1986)

An 'Engagement' account of this example is given in Table 3.7.

This account captures the fact that a debate is in progress, and that the results of experiments (in sentences 1–2) are presented dialogically, but in a way that closes the dialogic space leaving no space for argument, whereas the explanations for those results (in sentences 4–5) are presented in a way that leaves the space open. What is not indicated is the difference between sentence 1—findings that are time-limited and person-dependent, as indicated by *found* (Hunston 1993b)—and sentence 2—findings that are current and person-independent. The role of sentence 3 in confirming the likelihood of the proposition in sentence 2 is also not captured by this account. The account gives a good sense of academic debate but does not indicate what view the reader is positioned to take of the academic issue involved.

Table 3.7 'Engagement' Account of Example 3.17

Sentence	*Text*	*Choice from Engagement system*
1	*Greif and Gleason (1980) . . . found that preschool-aged boys and girls were equally likely to say* thank you *spontaneously.*	Heteroglossic: contract: proclaim: endorse
2	*The present results indicate that preschool-aged girls say* thank you *spontaneously more frequently than boys.*	Heteroglossic: contract: proclaim: endorse
3	*This finding is in accord with other research which supports sex differences in language development (Clarke-Stewart 1973; Maccoby & Jacklin 1974; Nelson 1973).*	Monoglossic
4	*Moss (1974) has suggested that girls receive more social training from adults than do boys and this may result in faster acquisition of the more social aspects of language.*	Heteroglossic: expand: attribute: acknowledge
5	*Sex differences in the spontaneous use of* thank you *by preschoolers may also have a cultural basis, with Western culture placing more emphasis on the use of politeness routines such as* thank you *by females.*	Heteroglossic: expand: entertain

A 'status' account of the same example is given in Table 3.8. In this case, as well as the status paraphrase, comments are added which suggest how the status categories in each sentence interact.

Table 3.8 'Status' Account of Example 3.17

Sentence	*Proposition*	*Status paraphrase*	*Comment*
1	*preschool-aged boys and girls were equally likely to say* thank you *spontaneously.*	'what previous researchers reported to be'	Reports results with limited validity
2	*preschool-aged girls say* thank you *spontaneously more frequently than boys.*	'what this experiment shows to be' → 'what is likely to be' (s3)	Reports results with wider validity
3	*This finding is in accord with other research which supports sex differences in language development (Clarke-Stewart 1973; Maccoby & Jacklin 1974; Nelson 1973).*	'what is'	Resolves the conflict in s1 and s2.
4	*girls receive more social training from adults than do boys and this may result in faster acquisition of the more social aspects of language.*	'what a researcher says may be'	Attributes a hypothesis to account for results in s2
5	*Sex differences in the spontaneous use of* thank you *by preschoolers may also have a cultural basis, with Western culture placing more emphasis on the use of politeness routines such as* thank you *by females.*	'what may be'	Offers a hypothesis to account for results in s2

This account prioritises the role of epistemology in building an argument and highlights the varying amounts of credence given to the different propositions. In this case, as noted, my view is that the Status analysis has more to offer, because the reader's role as interpreter of information is more significant than the role as interactant. What matters about the writer's persona (see Charles 2004), in this case, is the cautious assessment of how far results can be relied on.

I regard both analyses of both texts as valid, but I suggest that Engagement provides a more rewarding reading of the newspaper text while Status offers a more revealing interpretation of the academic text. Given the distinctive ways that the models were developed (Engagement in the study of news reportage, Status in the study of academic discourse), this is probably not surprising.

3.6 CONCLUSION

This chapter has presented a summary of the concept of evaluation of status, investigating the language resources it draws on, its connection with attribution and with multi-modal texts and its relationship to the Engagement systems that form part of Appraisal theory. The work presented in this chapter is text-based rather than corpus-based, but it forms a context for the corpus work on status nouns that appears in Chapter 6.

An interesting aspect of status is that the concept enters into a dialogue with the 'competing' notion of Engagement. As has been demonstrated toward the end of this chapter, this is not simply a difference of detail or nomenclature. Rather, it is disagreement concerning the nature of the phenomena under discussion and that, in turn, arises at least in part from the kind of language that gave rise to the theories of evaluative language being used. Both newspaper texts and academic prose serve a dual function of construing a kind of reality and constructing a persuasive argument. For newspaper texts, especially of the editorial kind, the argument function is prior, and the positioning of the reader is crucial to that argument. Complexities of attribution, therefore, serve not so much to nuance the certainty of the proposition as to enact a dialogue with the reader. For this reason, an Engagement analysis offers a satisfactory reading of that kind of text. In academic prose, although there is persuasion and dialogue, what is also construed is a process of knowledge-formation in which various discoursal objects, findings, hypotheses and so on, are the building blocks. A status analysis identifies these objects. In this respect, evaluation of status might be said to realise the Field of a discourse rather than Tenor. Engagement, on the other hand, realises Tenor and construes a relationship between writer and reader. The fact that either analysis can account for the same phenomena in a single text further suggests that Field and Tenor may be realised by the same language resources.

4 Evaluation, Quantity and Meaning

4.1 INTRODUCTION

In this chapter the contributions of corpus linguistics to the study of evaluation are made central. In particular we look at how three specific research questions have been asked and answered. These are:

- How much of what kind of evaluative meaning occurs in a given corpus of texts?
- Can algorithms be developed that will reliably identify evaluative meaning in a corpus of texts?
- What is the contribution of 'latent patterning' to evaluative meaning?

It might be argued that the first two of these belong to a broadly 'quantitative' tradition of corpus research and that the last one belongs to a 'qualitative' tradition. Quantitative corpus research investigates the distribution of sets of words and phrases across comparable corpora, often using calculations of statistical significance. It draws on theories of register and variability in language. Qualitative corpus research asks how the evaluative meaning and usage of individual words and phrases might be identified and described. It uses a corpus to carry out a detailed examination of individual wordforms and word sequences in context. It plays a substantial role in the identification of implicit or covert evaluation.

These two traditions are, however, by no means discrete. The work that I have labelled here 'qualitative' uses frequency information both to ascertain how a word or phrase is typically used and to generalise beyond the individual instance into theories of language. 'Quantitative' work generally takes comparative frequency of occurrence as the starting point for more intensive and qualitative work on (groups of) individual items. It would be inaccurate to see the qualitative and quantitative traditions as opposed to each other or as mutually exclusive.

Much of the work surveyed in this chapter is underpinned by the distinction between 'stance' and 'evaluation' that was mentioned in Chapter 2. Although both are considered in this chapter under the heading of

'evaluative language', evaluation itself refers to the ascription of a value to an entity, whether inside or outside the text, while stance, or interactivity more generally, refers to indications in the text that a human being, the writer, is communicating with another human being, the reader. All writers acknowledge that there is overlap between the two categories, but it is possible to have a text that is highly interactive but does not contain much evaluation, or one where indications of interactivity are largely confined to evaluation. It is particularly useful to consider the two separately in corpus studies, as markers of interaction are more easily identified in formal terms, and so are more amenable to the quantitative approaches mentioned earlier, while evaluation is more likely to be expressed implicitly and so to benefit from the qualitative approach.

4.2 STANCE: HOW MUCH AND WHAT KIND?

Research into the identification and quantification of stance is associated most prominently with Biber (e.g. Biber et al. 1999; Biber 2006a, 2006b; Conrad and Biber 2000) and with Hyland (e.g. Hyland 2000, 2009; Hyland and Tse 2004, 2005). For each of these writers register difference is a key research interest. Biber focuses most centrally on register, and particularly on the differences between spoken and written registers, as his work from 1988 onwards testifies. Biber et al. (1999), and in other work that draws on the same resources, distinguish between broad register types: news reporting, fiction, academic prose (all written) and conversation (spoken). Biber (2006a, 2006b), on the other hand, makes much finer distinctions between registers that university students participate in, differentiating between 'academic/instructional' and 'management' communicative purposes and between spoken and written modes. This gives a four-way distinction that groups classroom teaching, lab session and study groups under 'academic and spoken' while textbooks and course materials are 'academic and written'. Examples of management registers include service encounters (spoken) and university procedural documents (written) (Biber 2006a: 102). Hyland has a concern for the teacher of English for Academic Purposes and frequently compares texts from different disciplines and texts written by academics (for example, research articles) with those written by students (e.g. essays or dissertations). Much of the work of both writers rests on the interpretation of quantitative findings that compare the frequency of given linguistic features in two or more corpora.

Both writers can be seen balancing the desire to isolate a relatively limited number of features that can be quantified in large corpora with the need to be as inclusive as possible in terms of accounting for the aspect of discourse they are dealing with. Conrad and Biber (2000), following Biber et al. (1999), for example, discuss only adverbials as markers of stance whereas Biber (2006a; 2006b: 87–131) takes account of a broader range of stance markers,

including modal verbs and verbs, nouns and adjectives that have that-clauses and to-infinitive clauses as complements (Biber 2006a: 101–102). Examples include *conclude that*, *advise you to*, *obvious that, necessary to* and *assumption that*. Hyland and Tse (2005) go into considerable detail about one linguistic feature (that-clauses) that expresses stance, whereas Hyland and Tse (2004), conceptualising their focus as metadiscourse, cover a wider range of features including indicators of clause relations (e.g. *but*), evidentials (e.g. *according to*), hedges, boosters and attitude markers (e.g. *perhaps*, *definitely* and *unfortunately*). Hyland (2009) considers a fairly restricted range of features that realise what he terms 'engagement'; these include easily identifiable features such as personal pronouns as well as clauses that are identified through function rather than through form, such as 'asides'. As the number of features and the range of registers studied increases, it becomes more difficult to identify clear trends, but the scope of the work becomes more comprehensive. On the other hand, the intensive work on a smaller set of features allows the researcher to offer more qualitative detail.

From the work on academic discourses, a number of broad trends appear. Biber (2006a), for example, notes that stance markers are more frequent overall in spoken registers than in written ones, and are relatively infrequent in textbooks. It is perhaps not wholly surprising that instruction-oriented registers contain more epistemic stance markers in comparison with management-oriented ones that contain more directive stance markers. Hyland and Tse (2004: 171) note that metadiscourse is more prevalent in doctoral theses than in master's theses. Following earlier work by Hyland (1998, 2000) they distinguish between 'hard' and 'soft' subjects and note that metadiscourse as a whole is more frequent in the 'soft' disciplines (173).They also note, however, that there is considerable variation in features used by individual disciplines, which militates against drawing simple generalisations. They point out the need to go beyond a simple counting of instances of surface forms (174).

Hyland (2009) goes beyond surface form into more detailed description. He quantifies the directives used in both research articles and student reports in eight university disciplines, directives being one of the ways that writers acknowledge the presence of a reader of their texts. The frequency of directives used by students varies quite considerably, from over 20 occurrences per 10,000 words in Information Systems and Mechanical Engineering to fewer than four occurrences in Public Administration and Marketing. Hyland glosses the function of these directives as 'managing readers' and notes that the degree of imposition depends on the more precise function of the directive: from pointing the reader to a place in the text (e.g. *Please refer to the table below*), through indicating a physical action (e.g. *It is important to use . . .*), to the greatest imposition in directing the reader to carry out a cognitive act (e.g. *It is important to note that . . .*). Hyland goes on to link each kind of directive with the requirements both of the disciplines investigated and the students as novice entrants into the

academic world. This is done not only through investigation of the corpora but by interviews with student writers. Noting that directives concerned with physical actions are particularly frequent in the physical sciences and engineering; for example, he cites students who explain that this is an effective way of making the methodology of their research clear. Hyland (2009: 110) argues that the conjunction of corpus research and informant interview is beneficial in giving a holistic picture of writing practices. Corpus research, he points out, investigates community practice, 'dematerializing texts and approaching them as a package of specific linguistic features employed by a group of users'. It needs to be balanced by methods which '[rematerialise] those features to understand how and why writers make the choices they do when they write.' This is wholly in accord with a growing tradition of research that combines corpus and ethnographic approaches (e.g. Lillis et al, 2010) as well as corpus and discourse concerns.

The combination of quantitative and qualitative approaches is evident also in the work of Charles (e.g. 2003, 2006a, 2006b). Contrasting theses from a physical science discipline (Materials Science) with those from a social science (Political Science), she focuses particularly on the role of anaphoric nouns (Charles 2003, terminology from Francis 1986) and on citation practices (Charles 2006a, 2006b) in exploring the role of stance-taking in postgraduate student writing in those disciplines. Charles gives quantitative information. For example: stance-bearing anaphoric nouns are relatively more frequent in the politics corpus than in the materials one; reporting clauses with human subjects are more frequent in the politics corpus whereas those with *it* as subject are more frequent in the materials corpus; reporting clauses used to frame information sourced from the writer are more frequent in the materials corpus whereas those used to report 'real world' human beings are more frequent in the politics corpus. These figures form the backdrop to a more subtle discussion of the stance of the writers towards their material in each discipline. For example, Charles (2003) notes that writers in both disciplines choose anaphoric nouns to interpret and express a stance towards information given previously, but that whereas the materials writers usually indicate a positive stance towards that information—it is the work on which their own work builds—the politics writers are more likely to evaluate the information negatively—it is the work against which their own work reacts. She links this to the 'cumulative construction of knowledge' in Materials Science and the 'recursive knowledge-building' in Political Science (ibid.: 322).

4.3 SENTIMENT ANALYSIS: THE AUTOMATIC IDENTIFICATION OF EVALUATIVE MEANING

Sentiment analysis, also known as opinion mining, attempts to identify the expression of opinion in texts of various kinds with the aim of quantifying, for example, positive and negative opinions of particular products (Liu

2010) or views on each side of a controversy (Greene and Resnik 2009). At its most basic, the work simply counts instances of positive and negative words in a given corpus; Fletcher and Patrick (2006), for example, find that the 'bag of words' approach identifies instances of evaluative language more reliably than a more theory-driven approach based on Appraisal theory does. More generally, however, by attempting to derive algorithms for identifying evaluative language automatically, sentiment analysis adds greatly to our understanding of what is important for evaluative meaning. It also acts as a test bed for theory-driven approaches to evaluation. At the same time, the view of evaluation adopted tends to be relatively simple; the need for automation means that only the most apparent evaluation can be captured, although recent work by Greene and Resnik (2009) does attempt to capture more subtle indications of attitude.

One theme in sentiment analysis is how to distinguish between subjective and objective sentences in naturally occurring text (e.g. Riloff and Wiebe 2003; Wilson et al. 2005; Wiebe et al. 2005; Ruppenhofer et al. 2008; Somasundaran et al. 2008). Wiebe et al. (2005) and Somasundaran et al. (2008) propose an annotation system which is not unlike the FrameNet approach (see Chapter 7) and which even uses similar terminology. Wiebe et al. (2005) propose a number of 'private state frames' which specify the source, target, intensity and polarity of a subjective sentence. Somasundaran et al. (2008) use the term 'opinion frame' to describe a text in which two objects of evaluation are compared. The works just cited focus on mapping the elements of the frames on to text realisations as a precursor to the automatic processing of texts. They emphasise the importance of context in identifying meaning and, as was described in Chapter 3, are particularly useful in describing layering of attribution. Of the work in sentiment analysis, that undertaken by this group is probably the closest to a more traditionally linguistic approach to evaluation.

Many papers in sentiment analysis are concerned with how to identify subjective and therefore evaluative adjectives. Given that such adjectives form a very open set, the challenge is to train a computer to recognise them without being given a finite list to begin with. Turney and Littman (2003), for example, begin with a very small set of positive and negative words and use collocation (or 'Semantic Orientation from Association') to extend that list, using a corpus of 'opinion' texts. Whitelaw et al. (2005) and Bloom et al. (2007) begin with the adjectives listed by Martin and White (2005) and use thesauri and WordNet to extend the set.

Martin and White's work on Appraisal has influenced some work in sentiment analysis, and some researchers classify adjectives according to whether they realise Affect, Appreciation or Judgement and, correspondingly, whether the target of evaluation is the self, a thing or a person (Taboada and Grieve 2004). Fletcher and Patrick (2006) go further and place Martin and White's Appraisal network under scrutiny. It is not surprising, perhaps, that a system devised to assist a detailed, qualitative reading of given texts

is not entirely successful as a background for the quantification of evaluative meaning, but the Affect–Appreciation–Judgement distinction is clearly one that appeals to sentiment analysts as a useful additional distinction to be made.

All in all, the aim of sentiment analysis is not to achieve a perfect analysis of a text but to obtain 'good enough' coverage of a very large number of texts to get an overall sense of the evaluation therein. Turney (2002), for example, notes that the Internet contains at least 5,000 reviews of one particular holiday site in Mexico. The potential tourist, he suggests, needs a way of establishing quickly whether most of these reviews are positive or negative. It is of no importance to that kind of task whether evaluation in one text is implicit, hedged or expressed with irony; what matters is that a result such as '70% of the reviews are positive' is broadly accurate. Experiments are used to establish the accuracy of various methods of identifying evaluation targets and sources, and of measuring polarity and intensity (Bloom et al. 2007). Accuracy results of about 90 per cent are claimed, with the best results apparently occurring when a simple 'bag of words' is used but moderated by other factors such as position in the text (Taboada and Grieve 2004) and relation to the Martin and White taxonomy (Whitelaw et al. 2005).

4.4 UNITS OF MEANING AND IMPLICIT EVALUATION

4.4.1 Sinclair

Sinclair's ground-breaking work using a reference corpus in the service of lexicography (e.g. Sinclair 1987, 1991) demonstrated the interface of the practical (writing a better dictionary) and the theoretical (proposing a novel approach to word-meaning in English). Sinclair's ideas are by now very well known (see Stubbs 2009 for a concise summary); they include the observation that meaning cannot be said to belong to individual words, still less to lemmas that are the abstraction from a number of wordforms. Instead, meaning is expressed by 'units of meaning' or 'lexical items', which are sequences of words of varying degrees of fixedness, with flexible and permeable boundaries. Units of meaning are identified by observing what commonly occurs in the co-text of a given word or short phrase and depend on identifying what is similar in a number of unique instances. They differ from most linguistic units (such as clauses) in that they do not have a defined set of constituents and they have no defined boundaries. They have more in common with what Halliday calls prosodic meanings, where a meaning is realised by several items occurring in a context.

The identification of units of meaning is dependent on what Sinclair and Coulthard (1975: 125) call 'latent patterning' i.e. patterning in language that is not obvious to intuition or to language as it is observed in single texts.

Rather, the patterning, or repetition, is apparent only when several examples of the unit are perceived together, as happens when individual pieces of language are taken out of their textual context and placed adjacent to each other, that is, in concordance lines or similar. Although the work is qualitative and is concerned with description (of a unit of meaning) rather than quantity (how often that unit occurs), quantity is not irrelevant to Sinclair's work as patterning depends on frequent repetition and comparative frequency is the starting point for much work of this type. For example, the oft-cited example *visible / invisible to / with the naked eye* (Sinclair 2004: 40–48) has as its starting point the strange fact that whereas *eyes* usually indicates organs of sight (and the word is usually preceded by a possessive), *eye* rarely indicates this and is used in a variety of phrases (*eye-catching*, *eye contact*, *turn a blind eye*, *caught my eye*, *see eye to eye* and so on), of which *naked eye* is one. The words *naked* and *eye* occur together only about once in one million words in English, but their co-occurrence is statistically significant in terms of both t-score and MI score, with only *blind eye* scoring higher on both. What is more striking is that *naked eye* is practically always used in the unit of meaning proposed by Sinclair or with the derived scientific use of *naked-eye object* or *naked-eye brilliance*. It is possible to find other examples (e.g. *There is still a place in cricket for the naked eye and commonsense*), but they are comparatively rare. In other words, the identification of the *naked eye* unit of meaning is based on both quantitative and qualitative information, with the quantitative giving rise to the qualitative interpretation.

Sinclair (1991; 2004: 30–35) proposed that a unit of meaning (or lexical item) consists of a core word and four additional elements, or, as Stubbs (2001: 64) usefully describes them, 'four types of co-occurrence relations': collocation, colligation, semantic preference and semantic prosody. Collocation is the co-occurrence of the core or node word with another word. Where the co-occurrence is not with one or two words but with a range that nonetheless can be interpreted as belonging to a set, the relationship between the node and the set is termed 'semantic preference'. Colligation is the association of the node with a grammatical feature such as a word class, a clause type or negative polarity. Semantic prosody is the pragmatic meaning or discoursal function associated with the unit of meaning. There are two aspects of this work, and other corpus studies in the same tradition, that are particularly relevant to the study of evaluation. One is that the set of words included in 'semantic preference' often displays positive or negative evaluative polarity. Of these, Stubbs's (2001) example *CAUSE* is particularly well known. *CAUSE* co-occurs with a very wide variety of items including general ones such as *damage*, *despair*, *catastrophe* and specific ones such as *low IQ scores*, *a massive growth in big-city police expenditures*, *growth of algae* (to cite some attested examples). The second is that the semantic prosody, while expressing meaning that is too subtle and precise to be captured by a polarity label only, often does express some kind of affective meaning, such as reluctance or frustration or difficulty.

There is some debate over the appropriate nomenclature in this area (e.g. Hunston 2007b) and further debate over the mechanism at work in semantic prosody (e.g. Whitsitt 2005; Stewart 2009). I shall not repeat this debate here but shall focus on the issues of collocation, corpus observation and implicit evaluation.

4.4.2 First Point of Agreement: Collocation with Evaluative Words

The least controversial point of agreement is that, as noted earlier, collocates of particular node words often belong to sets, which often though not invariably have in common an evaluative meaning. The literature is full of examples (see, for example, Stubbs 2001 or Partington 2003). An implicit distinction is sometimes assumed between three different kinds of co-occurrence, exemplified by the words *terrible*, *alleviate* and *cause/bring about*. The first two of these are words that sentiment analysis might identify as indicating polarity, but the adjective *terrible* confirms the polarity of the following word whereas the verb *alleviate* reverses it. The second group (*cause*, *bring about*) are associated with polarity but might not be identified as evaluative out of context.

A list of collocates indicates that the nouns most frequently immediately following *terrible* are either negative in evaluative terms (e.g. *mistake*, *tragedy*, *pain*, *blow*) or neutral (e.g. *thing*, *time*, *state*, *events*). The range of uses is exemplified by ten randomly selected concordance lines, shown here in expanded form (Figure 4.1):

Figure 4.1 Examples of *terrible*.

1. . . .where women's every move toward self-determination was met with authoritative statements on the *terrible* prices to be paid for daring to break away from patriarchally prescribed roles.
2. I hope I'm wrong but I have a *terrible* dread that this may be the year when one finally does go to the wall.
3. Families were living in *terrible* conditions—on the pavement or in bombed-out basements.
4. Bethan came from the valleys, but she had never so much as mentioned these *terrible* things.
5. After this, those *terrible* feelings rapidly declined in intensity but I still have them, even now, like the after-shocks from an earthquake or those strange pains one has after giving birth.
6. There are many *terrible* things going on.
7. 'There is only one truth and everybody knows it: The event was a *terrible* thing,' said Li Yun Chou, 73, a devoted Communist Party member and pensioner in Beijing.
8. They continued examining the pictures, exclaiming, 'Oh, there he is with that *terrible* cap he used to wear everywhere!'

9. The brave children and professionals who spoke out about the *terrible* abuse in North Wales children's homes deserve, at the very least, an apology from the State for what was a national scandal.
10. Claire Cumins of Gorton, Manchester, had a *terrible* shock recently.

In some lines the adjective is followed by a negative evaluative noun—*dread* (line 2), *abuse* (9), *shock* (10)—in these cases *terrible* may be said to be an intensifier. In other lines the adjective is descriptive—it identifies one kind of entity from other possibilities: *conditions* (3), *feelings* (5), *cap* (8). The phrase *pay a terrible price* (line 1) will be discussed further below. The phrases *terrible thing* (7) and *terrible things* (4, 6) are endophoric and evaluate events described elsewhere in the text (before or after the noun phrase). It is possibly surprising that the phrase *terrible thing/s* is so frequent and that it contributes so markedly to textual cohesion, but in general the association of *terrible* with negative evaluation is so obvious as to be a trivial observation. In Martin and White's terms the evaluation is inscribed.

The second example, taken from Whitsitt (2005), is the verb *ALLEVIATE*. The items occurring to the right of the verb are mostly negative: *poverty*, *problem*, *suffering*, *pain*, *symptoms* are examples. Where the item is neutral, such as *situation* or *effects*, looking at a wider context reveals that the word refers to something undesirable. An example is cited here as Example 4.1:

Example 4.1

Fighting is continuing in the eastern suburbs between government troops and rebels of the National Patriotic Front. Aid workers are doing their best to *alleviate* the situation . . .

Again, this is hardly surprising, in the sense that it is obvious to intuition that *ALLEVIATE*, like *IMPROVE*, can be paraphrased as 'make a bad situation better'. As a consequence it is obvious that the thing that is alleviated must constitute a bad situation, whether or not the lexis with which it is described makes this apparent. Such instances are not restricted to verbs that 'reverse the polarity' of the situation indicated by their object. For example, *PERPETRATE* usually occurs with words that are not only negatively evaluative but indicative of excessive harm and criminal responsibility, such as *fraud*, *injustice*, *violence* or *atrocities*. Words such as these might be said to provoke evaluation (to use Martin and White's terminology again), but the evaluation is explicit. (As Whitsitt ibid notes, the difference between *ALLEVIATE* and *PERPETRATE* suggests that semantic prosody should not be interpreted as an automatic ascription of evaluative polarity. *PERPETRATE* maintains a polarity but *ALLEVIATE* switches it.)

The third example compares the words *CAUSE* (example from Stubbs 1996) and *BRING ABOUT* (from Tim Johns).[1]

Because one of these examples has been discussed extensively elsewhere (e.g. Stubbs 2001; Hunston 2007b), I will confine myself to showing ten concordance lines for each, as follows in Figure 4.2:

Figure 4.2 *CAUSE* and *BRING ABOUT*

people come to therapy because they cause *pain to others* and have become so
and distorted build ups of calcium cause *enlarged joints*. Many people suffer
on viruses that make eyes bleed and cause *lethal diarrhoea* in infants and on
for bikini lines because it does not cause *rashes*. Erma says that when the
There is nothing more likely to cause *trades unions to blow a fuse* over
a newspaper campaign to free him and caused *delayed verdicts* in last summer's
are not based on evidence, and have caused *distress* to mothers with peanut-
having to defend legislation which causes *unnecessary offence*. In so doing
ten or twenty years. The virus that causes *AIDS*, the Human Immunodeficiency
to the underside of the eyelid, causing *scarring and thickening* so that

people that will empower them to bring about *their own liberation*. Even
in the twentieth century was bound to bring about *a transformation of African*
the hope that mass action can bring about *a fairer society* is not just
Sihanouk, as part of efforts to bring about *peace*. He said this matched
s personal role in helping bringing about *the agreement in Berlin* and
his. Jesus also recognized that God brings about *justice* for his chosen ones (
that *the resolution of crisis* is brought about (that is, if it is) through
and new avenues and has brought about *a totally new situation*. We
and *heightened accountability*, brought about by such local initiatives as
Federation of Bosnia and Herzegovina brought about *a tentative peace*. But

The association between *CAUSE* and words indicating negative situations and between *BRING ABOUT* and words indicating positive ones is illustrated by the italics in these examples. The association does seem to be more consistent in the case of *CAUSE* than in the case of *BRING ABOUT*: there are examples such as *economic policies which brought about this disastrous situation* where no sense of the positive can be detected. (For some reason, these counter-examples seem to be more numerous with *brought about* than with the other forms of the lemma.) On the other hand, where the accompanying lexical signals are not strongly evaluative, as in *has brought about a change in the way . . .*, an assumption that the change is positive is corroborated by the rest of the sentence.

There are grounds, then, for saying that negative and positive evaluations are implied by the use of *CAUSE* and *BRING ABOUT* respectively. Corpus linguists tend to focus on examples such as these, which are less obvious to intuition and where corpus studies reveal patterns that might otherwise remain hidden.

4.4.3 Second Point of Agreement: Affective Meaning

The second point of agreement rests on the more subtle expression of semantic prosody. Most examples are Sinclair's: he cites frustration and irritation being associated with a lexical item that has *budge* as its core (2004: 144), and 'reluctance or difficulty' being associated with the lexical item that centres on *true feelings* (2003: 147). As noted earlier, whereas frustration, irritation, reluctance and difficulty may be said to indicate (negative) affect, they are both more subtle and more vaguely articulated than 'good' and 'bad'. Here I discuss two further examples.

My first example is the phrase *a shred of* (taken from Hunston 2002). Concordance lines for *a shred of*, shown in Figure 4.3, reveal a consistent unit of meaning:

Figure 4.3 a shred of.

1 Rugby Corporation had scarcely a shred of credibility. It is to the credit
2 he was outclassed, and if he has a shred of decency he will agree to a re-
3 for me to decide." <p> If he had a shred of decency in him, he'd plead
4 all no-no's if you want to retain a shred of dignity in the office the
5 are all in the audience. Lacking a shred of dramatic integrity, `Maverick"
6 being pilloried in leaks without a shred of evidence, really unforgivable
7 on their part since there is not a shred of evidence in the Gospels (and
8 Lybrand's knowledge there is not a shred of evidence of corruption in the
9 a psychiatrist who, "without a shred of evidence," compared a lust
10 igh places enabled Lew to retain a shred of honour despite the odds that had
11 position but so long as there's a shred of hope efforts should be made for
12 America would have even a shred of influence over how China treats
13 ble family", admit they have not a shred of information to suggest that she
14 sent, there seems to be at least a shred of truth in each of these
15 all out of proportion—there's a shred of truth in there but it gets
16 he film's problems. There is not a shred of urgency in this tale, even when

The nouns to the right of *a shred of* mainly belong to two sets: a group of moral values (*credibility*, *decency*, *dignity*, *integrity*, *honour*) and a group relating to factuality (*evidence*, *truth*). In addition, there is an association with the negative, both grammatical (*without* 6, 9, *not* 7, 8, 13, 16) and lexical (*scarcely* 1, *lacking* 5). In the other lines there is an implication that the quality may be lacking (*if he has* 2, *if he had* 3, and line 12 where the longer co-text is *Can anyone seriously believe that . . . America would have*) or that it is maintained with difficulty (*if you want to retain* 4, *enabled Lew to retain* 10). Overall there might be said to be a sense of pessimism relating to the lack or paucity of the quality concerned.

A second example comes from one of the lines used to illustrate *terrible* in Figure 4.1—*the terrible prices to be paid for* (line 1). Shown in Figure 4.4 are 23 randomly selected instances obtained by searching for 'PAY, price, for', with up to two other words occurring between PAY and *price*.

Figure 4.4 PAY . . . price.

1 after all, why would you pay full price for a seven-per-cent bond when you
2 understand that you—you paid a price for their lives by your endurance
3 the war. The British are paying a price for their *obstinacy*! The Luftwaffe
4 many Democrats they would "pay a price"for *straying on the issue*.)
5 , their customers are paying a big price for the *complacency* they have
6 The country was paying a heavy price for the *Faustian bargain* the Prime
7 lity to make everybody pay a heavy price for going to war in the Gulf. He
8 rs mining group, is paying a hefty price for propping up the value of the
9 e babies adopted. They paid a high price for their *mistakes* but nowadays the
10 donesia's economy is paying a high price for the *political mess* President
11 news, but Halifax is paying a high price for the volume gains. Overall pre-
12 ace, however, and are paying a high price for a *rushed build-up*. <p>
13 companies to pay an inflated price for the vehicle. NRMA state claims
14 COCKY GIBBS PAYS PRICE FOR *RASH SHOWMANSHIP*
15 making the Soviets pay a political price for the expansion of East—-West
16 e top flight, Swindon have paid the price for chasing the dream. The First
17 ayed some nice stuff—but paid the price for *failing to last the distance*.
18 1999 </dt> THE Millers paid the price for *missing early chances*. Gary
19 her do that." <p> Starling pays the price for dolling up—not only does she
20 </h> <p> Ottavio Bianchi paid the price for Internazionale's *poor start* to
21 pg> <hl> Hoddle may have to pay the price for his *verbal sin* </hl> <bl> By
22 expects. Mr Michael has paid the price for not appearing as a standard-
23 and my daughter paid the ultimate price for it. These people appear to be

Lines 1 and 13 will be ignored here as they refer to giving money in return for goods and are instances of a different unit of meaning from the one under discussion. It is worth noting in passing that *full* and *inflated* appear to influence the interpretation of these lines as being different from the rest. Line 2 is incoherent and is not interpretable even with a larger co-text. In all the other lines 'pay a/the price for' is used to indicate an undesirable situation. However, examination of the noun phrase or non-finite clause following *for* suggests a further similarity. In lines 3–6, 9, 10, 12, 14, 17, 18, 20 and 21 the action or situation indicated is one that is judged in negative terms: *mistakes*, *political mess*, *failing to last the distance* and so on (see words in italics). In lines 7, 8, 22 and 23 an expansion of the context shows that the action indicated is an undesirable one from the point of view of the person being reported. In 16 out of 20 instances, then, there is an undesirable consequence of an undesirable action. It might be hypothesised that 'pay a/the price for' expresses a sense of justice, of someone being punished for a mistake. This is certainly true in instances such as:

3 The British are paying a price for their obstinacy!

7 Iraq's ambassador to France . . . has said his country has the ability to make everybody pay a heavy price for going to war in the Gulf.

18 The Millers paid the price for missing early chances.

As with other instances of semantic prosody of this kind (see e.g. Hunston 2007b; also Stubbs 2001), the affective meaning comes from a particular point of view, not necessarily that of the writer. Of the three instances given, only line 18 indicates an averred point of view. In the other two cases the sense of justice is attributed—to a character in a novel in line 3 and to the Iraqi ambassador in line 7.

The sense of justice applies only in those cases where the person who pays the price is the same person as the one who has done the undesirable action. The most notable instance of the opposite situation—where the price-payer is not the perpetrator—is line 23:

23 Those charged with his supervision failed—and my daughter paid the ultimate price for it.

Here the message is one of injustice—one person should not have to suffer because of the misdeeds of another. My own reading of this example is that this is somewhat stronger than saying simply 'my daughter suffered as a result'. According to the speaker, it is not just that the daughter has suffered accidentally, as it were, but that others who should have suffered have not done so. This is not only 'not justice' but 'injustice', as when an innocent person is punished in place of a guilty one. In other words there is an enhanced sense of moral outrage, a counterbalance to the sense of moral justification found in examples 3, 7 and 18.

4.4.4 Third Point of Agreement: Implicit Evaluation

The third point of agreement is that where words or phrases are typically associated with affective meanings, an affective interpretation may be warranted even when there is no apparently affective lexis involved. The examples used earlier can be used to illustrate this point also.

It appears that the association between *ALLEVIATE* and a negative situation means that it can be used to imply a negative interpretation even of apparently positive situations, as in Examples 4.2 and 4.3:

Example 4.2

. . . the warm, sermon-alleviating glow that shone through the apostles in the great window over the church altar . . . (evaluates *sermon* as negative)

Example 4.3

. . . to alleviate natural selection through making the environment benign . . . (evaluates *natural selection* as negative)

PERPETRATE, too, where it occurs with something apparently positive or neutral as in Examples 4.4 and 4.5, implies that for the writers concerned Highland clearances or folk music are not agentless historical events or charming forms of entertainment (respectively) but criminal acts:

Example 4.4

The English perpetrated the Highland clearances.

Example 4.5

. . . probably folk singing would be perpetrated. (Pratchett 1993: 210)[2]

It was suggested earlier that the unit of meaning with *a shred of* as its core has a prosody of pessimism. Arguably, the pessimism is implied even in the three lines from Figure 4.3 that are grammatically and lexically positive:

11 Asked for clarification today a Foreign Ministry spokesman simply repeated China's position but so long as there's *a shred of* hope efforts should be made for a peaceful settlement.

14 At present, there seems to be at least *a shred of* truth in each of these hypotheses, reminding us of a theme from Chapter 1: for many questions in psychology, there are no simple answers.

15 The tabloids blow it all out of proportion—there's *a shred of* truth in there but it gets distorted.

In each case, it could be said that a negative affect is being implied—the writer does not believe much hope for a peaceful settlement exists (11), or that any of the hypotheses is correct (14) or that the tabloids are reporting the truth (15).

As is often the case, this kind of highly implicit affective meaning, and the dissonance between voices that it often accompanies, can be exploited for rhetorical effect (see Stubbs 2001 for further examples). An example can be taken from Figure 4.4 (line 19), of which Example 4.6 is a greatly expanded version:

Example 4.6

Today's work, with Foster and Hopkins facing each other down through the cell bars, is the very stuff of great filmmaking. To show Starling's resolution, says Foster, she forsook the trainee's natural shyness: 'It's the first time you sort of see her with her hair down and with a dress on. I wanted to do something where it was about her. She would never admit that to anybody, that she wanted this case so bad, she wore the outfit to get him to say things. I felt like it was important to see that, even though her training would never let her do that.' <p> *Starling pays the price for dolling up*—not only does she absorb the

vulgarities of the criminally insane inmates who share a corridor with Lecter, but he also carefully assesses her outfit, her shoes, even her scent . . .

This comes from a magazine article about the making of the film *Silence of the Lambs* which stars (Jodie) Foster as a trainee FBI agent, Starling, and (Anthony) Hopkins as the serial killer Lecter. Starling is described as undergoing humiliations because of the 'feminine' way she dresses when visiting Lecter, but is this a just repayment for 'dolling up' or an injustice? The phrase *dolling up* itself is perhaps ambiguous—a mark of solidarity when used among a group of friends but something of a put-down when used of an outsider, suggesting a deliberate use of femininity for gain. I would suggest that the choice of 'pays the price' in this instance, along with the choice of 'dolling up' indicates that this is Lecter's point of view rather than the writer's. Because Starling is the agent of both actions (paying the price and dolling up), this should be an example of 'justice', but to regard wearing a dress as 'deserving' humiliation is surely Lecter's worldview rather than the writer's.

In the examples quoted earlier, the argument is that evaluation is implied because of the way that word is typically used. A dictionary might include such information in its word definitions, and this was, indeed, one of the innovations of the COBUILD dictionaries associated with John Sinclair. One of the most interesting developments in this area, however, lies in the claim that a word or phrase may have evaluative associations only in certain types of texts or even in more restricted contexts. A compelling example is given by Coffin and O'Halloran (2006), who argue that newspapers sometimes use what they (citing Manning 2004) call 'dog-whistle politics', where words and phrases are used that appear neutral to the average reader/hearer but which have a particular evaluative meaning to a target group. Coffin and O'Halloran take as an example an article in the *Sun* newspaper about migration to the UK of people from countries newly welcomed into the European Union. Using a specialised corpus of *Sun* newspaper texts they show that some words and phrases, including *Eastern Europe*, *migrants* and *a better life*, are used consistently in the *Sun* in conjunction with negative evaluations. They argue that this consistency of experience by *Sun* readers will position them to interpret apparently innocuous sentences, such as Example 4.7 (Coffin and O'Halloran 2006: 94), as covertly negative:

Example 4.7

Many from countries like Poland, Slovakia, the Czech Republic and Hungary are expected to arrive seeking a better life in the wake of the EU expansion on May 1.

4.4.5 Fourth Point of Agreement: Irony and Insincerity

This brings us to the final argument, articulated most convincingly by Louw (e.g. 1993), that where counter-examples to semantic prosody are encountered

they may be interpreted in terms of irony or insincerity. Partington (2003: 19) takes up Louw's point that an example that runs counter to normal semantic prosody may indicate insincerity. He cites a White House press officer who is supposed to be on the side of the press but who uses the phrase *where we are dealing with you* when speaking to journalists. He points out that the phrase *DEAL with*, in the corpus concerned, is usually followed by nouns indicating people or groups who pose problems for the White House (e.g. *Milosovic*, *the Belgrade authorities*, *China*); thus the press officer's use of the same phrase followed by a pronoun referring to the press suggests that he sees the press as also constituting a problem. What is not entirely clear is how certain the diagnosis of insincerity is. Louw (1993) tends to be very firm in his interpretation. Partington (2003: 19), on the other hand, seems to leave the door open for dispute: '. . . but the phrase *dealing with* you *suggests* that he really sees the press as a problem not a client' (emphasis mine). Hunston (2007b) goes further and suggests that frequency of occurrence of this kind should be invoked as an explanation for a personal reaction to an example rather than be offered as evidence for the inevitable or 'correct' interpretation of that example. In other words, I suggest that if we intuit that the press officer is being insincere, examination of the corpus will serve to explain where that intuition probably comes from, but the corpus does not provide conclusive evidence of the press officer's state of mind.

4.5 CONCLUSION

This chapter has summarised work which is computational or corpus-based as opposed to the text-based work discussed in Chapter 2. This work encompasses a problem-solving approach to evaluation (do most reviews evaluate this product positively or negatively?), a corpus comparison approach to stance (how is stance expressed in this discipline or in this kind of document?) and a questioning approach to how phraseology in language works (what is fixed about this phrase?). What is intriguing is that all of this work combines the quantitative and the qualitative although the emphasis in each case is different.

The arguments presented in the following chapters similarly draw on both quantitative information and qualitative interpretation. For the most part, however, what is emphasised is the observation of latent patterning, with quantitative observations providing corroboration for the presence of pattern. It is argued that the combination of close attention to individual instances and interpretations of large quantities of data can bring about novel insights into how evaluative language works.

5 Modal-Like Expressions

5.1 INTRODUCTION

This chapter concerns those aspects of evaluative meaning which are often discussed under the heading of 'modality', including the expression of obligation or necessity, of possibility and of desirability or volition. Such meanings in English are typically described in relation to the grammatical category of modal auxiliaries, although it is commonly noted that modal meaning is expressed by a much wider range of items than this (e.g. Stubbs 1986; Coates 1989/1998; Conrad and Biber 2000). This chapter makes two points, or perhaps one point presented in two different ways: firstly, that some wordforms appear to occur consistently in contexts that express modal meaning, though that meaning may be realised in ways that are not normally considered under the heading of 'modality'; secondly, that some verbs act as 'attractors' of modal meaning, co-occurring with such meaning more frequently than other verbs do. The chapter also explores how corpus methodologies contribute to the formation of such arguments. A mainly qualitative methodology—reading concordance lines—is used to demonstrate the first point, while a mainly quantitative one—calculating comparative frequencies—is used to demonstrate the second (though see the discussion in Chapter 4 about the qualitative–quantitative distinction).

Example 5.1, from a corpus of issues of the British newspaper the *Guardian*, illustrates the phenomenon discussed in this chapter.

> Example 5.1
>
> On 'Personal Passions' [a TV programme], writer and broadcaster Paul Heiney explains his enthusiasm for heavy horses and talks about *the importance of preserving* rare breeds of farm animals like the Suffolk Punch.

In this example, the word *preserving* occurs in the context of an indication of necessity or obligation (*the importance of*); a very rough paraphrase might be 'Paul Heiney says we must preserve rare breeds . . .'. This apparently inconsequential observation gains more significance when

examination of a corpus indicates that *of preserving* most frequently follows items such as *aim*, *purpose*, *goal*, *necessity*, *task*, *interest*, *hope* as well as *importance*, and that another frequent two-word phrase, *in preserving*, most frequently follows *interest*, *interested*, *important*, *role*, *succeed* and *successful*. In other words, phrases such as *with the aim of preserving*, (be given) *the task of preserving*, *in the interests of preserving*, *have an interest in preserving*, *is important in preserving*, (be) *successful in preserving* are relatively frequent as part of the context of *preserving*. It appears that *preserving* expresses something that is desirable, something that is the object of volition ('we want to or will preserve') and necessity ('we must or should preserve'). These meanings might be said to be 'modal-like' and it might be noted that *preserving* frequently occurs in their immediate co-text.

Further quantitative investigation of the lemma *PRESERVE* (verb only) yields additional, apparently inconsequential, information. The base form, *preserve*, accounts for 41 per cent of the total, the -ed form, *preserved*, 30 per cent, the -ing form, *preserving*, 16 per cent, and the -s form, *preserves*, 4 per cent. (It should be noted that 'wordform' here refers only to the surface form and not to a classification of it. For example, in those verbs in English where two classes of verb form, past and past participle, have the same surface form, e.g. *preserved*, these are both referred to here as the '-ed form'. Similarly, 'base form' [e.g. *preserve*] refers to all classes of verb with this form, including the infinitive and the present tense.) Those percentages, compared with similar calculations for other verb lemmas, indicate that *PRESERVE* has a higher-than-average frequency of the base form and the -ing form. The most consistent phraseologies of *preserving* have already been noted. The base form *preserve* appears to occur infrequently as a finite verb (e.g. *they preserve*): a random sample of 100 instances contains only two finite examples. The others follow a modal verb (e.g. *can preserve*) or another verb (e.g. *help preserve*) or an expression with *to* (e.g. *wanted to preserve*, *whatever is necessary to preserve*). In other words, although it cannot be said absolutely that the lemma *PRESERVE* 'typically' occurs in the context of an indication of volition or necessity, it can be said that there is evidence that it does so more frequently than might be expected, because the wordforms that do typically occur in that context are of relatively high frequency .

In one sense we have simply observed that in the example sentence, the word *preserving* is found in a phraseology that belongs to a set of phraseologies in which the word often occurs. This is simple because, as noted in Chapter 3, consistency of occurrence is a recurring theme in corpus-based approaches to discourse. We have also proposed, however, that a range of different grammatical structures and lexis, which co-occur with *preserving*, actually belong to a single semantic set which might be described as 'modal-like'. This chapter will argue that the identification of such modal-like expressions, and of the words with which they co-occur more frequently than others, is an important aspect of the study of affective or evaluative language.

5.2 MODALITY AND MODAL-LIKE EXPRESSIONS

In this chapter an open set of expressions is proposed and given the label 'modal-like expressions'. The terminology is deliberately non-technical to emphasise the fuzziness of the set and the recognition that it is a functional rather than a formal grouping, and reliant on subjective interpretation. Briefly, modal-like expressions are expressions other than modal auxiliaries which express modal meaning. As will be demonstrated in the following, they comprise a wide range of lexical and grammatical forms. It is important to note that modal-like expressions were identified first, not as the result of a search for such items, but as a by-product of other searches (see below). Thus, the notion of these expressions is the outcome of the observation of occurrences rather than of a theory of language, although the concept of modality itself is familiar in linguistics.

Modal verbs in English have been extensively researched and described (e.g. Perkins 1983; Mitchell 1988; Palmer 1990; Halliday 1994; Mindt 1995; Kennedy 2002). It is well known that modal verbs have a different set of behaviours from other verbs[1] and that between them they cover a range of meanings. Halliday (1994: 357–358), starting from the meanings rather than the words that realise them, systematises these meanings into 'probability', 'usuality', 'obligation' and 'inclination' and further into degrees: 'high', 'median' and 'low'. Palmer (1987) similarly classifies modal meaning into 'epistemic', 'deontic' and 'dynamic'. Quirk et al. (1972), taking as their starting point each modal verb and its different uses, use the terms 'ability', 'permission', 'possibility', 'volition', 'obligation', 'probability' and 'logical necessity' among others. As is well known, there is no one-to-one relationship between a verb and an area of meaning; most modal auxiliaries in English are used to express a range of meanings. For example, Quirk et al. (1972: 101–102) note that the modal *must* can indicate both 'obligation or compulsion', as in *You must be back by 10 o'clock* and 'logical necessity', as in *There must be a mistake*.

Modal meaning is often described contrastively: a modal verb may have a different meaning according to its co-text (as in the *must* examples), or a clause may be modified differently according to the modal verb used (e.g. *He might be at home* compared to *He must be at home*). Modal meaning, however, is not as atomistic as this approach suggests, and is arguably located in the phrase rather than in the word. This is most clearly seen in the relatively fixed phrases such as *I must say/admit* (Palmer 1987: 125–126; Hunston 2000b), which are not found to alternate with other forms (**you must say, *I must not say, *I may admit*) and which act as a whole to modify a co-occurring clause. The meaning of the modal verb in those cases is much less important than the function of the phrase as a whole. Less extreme cases also, however, suggest that modal meaning is holistic rather than atomistic. Hunston (2000b), for example, notes that some sequences that include modals are consistent in their meaning. *You must be careful* almost

always indicates that care is important or obligatory, whereas *you must be mad* seems always to indicate a logical deduction. One way of looking at this phenomenon is to suggest that it is the phrase *must be careful/must be mad* that has meaning, rather than each word in the phrase. The prosodic nature of such meaning is worth bearing in mind when we turn to look at alternative realisations of the meaning.

The list of items classified as modal auxiliaries is a finite and familiar one: *can*, *could*, *may*, *might*, *must*, *shall*, *should*, *will* and *would* comprise the basic set of verbs. This is often extended to include two-word expressions of modal meaning such as *have to* and *ought to*, and may be further extended to a wider range of 'phrasal modals', including *be (un)able to*, *had better/best*, *be bound to*, *be going to*, *be liable to*, *be meant to*, *would rather*, *would sooner*, *be supposed to*, *be sure to* (Francis et al. 1996: 574).

It is also well known that English has numerous ways of expressing modality besides modal auxiliaries. Biber et al. (1999) identify a set of adverbials that express epistemic stance, including one and two word adverbs (*possibly*, *in fact*) and clauses such as *I think*. Coates (1989/1998: 244–248), discussing the importance of epistemic modality to supportive talk among women, identifies the following as contributing to such meaning: *perhaps*, *I think*, *kind of*, *sort of*, *probably* as well as tag questions. Stubbs (1986) identifies a wide range of lexical and grammatical resources that express and modify commitment to speech acts. These include the attribution of propositions to others, modifiers of promises such as *all being well*, *if I can*, *but don't count on it*, markers of detachment from precision of expression such as *so-called*, *so to speak*, *what they call* and much else. Stubbs's point, however, is not to itemise such phrases but to argue that modal meaning pervades all language use and all parts of the grammar and lexis of English. Choices between present and past tense, for instance, can modify commitment to a proposition (*I hear that Harry is dead* compared to *I heard that Harry was dead*) (Stubbs 1986: 16). Researchers into the language of learners of English, such as Aijmer (2002), report that native speakers of English consistently use fewer modal auxiliaries than learners do, suggesting that they frequently use alternative language resources to express modality.

The notion of modal-like expressions is consistent with Stubbs's work in that the category comprises a much wider set of expressions than those normally associated with modal meaning and includes many that are an integral part of the clause concerned, rather than a peripheral item such as an adverb. This breadth is a direct outcome of the method by which the expressions have been identified, as will be outlined in the following. Another direct outcome of this method is the hypothesis that modality in general and modal-like expressions in particular are not evenly distributed among verbs but are more frequently associated with some verbs than with others. This is in accordance with a view of language that sees lexis rather than grammar as the central organising motive behind language and individual lexical items as the locus of linguistic variation (cf. Sinclair 1991; Hoey 2005).

In the next section, the methodology which led to the identification of modal-like expressions will be described, and their significance in terms of corpus use and linguistic theory discussed. The hypothesis that some verbs are more frequently associated with modal-like expressions than others will then be illustrated.

5.3 CONCORDANCE LINES AND COLLOCATION

As noted earlier, the starting point for an investigation of modal-like expressions was not an attempt to identify the range of expressions that realise modal meaning, but the observation of the collocational behaviour of particular lexical items. It was found that when a set of concordance lines was selected to show *decide* followed by a wh-clause (Hunston 2003b), surprisingly few of the examples represented a finite verb *decide*, as in *I decide*, *they decide* and so on (see Figure 5.1). Instead, *decide* tended to be non-finite, following a modal auxiliary such as *can*, *must*, *could*, *should* and *will*, phrasal modals such as *have to*, and other instances of the infinitive marker *to*. Crucially, the whole range of items preceding *decide* were interpreted as showing a consistency in meaning. That meaning was interpreted as 'modal-like', as expressed by the modal auxiliaries and by the range of expressions also ending in *to* such as:

. . . *it's up to them to decide where* . . . OBLIGATION
. . . *forces the United Kingdom to decide whether* . . . OBLIGATION
. . . *help courts to decide how* . . . ENABLING/POSSIBILITY
. . . *it took several days to decide whether* . . . DIFFICULTY/POSSIBILITY
. . . *I've yet to decide whether* . . . DIFFICULTY/POSSIBILITY
. . . *attempting to decide what* . . . DIFFICULTY/POSSIBILITY

A subsequent study involving the sequences *distinguish between* and *distinguishing between* (see Figures 5.2 and 5.3) led to similar observations. *Distinguish between* occurs frequently after *to*, with phrases such as *ought to*, *need to*, *able to*, *important to* and *(not) easy to* predominating. *Distinguishing between* frequently follows phrases ending with prepositions including *methods for*, *way of*, *the importance of* and *difficulties in*.

Figure 5.1 Concordance lines for *decide wh* (from Hunston 2003b).

are not likely to be relevant. 1 Decide what to include in a working
that you can now name your MEP and decide whether or not he or she does a
has sprung to life again. Caste can decide who wins elections in India toda
Before you start to fight depression decide if you are truly depressed or ju
of March—and Mr Clinton must decide whether to reappoint him. That
until next summer and Leicester must decide whether to cash in now or lose h
larger than Virginia. I could not decide whether this recital was a comfo
not a few parliamentarians, should decide who is to lead the country. If y

(continued)

Figure 5.1 (continued)

But we make the clothes, we don't decide when they go into the stores." T
their Web page almost before they decide when to start the revolution, a
asylum. "Now it is up to them to decide where they want to go," he said.
which forces the United Kingdom to decide whether its destiny is European
gov.uk) to help courts to decide how to assess lawyers' charges
it will be a decision for him to decide what he wants to say. Thank you.
she says it took several days to decide whether to lodge with police the
district is an incongruous place to decide how and when peace will return t
fudge omelette; and then you have to decide whether you want your toast on
workers in the 1940s. They had to decide whether to spend their limited
I am feeling tired. I've yet to decide whether I will play in a reserve
up to the prosecution the police to decide what charges are appropriate rat
not a tool for attempting to decide what people cannot do before th
of a typical "cat". When we have to decide whether an animal we come across
purpose of a means inquiry was to decide whether wilful refusal or culpab
are pursuing their investigation to decide whether or not there are suffici
stud of Budgerigars. You have to decide what features are lacking in you
official representatives here and we decide who calls the shots? <p> Gross:
selectorial balancing act that will decide whether or not Vickery makes
than the authorities, who will decide how dangerous their experiments
of Public Prosecutions, who will decide if there is to be court action.
And the Supreme Court will decide whether a jury debating to impos

Figure 5.2 Concordance lines for *distinguish between*.

made known to man so that he can distinguish between the Straight P
re excellent. Only an expert could distinguish between a facsimile an
Lawrence III in the previous essay distinguish between the two? Expla
laden. In other words, one should distinguish between the linguistic
a small, messy person who couldn't distinguish between Richard James
in need of treatment. We ought to distinguish between work that is u
understand them fully. One way to distinguish between them is to rem
Eleven judges were asked to distinguish between six programs a
I think it's important to distinguish between two very diffe
on the matter, we need firstly to distinguish between communism in r
skill behaviours, and (b) to distinguish between appropriate an
that man is moral and able to distinguish between right and wron
At the very least, it wants to distinguish between unintended 'in
as they mature, so as to distinguish between resistant and
Francis Jeffrey would be able to distinguish between good governmen
the cognitive abilities to distinguish between fact and ficti
it is nevertheless helpful to distinguish between two traditions
status. It was not always easy to distinguish between household good
from what was mythical", and to distinguish between European know
of the inability of the Church to distinguish between the poetry of

Figure 5.3 Concordance lines for *distinguishing between*.

talk about this process by distinguishing between sex and gende
contenders. It rewards both-by distinguishing between a film and it
things simple for himself by distinguishing between just and unju
been most reassuring in clearly distinguishing between Islam and the
evolution. Methods for distinguishing between these two exp
are normative criteria for distinguishing between acceptable an
as well as mnemonics for distinguishing between "lama" and ll
then there is no allowance for distinguishing between the duty not
<p> A key factor in distinguishing between communication
found real difficulties in distinguishing between advice and ad
speakers were less clear in distinguishing between the Swiss and
little better than lay people in distinguishing between policy choice
will face a challenge in distinguishing between "genuine" ban
expectations. <p> That means distinguishing between "ripe" and `u
and vegetation all together, not distinguishing between them. <p> Q:
of the structure. 13 One way of distinguishing between them is in te
this work to the importance of distinguishing between constitution
USA, charged with the task of distinguishing between software prog
<p> There is currently no way of distinguishing between people who ar
the Burinmo had trouble distinguishing between blues and gre

To summarise: each of these examples led by chance to the observation that certain verb forms, when complemented in a particular way (i.e. *decide* + wh-clause, *distinguish* + *between*, *distinguishing* + *between*) tended not to occur in subject-verb combinations such as *I decide*, *we distinguish*, *they were distinguishing* etc., but following either modal auxiliaries, infinitive *to* or a range of prepositions. In turn, the items occurring to the left of *decide*, *distinguish*, *distinguishing* were interpreted as expressing modal-like meaning, especially obligation, possibility and necessity.

Examination of small sets of concordance lines is an accepted way of establishing how a word or phrase is typically used. Sinclair (1991: 84) reports examining the ways that *of* is used by looking at about 30 lines at a time (selected by taking every nth instance from a very large number of instances, to ensure spread throughout the corpus) and repeating this with successive sets of 30 lines selected in the same way until all the lines in each set exemplify what has previously been observed and no new observations are being made. An alternative method, however, is to use frequency information from a very large number of instances as a starting point. To extend the search for modal-like elements modifying *decide*, for example, all available instances of *to decide whether* can be obtained and the most frequent collocates of that phrase examined. (The sequence *to decide* is chosen because, as noted earlier, modal-like

expressions frequently comprise sequences ending in *to*; in addition, *to* is the most frequent word immediately preceding *decide*. The sequence *decide whether* is chosen because, as noted earlier, modal-like expressions are likely to precede *decide* when that word is followed by a wh-clause.) In calculating frequencies of collocation, it is assumed that *to decide whether* can be treated as though it were a single word for which frequent collocates can be identified.

The immediate left collocates of *to decide whether*, in order of frequency, are identified as:

> *have, has, yet, had, trying, is, today, need, asked, you, able, agreed, days, and, week, unable, expected, right, referendum, court, time, try, them, having, order, still, hard, chance, him, was.*

Some of these are clear candidates for modal meaning: *HAVE to*, *is to*, *need to*, *TRY to*, *(BE) able to*, *(it BE) hard to*, *(BE) asked/expected to* all express obligation, necessity, ability or difficulty. In other cases, more co-text has to be seen to test whether a modal interpretation is reasonable. The adverbs *yet* and *still* turn out to occur in phrases such as *is yet to decide whether* and *has still to decide whether*, indicating a difficult task not yet accomplished. The pronouns *you* and *them* occur as the objects of verbs or prepositions in a variety of phrases indicating difficulty, rights and obligation/necessity, such as *help you to decide whether*, *leave it to you to decide whether*, *it remains for you to decide whether*, *how long does it take you to decide whether* and *allow/enable/force you to decide whether.* The related nouns *days* and *week* occur with *HAVE* and *TAKE* to show difficulty or obligation, as in *have several days to decide whether* and *take a week to decide whether. Right* and *chance* occur with *HAVE* in the phrases *have the chance/right to decide whether.*

The verb *is*, occurring to the left of *to decide whether*, turns out to be a little more complicated than anticipated. As well as combining with *to* to modify the verb, as in *A higher court is to decide whether*, *is* occurs frequently as a copular verb following nouns such as *step*, *task*, *problem* and *job*, as in *our job is to decide whether . . .*, *the main task is to decide whether . . .* and so on. In this case, the modal meaning of obligation is located in the noun rather than in the phrase *is to.*

In short, almost all the most frequent immediate left collocates of *to decide whether* turn out to be part of phraseologies that have a modal meaning although the grammatical functionality is different in each case.

In a second example, the wordform *distinguish* occurs immediately following these words, in order of frequency: *to*, *can*, *not*, *that*, *n't*, *cannot*, *and*, *could*, *must*, *we*, *you, which*, *would*, *will*, *should.*

Table 5.1 shows these 15 words occurring most frequently immediate to the left of *distinguish*, together with the percentage of the total instances of *distinguish* that are accounted for by these forms. Together the forms account for about 80 per cent of the total instances of *distinguish.* The table shows

Table 5.1 Frequencies of Left Collocates before *distinguish* (Rounded Percentages)

Word sequence	*Frequency of word or sequence*	*Percentage of incidence of distinguish*
distinguish	4,154	100%
to distinguish	2,504	60%
can distinguish	175	4%
not distinguish	168	4%
that distinguish	142	3%
n't distinguish	94	2%
cannot distinguish	65	2%
and distinguish	60	1%
could distinguish	59	1%
must distinguish	48	1%
we distinguish	43	1%
you distinguish	41	< 1%
which distinguish	38	< 1%
would distinguish	24	< 1%
will distinguish	22	< 1%
should distinguish	21	< 1%

that *distinguish* most frequently occurs in the context of modal auxiliaries and of *to*, as well as in relative clauses (beginning with *that* and *which*), and less frequently following pronoun subjects such as *we* and *you*. The infinitive marker *to*, however, occurs with remarkable frequency and accounts for well over half the instances of *distinguish*. This confirms the information obtained from the concordance lines, that in terms of frequency, modal-like expressions are an important feature of this wordform. Further confirmation is obtained by a check of a random 100 concordance lines of *distinguish* (that is, lines obtained by selecting every nth instance in the corpus), in which only five are finite forms realising the present tense and only three are imperative. All the others are either negative (two) or are preceded by a modal auxiliary or by *to*.

The exercise of finding the immediate left collocates is then repeated with the sequence *to distinguish*, yielding the following items: *difficult*, *able*, *hard*, *important*, *unable*, *impossible*, *need*, *easy*, *ability*, *how*, *used*, *have*, *possible*, *and*, *is*, *way*, *learn*, *attempt*, *little*, *try*, *order*, *failure*, *failed*.

Together these words account for just under half (48 per cent) of the total instances of *to distinguish*. They can be examined in terms of the longer sequence each word is part of, the grammar patterns these sequences realise (see Chapter 7) and the meanings expressed. This is shown in Table 5.2.

Table 5.2 Sequences, Patterns and Meanings with *to distinguish*

Sequence	*Grammar pattern*	*Meaning*
. . . *it is difficult to distinguish* . . . *it is hard to distinguish* . . . *it is (not) easy to distinguish* . . . *it is impossible to distinguish* . . . *it is possible to distinguish*	*it be* + adjective + to-infinitive	degree of difficulty
. . . *x is easy to distinguish from y* . . . *x is difficult to distinguish from y* . . . *x is hard to distinguish from y*	link verb + adjective + to-infinitive	degree of difficulty
. . . *be able to distinguish* . . . *be unable to distinguish*	link verb + adjective + to-infinitive	ability
. . . *it's important to distinguish*	*it be* + adjective + to-infinitive	necessity
. . . *we need to distinguish*	verb + to-infinitive	necessity
. . . *there is a need to distinguish* . . . *the need to distinguish*	noun + to-infinitive	necessity
. . . *lose/improve the ability to distinguish*	noun + to-infinitive	ability
. . . *know/learn how to distinguish*	verb + wh word + to-infinitive	ability
. . . *be used to distinguish*	*be* + past participle + to-infinitive	purpose
. . . *have to distinguish*	phrasal modal + infinitive	necessity
. . . *there's no way to distinguish* . . . *the best way to distinguish*	noun + to-infinitive	degree of difficulty
. . . *any/no attempt to distinguish*	noun + to-infinitive	purpose
. . . *there's little to distinguish x from y*	pronoun + to-infinitive	degree of difficulty
. . . *try to distinguish*	verb + to-infinitive	purpose
. . . *in order to distinguish*	phrasal subordinating conjunction + verb	purpose
. . . *failure to distinguish*	noun + to-infinitive	degree of difficulty
. . . *failed to distinguish*	verb + to-infinitive	degree of difficulty

The range of grammar patterns here is easily observed; mostly, *to distinguish* is a to-infinitive that follows a noun, verb or adjective. There are other possibilities, however. In *in order to distinguish*, the phrase *in order to* is a subordinating conjunction. The sequence *be used to distinguish* is the passive form of the pattern **verb + noun + to-infinitive**, a pattern which is exemplified by other sequences not frequent enough to show up in the top 15, such as *enable us/them to distinguish* and *allow us/them to distinguish*. The

concordance lines for *decide wh* (Figure 5.1) also exemplify the **verb + noun + to-infinitive** pattern (e.g. *force the United Kingdom to decide*) and the **verb + to-infinitive** pattern (e.g. *attempting to decide*) as well as lexically more restricted patterns such as ***it* + verb + noun + amount + to-infinitive** (*it took us several day to decide*) and idiomatic expressions such as *it's up to you to decide* (coded in Sinclair et al. 1995 as **phrasal preposition + noun + to-infinitive**) and *I've yet to decide* (coded as **adverb + to-infinitive**).

The 'meanings' indicated in Table 5.2 are not intended to be definitive, and different terminology could be offered. The purpose is to highlight similarity between the identified sequences and concepts of modality and polarity. Mitchell (1988) suggests that modal meanings can be paraphrased using concepts of 'necessity' and 'possibility' together with positive and negative polarity, and these same concepts are found in the sequences preceding *to distinguish*. What are called in Table 5.2 'degree of difficulty' and 'ability' express different aspects of possibility, while 'necessity' and 'purpose' arguably both express necessity (or 'purpose' could be considered close to Halliday's notion of 'inclination'). Exact parallels should not be expected, however, as the sequences identified are not modals, either in form or meaning, but 'modal-like'.

Investigation of the wordform *distinguishing* suggests that it, too, is more likely to be part of a non-finite verb phrase than a finite one. The search prompt used is *distinguishing between*; this isolates verb uses as opposed to the modification of nouns such as *distinguishing feature*, *distinguishing characteristic*, *distinguishing mark* and so on. In addition, *between* is the most frequent immediate right collocate of *distinguishing*. Of 279 instances of *distinguishing between*, only eight (3 per cent) are preceded by a form of the verb *BE*.[2] The relative frequencies of the collocates of *distinguishing between* are shown in Table 5.3.

Table 5.3 Frequencies of Left Collocates before *distinguishing between* (Rounded Percentages)

Word sequence	*Frequency of word sequence*	*Percentage of total occurrences of distinguishing between*
distinguishing between	279	100%
of distinguishing between	62	22%
in distinguishing between	37	13%
by distinguishing between	27	10%
for distinguishing between	20	7%
difficulty distinguishing between	11	4%
trouble distinguishing between	5	2%
to distinguishing between	4	1%
at distinguishing between	4	1%

Taking each of these sequences in turn and identifying collocates occurring to the left, broadly defined meaning groups can again be identified:

Possibility

way/s of distinguishing between
means of distinguishing between
method of distinguishing between
basis for distinguishing between
criterion/a for distinguishing between
rules for distinguishing between

capable of distinguishing between
incapable of distinguishing between
good at distinguishing between
adept at distinguishing between

difficulty of distinguishing between
task of distinguishing between
difficulty/ies in distinguishing between
problems arise in distinguishing between
have difficulty distinguishing between
have trouble distinguishing between

Necessity

importance of distinguishing between
important for distinguishing between

Again, also, there is a range of grammar patterns, including **noun + *of* + -ing form** (*difficulty of distinguishing*), **noun + verb + *in* + -ing form** (*problems arise in distinguishing*), **noun + -ing form** (*have difficulty distinguishing*), **adjective + *of* + -ing form** (*capable of distinguishing*) and **adjective + *at* + -ing form** (*good at distinguishing*).

It will be noted from Table 5.3 that the number of instances of each individual preposition before *distinguishing* is relatively low. It is therefore likely that the *distinguishing* search does not yield anything like the full range of possible examples of modal-like expressions preceding -ing forms. To compensate for this, a further, admittedly somewhat subjective, search was done comprising *of* followed by any -ing verb form. Ten thousand of the resulting lines were selected and those lines were scanned to find potential modal-like expressions.[3] The results are shown in Table 5.4. This is not intended to be a definitive list; as has been noted, what belongs and does not belong to the set of modal-like expressions will to a large extent depend on the verb following the preposition. It is, however, indicative of the range of items that in some contexts might be considered to have a modal meaning.

Table 5.4 Modal-Like Elements Identified by Selecting ***of* + -ing form**

Phrase	*Example*	*Meaning*
the advisability of the advantages of the benefits of the desirability of the disadvantage of	*What is the advisability of establishing separate accounts. . .?*	obligation
aim of goal of hope of wouldn't dream of had enough of be hopeful of have the/no intention of in the interests of the objective of policy of for the purpose of do not think of	*. . . one treatment targets the microscopic blood supply of the tumour with the aim of starving it to death*	inclination
be on the brink of be certain of be confident of have a chance of in danger of for fear of in the event of in the expectation of the feasibility of the likelihood of have the opportunity of have the option of possibility of face the prospect of run the risk of at the risk of be sure of be suspected of on suspicion of the threat of be on the verge of	*. . . a woman who has to suppress her femininity for fear of being seen as the weaker sex*	possibility
the difficulties of the impossibility of have the job of the problem of the task of	*It leaves us with the problem of finding free dates . . .*	possibility/difficulty

(continued)

Table 5.4 (continued)

Phrase	*Example*	*Meaning*
be capable of be incapable of have a knack of a means of method of way of	*. . . she does seem to possess a knack of sailing into troubled waters*	ability
the importance of the necessity of the need of	*I can understand the importance of paying any further pension contributions on a one-off basis . . .*	necessity

5.4 THE DISTRIBUTION OF WORDFORMS ACROSS LEMMAS

In section 5.3 a methodology was demonstrated which involved concordance lines and lists of collocates. This represents one way of 'doing corpus linguistics': finding patterns in the behaviour of an individual lexical item by observing instances of its use. Corpora also, however, produce quantitative information which can be processed to make generalisations about language use. An example will be given in this section.

Sinclair (1991: 44–51) has noted that the wordforms that constitute a lemma may behave very differently from one another in terms of phraseology, and that in those cases where a lemma may be said to represent a number of different meanings, each wordform will be disproportionately associated with one or other of the meanings. He notes, therefore, that the assumption that wordforms such as *accept*, *accepted*, *accepting* and *accepts* represent 'the same word' may be premature. This in turn is part of his argument that a corpus-driven view of language will focus on individual wordforms rather than on abstractions such as lemmas or grammatical categories. For Sinclair, the locus of enquiry should be, for example, the wordform *accept*, rather than either the lemma *ACCEPT* or, say, the present tense. If it emerges that generalisations can be made about *ACCEPT*, or about tenses, so be it, but such generalisations should not be assumed.

While taking Sinclair's point, I find it useful for present purposes to assume that lemmas have some validity because it allows us to compare the relative frequency of the wordforms making up the lemma.[4] One of the observations made as by-products of an investigation into the co-occurrence of wordform and complementation pattern (Hunston 2003b) is that verb lemmas that comprise four wordforms (typically: base form, -s form, -ed form and–ing form, and bearing in mind that these are surface form categories only) are not consistent in the order of frequency with

which those forms occur. For example, comparative figures for *FILL* and *EXPLAIN* are: *fill 35%*, *filled 47%*, *fills 5%*, *filling 14%*, *explain 41%*, *explained 37%*, *explains 13%*, *explaining 9%* (based on data from Leech et al. 2001). That is, whereas the -ed form of *FILL* is more frequent than the base form, the reverse is true for *EXPLAIN*, and whereas the -ing form of *FILL* is more frequent than the -s form, again the reverse is true for *EXPLAIN*. Differences between these verbs are to be expected, as, in Halliday's (1994) terms, *FILL* realises a material process whereas *EXPLAIN* realises a verbal one. As will be seen in the following, such explanations account for some but not all of the observed differences in frequency.

The proposal that such proportional differences are worth investigation owes much to Systemic Functional Linguistics, and in particular to the perceived importance of the relative probabilities attached to the various choices embodied by systemic networks. According to Halliday (1993: 3), a complete description of the systemic networks of a language involves not only an account of the (mostly binary) choices involved but also a calculation of the probability of each choice being realised. Accurate assignments of probability have had to wait for the availability of corpora. Halliday (1993) calculated the proportions attached to the polarity system (positive/negative) and the tense system (present/past) in the Bank of English; Matthiessen (2006) has taken this work much further by calculating the relative frequency of the choices in various systems in different registers, using corpora annotated for the occurrence of systemic features. One aspect of this work that is important is the possibility of identifying individual texts or indeed lexical items that are unusual in terms of the frequency of occurrence of associated features. For example, Matthiessen (2006: 106) demonstrates that overall texts belonging to the register of 'interviews' have a slightly higher proportion of free clauses than texts belonging to the register of 'news reports' (just over and under 80 per cent, respectively). One interview text that he cites, however, has a proportion of free clauses of about 90 per cent while one news report text has a relatively low proportion of about 70 per cent. The point is that an individual text, although it might have many more free than bound clauses, may still appear to have an unusually high proportion of bound clauses because comparison is made with the overall unequal frequency of occurrence rather than with a notional fifty-fifty split. A similar point can be made about lexical items. As Hunston (2006a: 67) notes, the verb *AFFORD* intuitively seems to be frequently negative (and is defined in the negative in Sinclair et al. 1995). Examination of corpus evidence suggests that in fact the verb occurs equally frequently in positive and in negative clauses. However, according to Halliday (1993) and Matthiessen (2006), the overall ratio of positive to negative clauses is about 9:1. Thus, an individual verb that is just as likely to appear in the negative as in the positive is in fact highly

skewed towards the negative. Even though the word cannot be said to be 'typically' negative, the relatively high proportion of negative instances warrants explanation. In the same way, the relatively high proportion of the base form of *EXPLAIN* warrants further investigation (see below).

It would be wrong, however, to see a strict parallel between the probabilities attached to choices in system networks and the relative frequency of wordforms in a lemma. Categories such as present and past tense represent alternatives within a grammatical system. At each point in a text, speakers are constrained to select one or other of a particular set of alternatives by global and local contextual factors. Although speakers do not make conscious clause-by-clause choices, it does make sense to say that a text is the outcome of a series of such choices. Recent work in corpus linguistics, however, has stressed that if what is chosen is both grammar (present tense or past tense, for instance) and lexis (one word instead of another), those choices interact with each other in extraordinarily complex ways (e.g. Hoey 2005). It is indeed possible to see the grammatical structure of each clause in a text as being the outcome of a series of lexical choices, each with its own preferred phraseology, rather than the outcome of the influence of contextual factors. However, it should be clear that the parts of the lemma themselves are not chosen. There is no sense in which a speaker selects between the base form, -ed form, -s form or -ing form, in an abstract sense, of a given lemma. Rather, the occurrence of the form is the consequence of a range of other choices, both, for example, the choice to tell a story using present or past tense and the choice to express a given idea using a particular lexical item. Thus, the proposal to use relative frequency as a starting point for investigation should not be taken to imply that the wordforms in a lemma form a system network.

In methodological terms, then, I am adopting from Sinclair the concern for the phraseological behaviour of individual lexical items, but I am in this discussion replacing the notion of 'typicality' (Hanks 1987: 121) with the notion, adopted from Halliday, of 'atypical probability'. Looking at the wordform *afford*, for example, it cannot be said that the word is 'usually negative' or 'typically negative', but it can be said that it is 'more-often-than-the-average negative'.

A number of explanations may be offered for the relative difference in the frequency of wordforms. Halliday and Matthiessen (2004: 301), for example, suggest that process types are distinguished by their unmarked aspect: progressive for material and behavioural processes and simple for verbal and mental processes. Thus we would expect verbs such as *WALK* or *CARRY* to have a higher proportion of -ing forms than ones such as *THINK* or *CONSIDER*. Table 5.5 shows the actual proportions in the British National Corpus (raw data taken from Leech et al. 2001).

Table 5.5 Rounded Percentage of Each Wordform in Four Lemmas

Lemma	base form	-ed form	-ing form	-s form	Total
WALK	31%	44%	22%	3%	100%
CARRY	32%	44%	18%	6%	100%
THINK	60%	31%	6%	3%	100%
CONSIDER	40%	46%	9%	4%	99%

As expected (if 'unmarked' can be taken to imply 'most frequent'), the -ing forms of *WALK* and *CARRY* are proportionally much more frequent than the same forms of *THINK* and *CONSIDER*, while the base forms of *THINK* and *CONSIDER* are proportionally more frequent than the base form of *WALK* and *CARRY*. However, the -s form of each verb occupies a similar proportion of the total of each verb, and the base form *think* is proportionally much more frequent than the base form *consider*.

To investigate this further, Table 5.6 shows the frequency of the immediate left collocates of each of the wordforms *walk*, *carry*, *think* and *consider*, taken in each case from a sample of 200 concordance lines from the Bank of English. The table confirms that the material process verbs *WALK* and *CARRY* are relatively infrequently found in the simple present, in that more of the base forms are non-finite, following *to* or a modal, than are finite following a pronoun. On the other hand, the same can be said for the mental process verb *CONSIDER*. In fact, Table 5.6 suggests that *think* is the anomaly, following *I* and *not* more frequently than any of the other verbs.

Table 5.6 Left Collocates of Four Base Forms (200 Lines Each)

Left collocate	*walk*	*carry*	*think*	*consider*
I	9	2	88	6
we	3	2	3	8
they	2	7	4	6
you	15	1	18	11
not / n't	7	11	20	9
modal verb	31	21	9	34
to	65	87	17	59

None of this contradicts Halliday and Matthiessen, who are specifically concerned with aspect rather than with wordform. However, it does suggest that explanations for the unequal distribution of wordforms may need to involve phraseology as well as grammar. For example, *you walk* is, from the evidence of the 200 lines of each verb shown in Table 5.6, relatively more frequent than *you carry*. In a sample 100 concordance lines of *you walk*, about half (52) are in the phrases *as you walk*, *if you walk* and *when you walk*. In a similar sample for *you carry*, the same phrases account for only 24 lines and there is no other very frequent phraseology. Thus, an explanation for the relative frequency of *you walk* would be incomplete without an explanation of the relative frequency of *as/if/when you walk* compared with the frequency of such subordinate clauses in general.

To take another example, Table 5.7 shows the comparative wordform frequencies for two sets of 'opposite' verbs. In each case where there is a wide difference in the relative frequencies, a difference in phraseology is also apparent. For example, *agreed* (-ed form) comprises more than half of the lemma *AGREE*, whereas *disagreed* (-ed form) comprises less than a quarter of the lemma *DISAGREE*. Further investigation suggests that whereas *disagreed* is usually past tense (about 92 per cent), *agreed* is both past tense and a past participle forming part of a perfective aspect verb phrase (22 per cent). More importantly, *agreed* is frequently followed by *to* or *that*, whereas *disagreed* is more frequently followed by *with*, *on* or *about*. Typical phraseologies, therefore, are *has agreed to* (*meet them*) and *we disagreed on* (*many things*). Similarly, the relatively frequent *appears* (-s form of the lemma *AGREE)* is frequently followed by *that* or *to* (also by *in* and *on*) whereas *disappears* (-s form of the lemma *DISAGREE*) more frequently ends a clause.

To summarise: it can be observed that wordform distribution is not consistent across lemmas. Some of the differences in relative frequency are explained by differences in marked and unmarked aspect, as predicted by Halliday. Most, however, are more easily accounted for in terms of phraseology. That is, an unusually frequent wordform will often occur frequently in a particular set of phrases (e.g. *I think*, *I don't think*, *as you walk*), and

Table 5.7 Relative Frequency of Wordforms for Four Lemmas

Lemma	base form	-ed form	-ing form	-s form	Total
AGREE	35%	58%	3%	4%	100%
DISAGREE	54%	23%	8%	8%	93%[5]
APPEAR	36%	35%	5%	25%	101%
DISAPPEAR	25%	59%	9%	7%	100%

proportionally different wordforms will often evince very different phraseologies (e.g. *have agreed that*, *disagreed about*).

This argument is relevant to the notion of modal-like expressions because where a base form or an -ing form has an unusually high frequency one reason may be a high proportion of modal meaning preceding that form. For example, the base form *explain* which, as noted earlier, appears to constitute a relatively large proportion of the lemma *EXPLAIN* occurs frequently in phrases such as *this could explain*, *we'll explain*, *trying to explain*, *at a loss to explain*, *never had the time to explain* and *how do you explain*. These are modals or express modal-like meaning.

The argument becomes stronger if complementation pattern is taken into account (as is the case with verbs followed by wh-clauses). There seems to be an association (a colligation) between modal-like elements and wh-clauses. For example, the ten most frequent verbs (wordforms) followed by *whether* are: *know*, *decide*, *determine*, *see*, *asked*, *wonder*, *say*, *wondering*, *wondered*, *deciding*; Table 5.8 shows the proportion of each phrase (*know whether*, *decide whether*, *determine whether* and so on) preceded by a modal auxiliary (excluding *ought to* and *have to*) and by *to* or a preposition. (In each case information is taken from 100 lines of the target phrase from the Bank of English. There is considerable variation in how many of the 100 lines are accounted for by the items mentioned.)

Table 5.8 Modal Meaning and *whether*

Phrase	*Preceded by modal (excluding modal + negative)*	*Preceded by to*	*Preceded by preposition/ conjunction*	*Preceded by* not / n't	*Total*
know whether	2	19	0	67	88
decide whether	21	60	0	6	87
determine whether	21	63	0	1	85
see whether	3	84	0	0	87
BE asked whether	9	7	1	0	17
wonder whether	7	32	0	0	39
say whether	8	56	0	29	93
wondering whether	0	1	8	0	9
BE wondered whether	-	-	-	-	-
deciding whether	0	2	64	0	66

(Note: there are too few instances to *BE wondered whether* to perform the calculation.)

Although there are some exceptions (*BE asked whether* is normally finite, and *wonder* is normally used in the phrase *I wonder whether*), most of these phrases show a high proportion of either the negative (especially in the phrase *I don't know whether*) or modals and modal-like expressions. This suggests that wh-words such as *whether* are strong attractors of modal meaning (and therefore, mostly, of the base forms that allow such meaning).

5.5 WHY DO MODAL-LIKE EXPRESSIONS MATTER?

So far it has been argued that some verbs, but more importantly some sequences, occur frequently both with modal auxiliaries as traditionally defined (*may*, *must*, *should*, etc.) and with a wide range of other sequences, often ending in *to*, *of* or *in*, that express forms of modal meaning such as possibility, ability and obligation along with related concepts such as degree of difficulty and importance. These have been called modal-like expressions. These expressions demonstrate the pervasiveness of evaluative or affective meaning in language and the importance of recognising such meaning as an integral, rather than adjunct, aspect of any clause.

Part of the motivation for discussing modal-like expressions has been to demonstrate one kind of corpus-driven methodology: not only were the expressions first identified by chance, the by-product of a very different study, but the category itself is identifiable only via concordance lines and sets of co-occurring words. Indeed, it might be said that the category exists only in the context of certain verbs or sequences. Whereas some phrases, such as *with the aim of*, might consistently be interpreted as modal-like, others are likely to be so interpreted only in the context of the relevant concordance lines. For example, a sequence such as *forced him to* is not in itself a modal-like expression; it becomes so only when it occurs before a verb which is otherwise frequently modified by other such expressions.

In this and the next section I wish to consider other reasons why the identification of modal-like expressions is of interest. I shall argue that in looking at the expressions themselves we get an insight into the complexity of evaluative meaning in English discourse; and in looking at verbs frequently modified in this way we see the importance of phraseology to that evaluative meaning.

I would like to offer three reasons for the importance of modal-like expressions:

- First, they represent some of the many ways in which modal meaning in English is expressed. They demonstrate the pervasiveness of such meaning in English discourse, far beyond what is apparent from the frequency of modal auxiliary verbs themselves (cf. Stubbs 1986).

- Second, they comprise additional evidence for the significance of phraseology in the way that English works. As argued earlier, it appears to be the case that some words, phrases and sequences are frequently preceded by modal-like expressions. In many cases it can be shown that such words etc. are frequently modified, in the grammatical sense, by modal phrases (e.g. *it is necessary to*), as well as by modal verbs (e.g. *must*). In others, however, the grammatical relation is not one of modification, and it may not be obvious from an individual example that any kind of modality is present (e.g. *he was told to*). Modal-like expressions are therefore a feature of discourse, and indeed of intertextuality, as much as of lexis or grammar.
- Third, it will be argued in the following that modal-like expressions are not simply an alternative to modal verbs but that they construct a more complex interaction between speaker and hearer, expressing other kinds of evaluative meaning and creating potential conflicts between points of view.

One of the points made at the beginning of this chapter was that learners of English consistently use modal auxiliaries more frequently than native speakers of English do (Aijmer 2002). A possible explanation is that there are contexts where a native speaker will use a modal-like expression in preference to a modal verb. Modal-like expressions of obligation, such as *it's essential to*, for example, may be regarded as less personal and so less face-threatening than an expression with a modal, such as *you should* or *you must*. There are, however, more interesting possibilities. The phrase *It's up to [you] to*, for example, can be seen as an alternative to *you must*. Like all structures with an anticipatory *it*, however, it also performs a function in organising information (Francis et al. 1996: 518). Both the pronoun (or other noun phrase occurring after *it's up to*) and the to-infinitive clause, occur in the New-information position in the clause, allowing emphasis to be placed on both. Thus, in *It's up to me to help her learn English*, both *me* and *to help her learn English* are presented as new information. The fact of the obligation is the 'point of departure of the message'—using Halliday's (1994: 37) glossing of Theme—with both the obliged person and the obliged action as new information. In the alternative *I should help her learn English*, the obliged person is the point of departure and the obliged action the new information.

A more complex interpretation is possible for the phrase *for fear of*, which presupposes the possibility of a future action or event, something which may be expressed also by the modal auxiliary *might*. Table 5.9 shows some examples with *for fear of* together with possible paraphrases that focus on the presupposed possibility with the word *might*.

Table 5.9 Sentences with *for fear of* and Their Paraphrases

Original sentence	*Paraphrase*
She did not cry out for help for fear of being further assaulted.	If she had cried out she might have been assaulted again.
I couldn't even go for a walk for fear of being mobbed.	I did not go for a walk because I might be mobbed.
I didn't say anything for fear of sounding naïve.	If I had said something I might have sounded naïve.
She never goes to the cinema for fear of picking up germs.	She does not go to the cinema because she might pick up germs.
Would witnesses do nothing for fear of getting involved?	Would they avoid reporting the crime because they might get involved?

In these examples, it is obvious that *for fear of* expresses not only possibility, but also, because of the word *fear*, a further negative evaluation. Use of the phrase stresses that events such as being assaulted or mobbed, sounding naïve or picking up germs are undesirable. In the case of the final example in Table 5.9, where 'getting involved' could be seen as either good or bad, *for fear of* indicates which is the correct interpretation.[6] In addition, however, the phrase indicates point of view, that is, who is making the judgement of possibility. For instance, in the first example, 'she might have been assaulted again', which in Halliday's (1994: 358) terms includes a modal with a subjective/implicit orientation, expresses a judgement made by the writer, whereas *for fear of being further assaulted* places responsibility for the judgement on the 'she' who is being reported. This is particularly significant in the final example, where the writer ascribes a negative interpretation of 'getting involved' to the hypothetical witnesses, reserving an alternative interpretation (that good citizens do get involved) for himself.

Thus it can be argued that modal-like expressions are an important resource in the expression of evaluation in English, expressing a complex interaction between assessment of certainty and assessment of value, and exploiting the potential distance between points of view.

5.6 WHICH VERBS ATTRACT MODAL MEANING?

It was asserted earlier that some verbs or sequences seem to attract modal meaning. The implication is that modal meaning, expressed both by modal verbs and by modal-like expressions, occurs more frequently with some verbs, or sequences, than with others. A challenge to the researcher, then, is to identify which verbs those might be. Whereas it is not likely that any

lexical item will always or never be modified modally, it is possible that there will be a 'norm' in the sense of a typical proportion of any verb that is modally modified, and that some verbs will fall outside that norm. The earlier examples were identified by serendipity, as a by-product of other studies. What might a more principled search of a corpus involve?

One way of identifying consistently modalized verbs is to exploit what is known already about modality, pattern and meaning. The study of the wordform *decide*, reported earlier, indicated that the sequence *decide* + wh-clause attracts expressions of modal meaning. It may be hypothesised that what attracts these expressions is the presence of a 'concerted mental process' (such as taking a decision about a possible course of action). If that is the case, then other similar verbs, together with their complementation patterns, will also attract modal meaning. To identify such verbs, a search was made of Francis et al. (1996) to identify likely verb groups from different patterns, and one example of each group was taken for further investigation. For example, the pattern 'verb *about* noun' includes a meaning group glossed as 'concerned with thought or feeling, or the expression of thought or feeling' (Francis et al. 1996: 148). It includes 31 verbs, such as *agonise*, *brood*, *cogitate*, *deliberate*, *muse*, *philosophise* and *worry*. Of these, one (*worry*) is chosen for further study. Ten such sequences were identified and investigated. These were:

speculate wh clause
speculate as to wh clause
persevere in n/-ing
wonder about n/-ing
agonise over n/-ing
ponder on n/-ing
look into n/-ing
grapple with n/-ing
turn attention to n/-ing
consider n *as* n/adj

For each sequence, a random 40-line sample was obtained. The 400 lines are divided into groups as follows:

1. finite verb form
2. with operator *DO*
3. modal meaning: obligation/inclination
4. modal meaning: necessity/importance/interest
5. modal meaning: possibility/ability/difficulty
6. other lines

The results are shown in the Appendix to Chapter 5 and are summarised in Table 5.10. As might be expected, there is considerable variation between

the verbs, as a result of the preferred phraseology of each. For example, *speculate as to* occurs very frequently in the sequence *we can only speculate as to* and *look into* occurs frequently with *be asked to* or *be set up to*. There is also consistency, however. Most of the verbs, as expected, have half or more of their occurrences accounted for by columns 4, 5 and 6. Only *wonder about* and *agonise over* occur with considerable relative frequency as finite verbs (column 2). It might be noted, too, that the 'other' category has been interpreted conservatively, and that some of the lines in there might be argued to have a touch of modal-like meaning (e.g. *Let's*, *his failure to*). In short, there is evidence that most of these verbs in these patterns attract modal meaning.

It is tempting to consider ways in which the predominance of modal meaning might be identified algorithmically. One possibility is to link the proportional frequency of base forms to relative frequency of modal meaning. As a pilot study, 30 verbs from a set of 101 meeting the criteria of four differentiated forms and a normed frequency of between 60 and 100 in Leech et al. (2001) were selected. For each of the 30 verbs, the percentage of the lemma accounted for by each form was calculated, using Leech et al.'s figures. From this, six verbs were identified with a 'high' frequency of more than 40 per cent base form, along with a further eight verbs with a 'low' frequency of less than 20 per cent. Concordance lines for those 14 verbs confirmed that their phraseology was highly patterned. For example, in 100 instances of *prefer*, 21 are *I prefer*, 16 are *would prefer* and 9 are *if you prefer*. Together these account for 46 per cent of the lines. In the case of *care*, polarity is important. Phrases such as *don't care*, *didn't care* and

Table 5.10 Summary of Uses of 10 Selected Verb Sequences

	Finite	**'do'**	**Obl/inc**	**Nec/imp**	**Poss/abil**	**Other**	**Total**
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Speculate wh	2	2	11	9	12	4	40
Speculate as to	4	0	12	1	21	2	40
Persevere in	7	0	18	2	6	7	40
Wonder about	16	3	9	0	5	7	40
Agonise over	16	4	3	4	2	11	40
Ponder on	6	4	10	0	9	11	40
Look into	1	1	25	0	2	11	40
Grapple with	8	0	13	4	11	4	40
Turn attention to	5	1	17	1	6	10	40
Consider as	6	5	7	1	6	5	30
Total	71	20	125	22	80	72	390

couldn't care less account for 41 of the 100 lines. In some cases, the phraseology is modal. In the case of *to link*, for example, meanings of inclination and ability dominate: e.g. *attempt to link*, *eager to link*, *keen to link*, *tries to link*, *able/ability to link*. However, frequency of *to* preceding the base form, and therefore occurrence of modal-like expressions, did not correlate with the relative frequency of the base form in comparison with the lemma as a whole. Rigorous ways of identifying high proportions of modality would clearly need to be rather more sophisticated than the one attempted here.

5.7 CONCLUSION: MODAL-LIKE EXPRESSIONS AND SEMANTIC SEQUENCES

This chapter has proposed a category of linguistic items called modal-like expressions which, it is claimed, can be seen to occur in concordance lines along with ordinary modal verbs. They co-occur mainly with the base form and the -ing form of verbs and seem mainly to consist of phrases ending with either infinitive *to* or a preposition. It is suggested that they are an important addition to our knowledge of modal meaning, an aspect that has been overlooked because they do not fall into any one grammatical category and are difficult to observe except in the context of concordance lines. In addition, they tend to imply modality in an oblique way rather than realising modal meaning atomistically.[7] It is hypothesised that some verbs are attractors of modal-like expressions, that is, that they co-occur with modal meaning in general more frequently than the average. Some illustrations of this have been given but at present no fool-proof way of identifying such verbs has been found.

In addition to demonstrating modal-like expressions, this chapter has concerned method: how linguistic features are discovered in corpora. It demonstrates some of the issues involved in moving from a corpus-driven, serendipitous method (just 'noticing' something that occurs frequently in concordance lines) to a corpus-based, rigorous method (attempting to quantify a concept that has no single realisation).

The basic method of investigation in the chapter has been the scrutiny of concordance lines. What concordance lines demonstrate is the way that multiple instances of a given sequence occur in broadly similar contexts, even though each instance has been produced in very different circumstances. Those 'broadly similar contexts' might be described in terms of semantic sequences. A word such as *distinguishing* is frequently followed by a prepositional phrase beginning with *between* and containing a plural noun phrase. This is grammatical information. It is also true that the sequence *distinguishing between* frequently follows an expression of difficulty or importance. It is therefore the case that we can identify something that is often said and which we can characterise as a sequence of semantic elements: 'difficulty or importance' + *distinguishing between* + 'two or more similar things'.

It will be argued in chapter 6 that semantic sequences are a broader concept than that of 'phrase' or 'unit of meaning' and that they are particularly useful in characterising given discourses. For the moment we will simply note that modal-like expressions form a recognisable part of many such sequences.

It is clear that this chapter has only scratched at the surface of what may be a much more pervasive phenomenon. For example, in a study of the wordform *gene* in a corpus of academic papers in Genetics, Plappert (in preparation) has noted that the phrases such as *candidate . . . gene* and *putative . . . gene* are ways in which geneticists appear to hedge or modalize their identification of particular gene functions. This suggests that modal meaning may be expressed in discipline-specific ways that do not involve modal verbs or adverbials. It is interesting, though probably not surprising, that the studies reported in this chapter, focusing on verbs, have mostly identified deontic modal meaning, whereas Plappert's study, focusing on nouns, has identified epistemic meaning.

6 Corpus Approaches to Investigating Status

6.1 INTRODUCTION

In Chapter 3, 'status' was defined conceptually as the averred degree and type of alignment between a text or proposition and the world. It was noted that status could be marked linguistically by a variety of lexico-grammatical features or left unmarked. It was also noted that the same concept could be applied to visual as well as to verbal texts.

Corpus studies of status are necessarily impoverished, from this point of view, in the sense that methodologies based on searching for or quantifying features rely on those features being identifiable without human intervention. For example, a computer cannot identify 'discoveries' but can identify instances of *discover*, *discovery*, *discovered* and so on. In compensation, however, corpus studies can reveal recurrences that are not visible when examining single individual texts or even many texts individually.

In this chapter I shall examine expressions that evaluate status in one corpus, looking at the contexts in which they typically appear. The aim is to build up a picture of common epistemological practices in the corpus under investigation. The methodology used will be similar to that described in Chapter 5, in that the starting point is the items such as prepositions that frequently co-occur with a given word or phrase and the focus is the semantic similarity of the phrases incorporating those prepositions. What is discovered using this method is 'what is often said'; in this chapter, we look at what is often said about propositions identified explicitly as being hypotheses, discoveries, assumptions and so on.

The chapter begins with a study of a number of status-indicating nouns, first reported in Hunston (2008a). It continues with more detailed case-studies of *assumption* and *fact*, both compared with *discovery*. The chapter ends with a discussion of how work of this kind might be of use to researchers in other disciplines.

6.2 STATUS-INDICATING NOUNS

Hunston (2008a) reports a study of 11 status-indicating nouns (such as *idea*, *assumption*, *conclusion*) occurring in a corpus drawn from the

publication *New Scientist*.[1] Each of the nouns concerned sometimes occurs in the pattern **N that**, that is, followed by an appositive that-clause containing the proposition given status by the noun. For example, in Example 6.1 the proposition expressed as 'we live in an ordered universe that is subject to precise mathematical laws' is evaluated as an *assumption*, that is, something that is not subject to immediate verification.

Example 6.1

Science relies on the assumption that we live in an ordered universe that is subject to precise mathematical laws

The phrase *on the assumption that* is particularly frequent: of the 63 instances of *the assumption that* in the corpus in question, 26 (41 per cent) are preceded by *on*. Most frequently (in all but six cases), the *on* is itself part of a verb + preposition combination such as *is based on*, *rest on* and *relies on*. We may conclude that 'assumptions' are most often construed as the foundation of other ideas. This is corroborated by other relatively frequent phraseologies, such as *START/SET off with the assumption that* (three instances), and *arises/starts from the assumption that* (two instances). Although no other phraseology is anywhere near as frequent, other noticeable phraseologies include *MAKE the assumption that* (five instances) and a set of instances that indicate a negative evaluation of the assumption. Three of these have *the assumption that* as a Complement following *is*, as in Example 6.2.

Example 6.2

Of rather more concern . . . is the assumption that because scientists cannot fit the reports of people who actually experienced the event into their theoretical model, those people must therefore be wrong.

Other examples where the assumption is evaluated negatively include:

warned against the assumption that . . .; *he then slides illogically . . . into the assumption that . . .*; *her results turn on its head the assumption that . . .*; *challenge/s the assumption that. . . .*

In the minority of cases (14, shown in Figure 6.1) where *assumption that* is preceded by an adjective, for the most part either that adjective indicates falsity (*dubious* or *unlikely*) or the phrase is in a context of challenge and opposition (*the seemingly innocent assumption*; *the folly of the old assumption*; *contradicting the standard assumption*). Line 10 (*it is a safe assumption that . . .*) is an exception to this.

Figure 6.1 Concordance lines for *assumption that*.

1 of the problem is the apparent assumption that, as the backdrop is so
2 s women may resent the automatic assumption that they are victims, it has
3 even if one makes the dubious assumption that the race will never become
4 comes from the seemingly innocent assumption that electromagnetic waves insid
5 <p> Unfortunately, it is this key assumption that seems to make Hsiang's proo
6 and exposed the folly of the old assumption that oceans mopped all of it up.
7 appears based on the hoary old assumption that anything which can make
8 there. This challenges the popular assumption that humans gradually increased
9 was based on the very reasonable assumption that the amount of information
10 their luggage. (It is a safe assumption that modern terrorists are not
11 by contradicting the standard assumption that neutrinos have no mass.
12 solution. He makes the standard assumption that the Universe began with the
13 values are based on the unlikely assumption that the observed X-ray source
14 of London, there is an unspoken assumption that those who are about to die

In short, a brief examination of the most frequent phraseologies of *the assumption that* indicates that where this phrase occurs the assumption is most frequently construed as the basis for scientific endeavour, with other examples indicating only that the assumption is made, or that the assumption is in question. Hunston (2008a) extends this methodology to other status nouns followed by that-clauses. Five main discourse functions in the context of the **N that** pattern are proposed:

1. Existence: assertion that the idea, suggestion, assumption etc. exists
2. Evaluation: evaluation of the idea, suggestion, assumption etc., either positively or negatively
3. Cause: identification of the idea, suggestion, assumption etc. as the cause of something else
4. Result: identification of the idea, suggestion, assumption etc. as the result of something else
5. Confirmation: recognition of confirmation or disconfirmation of the idea, suggestion, assumption etc.

Examples of phrases exemplifying each of these, taken from Hunston (2008a), are as follows:

1. Existence

There is a view that . . .
Finlay proposed his theory that . . .
assert the view that . . .
One woman made the discovery that . . .
make the assumption that . . .
take the view that . . .

come to the conclusion that . . .
come round to the view that . . .
come away with the impression that . . .
be under the impression that . . .

2. Evaluation

This in turn is divided into three sets of examples. The first set ('agreement') indicates that someone—either the writer or someone else—agrees or disagrees with the idea etc. The second set ('affect') indicates a personal and affective response—positive or negative—to the idea etc. For the third set I have adopted Martin and White's (2005) term 'appreciation' for a group of examples that indicate an assessment of the value of the epistemic object.

agreement:

The idea that . . . is accepted by . . .
still cling/hold to the notion . . .
Armstrong refutes our suggestion that . . .
supporter of the view that . . .
sway from the view that . . .
held firm to the belief that . . .
abandoned the belief that . . .
We should resist the claim that . . .
She denied the claim that . . .
concern is to debunk the theory that . . .

affect:

be attracted by the notion that . . .
be happy/have no problem with the idea that . . .
The discovery that. . .has rekindled fears . . .
More puzzling was the discovery that . . .

appreciation:

The notion that . . . is incorrect/abhorrent intriguing
The claim that . . . is questionable
The suggestion that . . . is misconceived
false/distinct/misleading impression that . . .
standard/popular/unlikely/mistaken assumption that . . .

3. Cause:

is based on the idea that . . .
rests/is founded on the notion that . . .

based/rests on the assumption that . . .
stem from the discovery that . . .
following the discovery that . . .

4. RESULT:

leads to the notion that . . .
give the impression that . . .
led to the discovery that . . .
has led to the claim that . . .
leads to the conclusion that . . .
will lead to a new theory that . . .
prompt the suggestion that . . .
has led to the suggestion that . . .

5. CONFIRMATION:

Again this group is divided into three based on the lexis used. EXPLANATION uses *explain*, CONSISTENCY uses (*in*)*consistent* or the near synonym *fit* and SUPPORT uses *support* and near synonyms such as *reinforce* or antonyms such as *undermine*.

EXPLANATION:

The idea that . . . may explain why

CONSISTENCY:

be consistent with the idea that . . .
inconsistent with the view that . . .
fits Lakes's theory that . . .

SUPPORT:

supports the notion that . . .
lend support/add weight to the idea that . . .
support the idea that . . .
The idea that . . . is supported/reinforced/undermined by . . .
Developments have strengthened the claim that . . .
to support the claim that . . .
lend support to the theory that . . .
has long since discredited the theory that . . .

It is further observed that the five functions do not occur with equal frequency with each of the nouns concerned. The following generalisations come from Hunston (2008a), but with additional commentary in each case. Data is taken from the *New Scientist* corpus throughout.

The CONFIRMATION category (number 5 in the previous list) is used with *idea*, *view*, *theory*, *claim*. For example, out of 100 instances of *the idea that*, 31 occur in contexts where the idea occurs in relation to another concept, a relationship which supports or challenges the idea. Examples include 6.3–6.5:

Example 6.3

The idea that centrioles may act as 'eyes' *would explain why* the centrioles in a pair almost always lie at right angles to each other.

Example 6.4

His new study challenges **the idea that** colour perception is universal.

Example 6.5

Evidence for a star orbiting a massive black hole in the centre of a galaxy has been revealed by X-ray observations. *The discovery supports* **the idea that** quasars and other bright galaxy centres are powered by gas swirling towards a black hole containing as much matter as a million or more Suns.

Most of the nouns—*idea*, *notion*, *assumption*, *discovery*, *impression*, *claim*, *conclusion*, *theory*, *suggestion*—are associated with CAUSE and RESULT (numbers 3 and 4 in the earlier list). *Discovery* is most usually construed as being the cause, either of an emotion or of an idea. Of the 54 instances of *the discovery that*, 28 fall into this category, of which the following (6.6–6.8) are examples:

Example 6.6

The discovery that mouse bone marrow cells can be turned into heart cells *could lead to* new treatments for human heart disease, say Japanese scientists.

Example 6.7

Researchers are eager to test the vitamin's effectiveness *following* **the discovery that** large doses can relieve the symptoms of a rare genetic disorder which causes a similar form of blindness.

Example 6.8

In recent years, asteroid hunting *has been given new impetus with* **the discovery that** there are far more asteroids close to the Earth than previously believed.

On the other hand, *impression*, *claim*, *conclusion*, *theory* and *suggestion* are more usually construed as being caused by something else. The most

straightforward of these is *conclusion*. Of the 49 instances of *the conclusion that*, most report that a conclusion has been 'reached' as an end goal, including 17 *come/came to the conclusion that*, four *arrived at the conclusion that* and three *reach/es the conclusion that*. There are a few evaluating the inevitability of the conclusion: *it is difficult/hard to avoid/escape the conclusion that* (five instances). Of the rest, most (eight) are *lead to/be drawn to the conclusion that*. Mostly, then, a conclusion is construed as the goal or consequence of an enquiry. An example is:

Example 6.9

Walker's research *leads him to* **the conclusion that** there was no 'failure' of German effort in nuclear fission research.

Another observation is that 'conclusions' are rarely evaluated or (dis)confirmed, Example 6.10 being atypical:

Example 6.10

The experiment *tears away the last arguments against* **the conclusion that** developing animals rely on counting cells to measure their size.

Other status nouns, however, most usually occur in the context of EVALUATION (number 2 on the earlier list). One instance is *notion*. Taking only those instances where *the notion that* is sentence initial (19 in total), two main groups of instances are observed:

- Accounts of the role a notion has played in intellectual debate (six instances) i.e. . . . *was the Holy Grail of the 1950s*; . . . *has been around since early this century*; . . . *was banned from discussion*; . . . *was bounced around for a while and sank*; . . . *is seen in the West as a piece of Stalinist propaganda*; . . . *was an entrenched part of the geophysical canon*.
- Evaluations of or reactions to a notion (nine instances) i.e. . . . *is profoundly dangerous*; . . . *is not as crazy as it may seem*; . . . *is intriguing but somewhat hard to believe*; . . . *is really abhorrent to some*; . . . *is incorrect*; . . . *invites myth-making about*; . . . *seems absurd*; . . . *was particularly significant*; . . . *misses the point*.

As a footnote to the contrasting descriptions of *conclusion that* and *notion that*, a somewhat unexpected finding is that *conclusion that* is more likely to be preceded by an adjective (11 of 80 instances, or 14 per cent) than *notion that* is (12 of 140 instances, or 9 per cent). Both sets of adjectives include those that indicate an affective reaction: *depressing/painful conclusion* and *depressing/reassuring notion*; both include indications of degree

of normality: *bizarre/surprising conclusion* and *paradoxical notion*; both include indications that the conclusion or notion is false or doubtful, although *notion* has a greater number of types in this group: *erroneous conclusion* and *absurd/erroneous/foolish/intuitive/speculative notion*. In addition, *conclusion* is used with adjectives with a positive evaluative sense, including those indicating a science-based epistemology: *fair*, *logical*, *mathematical* and also with *important*, whereas *notion* is used with adjectives indicating the place of the notion in (folk) epistemology: *ancient*, *common*, *cosy* and *popular*.

To conclude: this section has demonstrated that close attention to the co-text of a set of nouns with evaluative significance can show both similarities and differences in the ways they are used, which in turn throws some light on the role of language and of ideas in a given corpus.

6.3 CASE-STUDY: ASSUMPTIONS AND DISCOVERIES

In this section I wish to extend this study in another direction and look in more detail at 'assumptions'. It must be acknowledged that the sub-set of examples picked up by a search for *the assumption that* may not be representative of the whole. It is also true that not all assumptions are labelled as such; on the other hand, it can be argued that what we are looking for is not 'assumptions' per se but propositions which are explicitly labelled as assumptions. Possible searches, then, centre on the noun *assumption* and the verb lemma *ASSUME*.

In the *New Scientist* corpus I am using, there are 204 instances of the noun *assumption*, of which 95 are followed by *that* and a further 28 are preceded by *this* or *that*. There are 778 instances of *ASSUME* including 441 followed by *that*. In addition to the study of *the assumption that* reported in the preceding section, the following sets of concordance lines have also been examined:

- All instances of *this/that assumption*.
- All three or four word clusters occurring more than six times and ending with a form of the verb *ASSUME* followed by *that*—these include *to assume that*, *have assumed that*, *by assuming that*, *they assume/d that*, *I assume/d that*, *can only assume that*.

The aim was to test whether the results for *the assumption that* were born out on this wider scale: that the propositions labelled assumptions occurred in the context of (i) their link to other theoretical concepts, specifically that the assumption would be the basis for other research, and (ii) indications that the *assumption* is probably erroneous.

In both cases, the answer is in the affirmative. The study of *this/that assumption* finds five instances where the assumption is the basis for another theory

and only one where the reverse is true. Phraseologies associated with 'assumption as basis' are: *reflected that assumption*, *rely on this assumption*, *incorporate that assumption*, *this assumption was affecting*, *from that assumption they deduced*. In addition, the phrase *by assuming that* often indicates that the assumption is a means towards a theoretical goal, as in Example 6.11:

Example 6.11

Physicists thought they could explain the phenomenon **by assuming that** atoms were built from alpha particles bound together.

There are some counter-examples, that is, instances where the assumption is construed as being the result of observation. Examples include: *they based this assumption on . . .*; *led scientists to assume that . . .*; *as a result, physicists have assumed that. . . .* These counter-examples are relatively infrequent in this corpus, however.

In the larger study it becomes apparent that an assumption is far more likely to be evaluated negatively than positively. Examples of the phraseologies most associated with the negative evaluation are shown in Figure 6.2 (though in many cases corroboration of the negative evaluation comes from a wider context than can be shown here).

Figure 6.2 Phraseologies associated with the negative evaluation of assumptions.

in February. 'We can't make that assumption any more.' <p> In the 1980s, m
dense, star-like objects. <p> That assumption has been challenged by Edward
and colleagues have disproved that assumption (Annals of Internal Medicine,
are not touch-sensitive. But this assumption is wrong, as one American pla
mental activities? Or is this assumption wrong, and that's why Mr and M
The May inquiry has dealt this assumption a serious blow. <p> It could
applications see This Week), this assumption needs to be questioned. <p> T
turns out to be a child this assumption can be lethal, as it leads
to be impossible, but is this assumption correct? <p> Following the
more rain. What is wrong with this assumption? Michael da Silva (age 12)
Food (MAFF) have found that this assumption was mistaken. Although levels

surely, the Classic mistake is to assume that all scientists are male? So
enerations, it may be a mistake to assume that tomorrow's humans will stay
researchers have been too ready to assume that this is the reason they all
surprise us. There is no reason to assume that, for most of us at most times
and that there is no reason to assume that its 1930 level was any more
blooms. <p> It would be wrong to assume that rich countries will always be

Before Lockerbie, experts had assumed that it would take a sizeable bom
<p> Most scientists had assumed that osteoporosis was caused by
of depth. <p> Researchers had assumed that to produce a realistic 3D

Figure 6.2 (continued)

prevailed. <p> It has been assumed that marsupials couldn't hack it
nnes in all. Until now it has been assumed that the destruction of forests -
each other. <p> It has always been assumed that chimpanzees and gorillas in
d realistic. <p> It has often been assumed that 'real' dreams occur in REM
bstances at work, it has long been assumed that women should automatically

In the past, scientists have assumed that the collapse phase, which wa
females. Most anthropologists have assumed that it evolved in the context of
<p> Previously, researchers have assumed that the differences in web
<p> Until now, cosmologists have assumed that the big bang would have left

has two fallacies. First, it assumes that the guards are infallible,
of their weapons. Secondly, it assumes that their loyalty is absolute,
turns out to be too simplistic. It assumes that the surface is perfectly

his century" who made the error of assuming that rotational motion can be
aw, that is the classic mistake of assuming that all scientists are from the
into the philosophical trap of assuming that if something is affected by
People still fall into the trap of assuming that the laws of physics existed

The picture is not, however, uniform. The first major counter-example is the sequence 'adjective *to assume that*', which is often used to evaluate an assumption as 'reasonable' (Figure 6.3):

Figure 6.3 adjective to *assume that*.

very similar, it is reasonable to assume that they have a common source.
he earthquake, it is reasonable to assume that the fault follows this rout
dock, it would seem reasonable to assume that the breeders of puppies of
<p> The doctors were right to assume that the yolk was heated above i
term memory. But it seemed safe to assume that all these parts evolved
argues that it is only sensible to assume that they did. 'The megafauna were

The second exception is the phrase *can only assume that*, which is used in letters and editorials in the *New Scientist* with particular pragmatic effects, such as a covert criticism of a previous writer, as in Examples 6.12 and 6.13:

Example 6.12

The article by Terence Kealey . . . makes privatisation sound like a blessed relief, escaping the miseries of state support for the joys of the free market. **I can only assume that** the author has never actually undergone the experience.

Example 6.13

Your recent letter on Feynman's joke (14 January) reminded us of . . . Feynman's experiments with spaghetti: "If you get a spaghetti stick and you break it, it

> turns out that instead of breaking in half, it will almost always break into three pieces. Why is this true—why does it break into three pieces? Well we ended up at the end of a couple of hours with broken spaghetti all over the kitchen and no real good theory about why spaghetti breaks in three." **We can only assume that** Feynman was not really trying, since when we investigated this profound and fundamental problem in our own kitchen laboratory, not only did we quickly establish the underlying mechanism, but we even went on to formulate the following general rule for linear spaghetti structures . . .

The final hypothesis to be tested is that assumptions ascribed to the writer will be assumed to be true, whereas those ascribed to others will be more likely to be challenged. In other words, the hypothesis is that Example 6.14 will be typical of 'my' assumptions where Example 6.15 is typical of 'their' assumptions:

> Example 6.14
>
> [Speculating on the cause of the fracture of a glass cupboard in a laboratory] In our case the handle was bonded to the glass and **we assume that** the failure was caused by a flaw in the vertical hidden edge of the door which grew to a critical size by fatigue from the stresses of repeated operation of the door . . . [assumption is neither verified nor challenged]

> Example 6.15
>
> By contrast, Gross and Levitt reassert 'the fundamental methodology' of science; **they assume that** some neutral universal method can offer privileged access to higher truths. What lies behind this fervent belief? [assumption is ridiculed]

This turns out not to be the case, or at least not in so simple a fashion. Assumptions marked as belonging to a previous time, either by the tense or aspect of the verb phrase or by other markers around the noun, are generally questioned, as in Figure 6.4:

Figure 6.4 Writer's assumptions.

> This does not fit in with our assumption that each electron goes throug
> to be typed into databases. <p> My assumption has always been that the entir
> not clear, but it put paid to our assumption that upstream journeys by

> Radio 4 pips or Ceefax I naturally assumed that the broadcasters would set
> had made the same observation. I assumed that a much more sophisticated
> hat there was an "epidemic". I had assumed that TB was endemic (it has been
> reading Aldhous's article I had assumed that this was an accepted

(continued)

Figure 6.4 (continued)

of descent of organisms. We have assumed that genomes change only
World should be opposed. `We have assumed that it is bad, but we haven't
s worked so well for a while we've assumed that it has to be that way." <p>
eir failure shocked us. We had all assumed that our futures didn't require
ver mountainous areas, we'd always assumed that that's where we would see th
nnecessary drugs. Up to now, we've assumed that the two more or less balance

Where the assumption is 'present', a distinction needs to be made between those cases where *we* means 'people in general' and those where it means 'my team of researchers'. In the first instance, the assumption is generally challenged, as in Figure 6.5 and Examples 6.16 and 6.17:

Figure 6.5 we assume that.

be extremely doubtful, unless we assume that all life everywhere follows,
itnessed in a human population, we assume that it is cultural but ascribe th
to humans," says Shapiro. `Weassume that bacteria are rather simple,
ould normally have laughed off, we assume that some particular event has
as error detection. `Generally, we assume that people err because they act

Example 6.16

He expressed the hope, for example, that future scientists will be less dependent on mathematics for modelling reality and will draw on new sources of metaphor and analogy. '**We have an assumption** now that's getting stronger and stronger that mathematics is the only way to deal with reality,' Bohm said. 'Because it's worked so well for a while we've assumed that it has to be that way.'

Example 6.17

We have this assumption about ourselves that mind and consciousness are synonymous, that we're aware of everything that's important . . .

In the second instance, it is not challenged, though being an assumption it is not verified either, as in Example 6.18 and Figure 6.6.

Example 6.18

We made the assumption that people store names in the same way that they use them.

Figure 6.6 we assume that.

And, returning to EST : if we assume that he is generally less able to
or computer to do the sum. <p> We assume that our users are mathematically
was bonded to the glass and we assume that the failure was caused by a
ninhibited atmosphere. Even so, we assume that an errant spellchecker was
the sperm get a second chance. `We assume that it's adaptive. Every male is

The phrase *I assume that* is infrequent and, in this corpus, usually occurs in letters. Some are jokey, e.g. a writer reporting seeing shopping trolleys stranded like fish on a river bank and commenting *I assume that the trolleys are voracious feeders*. . . . In some cases, the writer is responding to another correspondent and states the premise of the reply, e.g. in response to a question about why portholes are round, a writer begins, *I assume that your correspondent is referring to old pictures of wooden ships*. . . . In other cases, the assumption appears to invite confirmation or correction from other writers e.g. in a letter about garlic capsules: *I assume that they carry the same benefits as fresh garlic*.

To summarise this study of the noun and verb that indicate the status of 'assumption': assumptions do seem to be a site of contention—indeed, it seems that challenged propositions are quite likely to be labelled as 'assumptions'. In addition, the 'assumption' label is often used to indicate changes in scientific belief, as old assumptions are overturned by new discoveries. This is not always the case, however. Current assumptions can be identified as *safe* or *reasonable*, and writers may label their own currently speculative propositions as assumptions without subsequently challenging them.

These observations might be set in context by comparing them with the lemma *DISCOVER/DISCOVERY*. There are a number of ways in which a proposition labelled *discovery* differs from one labelled *assumption*.

As noted earlier, the sequence 'adjective *to assume that*' is used to evaluate the assumption in terms of its reasonableness or correctness. Typical adjectives used in this context are: *appropriate*, *correct*, *too early*, *logical*, *natural*, *necessary*, *too ready*, *reasonable*, *right*, *safe*, *sensible* and *wrong*. In the equivalent sequence 'adjective *to discover that*', the adjectives are mainly those indicating a reaction on the part of the discoverer: *alarmed*, *amazed*, *astonished*, *upset*, *delighted*, *disappointed*, *interested*, *shocked* and *surprised*. Other sequences include '*it took* amount of time *to discover that*' and '*it is disheartening/interesting to discover that*'. In other words, whereas the adjectives with *assumption* assess the status of the assumption, those with *discovery* take the status of the discovery for granted and instead assess its affective impact. This coincides with a grammatical difference: most of the instances with *assume* have anticipatory *it* as the clause subject (e.g. *It is correct to assume that* . . .), whereas those with *discover* mostly have a subject which indicates an individual (e.g. *I was delighted to discover that* . . .). In the terms used by Mindt (forthcoming), in *it is correct*

that the adjective *correct* is an evaluator whereas in *I was delighted to* the adjective *delighted* is experiential.

Another distinction is useful here, one that will be mentioned also in the following. This is between 'personal' knowledge, which is different for each individual, and 'public' knowledge, which is the accepted epistemology of the relevant scientific community. In most cases, the sequence 'adjective *to discover that*' relates to a personal discovery, as in Examples 6.19 and 6.20:

Example 6.19

I was *upset* one day **to discover that** the flourishing patch of ivy-leaved toadflax (Cymbalaria muralis) growing on it had been sprayed by the parks department.

Example 6.20

Wilson was *shocked* **to discover that** in Cuba the natural environment scarcely existed.

Most instances of the sequence 'adjective *to assume that*', however, relate to public knowledge, as in Example 6.21.

Example 6.21

If two pieces of DNA are very similar, it is *reasonable* **to assume that** they have a common source.

It was noted earlier also that where an assumption was marked as belonging to past time, through verb phrase tense or aspect, for example, it was most frequently then said to be no longer believed. This is not true for discoveries. None of the instances of *discovered that* in the *New Scientist* corpus show any sign of the discovery being contested.

There are, however, a few additional points that might be made. Firstly, the proposition that is marked as a 'discovery' may itself also be marked as only tentatively aligned with the world. In Example 6.22, *may* is the marker:

Example 6.22

At the same time, AIDS researchers **have discovered that** new, lower-cost treatments to reduce the risk of pregnant women passing on HIV in the womb and at birth *may* increase the dangers of mothers transmitting the virus . . .

Secondly, the existence of the discovery itself may be marked as not aligned with the world. The *New Scientist* corpus contains ten examples of *CLAIM to have discovered*, although only one (Example 6.23) is followed by a that-clause:

Example 6.23

For it was Marston who in 1915 *claimed to* **have discovered that** blood pressure often goes up when people lie.

A similar example, though without a that-clause, is 6.24, where again the status of an event as a discovery is brought into question:

Example 6.24

The idea that Columbus '**discovered**' America is now being replaced by the less contentious description of his voyage as one of the first European 'encounters' with the continent.

Finally, some of the propositions marked as discoveries are in contexts where it is apparent that the writer regards the alignment of proposition and world as tentative at best. This is not a frequent occurrence, and the two examples in the *New Scientist* corpus are both in non-finite clauses. In Example 6.25 the writer argues that current practices in teaching science in schools are counter-productive and lead to a reluctance to engage with scientific issues:

Example 6.25

Having discovered that science is boring, remote, fragmented and hard, millions of people are effectively programmed to switch off when the subject comes up.

The proposition 'science is boring, remote, fragmented and hard' is attributed to 'millions of people'. Although marking the proposition as a discovery would normally mean that the writer agrees with the evaluation of alignment, the wider context indicates that this is not the case. This difference in point of view is often termed irony. Another way of looking at this is to call on the difference between a 'personal discovery' and a public one.[2] Most of the 'discoveries' in the *New Scientist* corpus are propositions which have become accepted as knowledge by the scientific world. In this case, however, it is a personal discovery that is at odds with the knowledge-base of other individuals.

The second example similarly plays with personal and community discoveries. It comes from an article in which the writers effectively evaluate the status of various well-known constants, such as the speed of light (c) or Planck's constant (h). They suggest that these constants are merely conventions and that calculations can be, and are, regularly performed without them. They talk about scientific units being subject to fashion and describe their own manipulations of the calculations as *tricks*. In this context of challenge to received knowledge, the sentence shown here as Example 6.26 appears mildly humorous.

Example 6.26

Having discovered that electromagnetism does not need epsilon, it may not come as too much of a shock to learn that gravitational theory does not need G.

The writers have argued, in effect, that alignment between proposition and world is contingent only, that what might have been thought to have been a 'discovery' (Newton's gravitational constant, for example) is only convenience. The new 'discovery', then, can only be a personal one ascribed to the reader.

The final study is of *this/that discovery*. There are two main contexts for these phrases, although sub-divisions can be identified. One context is a narrative—an explanation of how the discovery was made, as in Example 6.27:

Example 6.27

Doctors made **this discovery** when they applied a tiny current to a woman's brain and found she developed severe depression, which lifted soon after the current was switched off.

The other, much more frequent context, links the discovery to other ideas, usually with the discovery as the cause or starting point, as in Figure 6.7.

Figure 6.7 Concordance lines for *this discovery*.

However, Diamond thinks this discovery about ants may influence the
hair cell loss (see Box 1). This discovery brings hope that some day there
is expanding. Following this discovery, Einstein disowned the
in the impact plume. <p> Until this discovery, geologists thought that crater
myocardium). <p> Prompted by this discovery, Hariclia Karikouris and his
cent of the Sun's mass. <p> This discovery has shed light on the second,
molecular time bomb. <p> Does this discovery have any relevance for the
equipment needed to exploit this discovery in diagnosing Alzheimer's
<p> The key step in turning this discovery into a new lead for coronary
tongue and other muscles. But this discovery is unlikely to resolve the fierce
We don't know how important this discovery is yet, but it's going to change
be dangerous as well. <p> This discovery led the researchers to look at
the well-known Doppler effect. This discovery led to the theory of the
be isolated and manipulated, this discovery may open the gates to a flood of
as well as healthy snakes. This discovery may be the first direct evidence
Science, vol 267, p 217). <p> This discovery means that polypropylene can now
Edinburgh. On the basis of this discovery, most palaeontologists agreed
behaved as a semiconductor. This discovery opened up the prospect of
as long as the exons. <p> This discovery solved a couple of questions that
journey to Earth. Prior to this discovery, the most distant known supernova
of a cluster of stars). Before this discovery, the fastest spinning red dwarf
of cancer. The implications of this discovery will go far beyond those few

There is no equivalent of the use of *this/that assumption* in the context of negative evaluation. It might be noted, too, that a frequent use of *had assumed* is in the context of change ('scientists had assumed something until something else happened'). As the previous concordance lines show, *this/that discovery* is also used in the context of change, but this time as 'after' rather than 'before'.

6.4 CASE-STUDY: FACTS

The study reported in Hunston (2008a) avoided examination of the most frequent status noun, *fact*. There are a number of reasons for this. One is that the word is often used in phrases that perform functions other than the evaluation of status, such as *in fact*. Another is that this word potentially leads to contentious discussions about the nature of facts and reality. Related to this is a suspicion that when writers use a phrase such as *the fact that* they are not assessing the status of the proposition in quite the same way as they are when they choose a similar phrase such as *the hypothesis that*. In other words, the word *fact* has a different relationship to the things called 'facts' than the word *hypothesis* has with the things called 'hypotheses'. Calling something a hypothesis makes it so, and a hypothesis has to be so labelled, in academic discourse at least, to have that status. Propositions can have factual status without being labelled. In addition, as Francis (1993: 154) has noted, the propositions following *the fact that* sometimes contain a modal verb, suggesting that something labelled *fact* may not actually be a 'fact', rather as saying that a proposition is 'certain' actually detracts from the absolute certainty of the proposition (Halliday 1994: 89). In short, the label *fact* does not necessarily have a direct relationship with propositions considered by a discourse community to be 'facts'.

Partial corroboration of this comes from quantitative data relating to the word *fact* in the Academic sections of the British National Corpus (BNC). Table 6.1 shows the figures for the six discipline groupings identified. The first column shows the frequency of *fact* per million words. Because it is possible that these figures are skewed by high frequencies of *in fact*, figures for this phrase are given separately in the second column. The net figures for *fact*, excluding *in fact*, are given in the final column.

The table suggests a split between, on the one hand, Politics, Law, Education, Humanities and Social Science (with a higher occurrence of *fact*), and on the other hand Natural Science, Engineering and Medicine (with a lower occurrence of *fact*). It seems that disciplines such as those grouped as Natural Science, which might aver a more direct alignment between proposition and world, label that alignment with *fact* less frequently than do disciplines such as those in the Humanities, which might be expected

Table 6.1 Frequencies of *fact* in the BNC

Register	*Per million*	*'in fact' per million*	*'fact' less 'in fact'*
Politics, Law, Education	597.58	184.90	412.68
Humanities	643.61	294.91	348.7
Social Science	572.57	244.84	327.73
Natural Science	339.98	148.40	191.58
Engineering	341.11	189.50	151.61
Medicine	191.99	74.55	117.44

to propose a less direct alignment. Put rather crudely, it appears that the disciplines with the 'most facts' use the word *fact* least. For comparison, Table 6.2 gives the equivalent figures for *hypothesis*. Medicine, Natural Science, Social Science and Engineering use the word most, Humanities and Politics, Law and Education use it least.

Table 6.2 Frequencies of *hypothesis* in the BNC

Register	*Per million*
Politics, Law, Education	16.0
Humanities	36.1
Social Science	79.5
Natural Science	79.7
Engineering	91.4
Medicine	122.5

Further corroboration of the negotiability of propositions labelled with *fact* comes from looking at the collocations and colligations of *fact*, that is, the lexical and grammatical environments in which the word appears, and contrasting these with the similar contexts of *discovery*. Firstly, the most frequent adjectives preceding *fact* can be identified and grouped as follows:

Discipline: *scientific*, *historical*, *mathematical*, *technical*, *biological*, *behavioural*

Importance/size: *important*, *crucial*, *key*, *main*, *small*

Clarity/truth: *inescapable*, *actual*, *obvious*, *solid*, *unassailable*, *apparent*, *simple*, *plain*, *mere*, *brute*

Positive affect: *illuminating*

Negative affect: *sad*, *uncomfortable*, *shocking*, *sorry*, *distasteful*, *regrettable*, *hard*

Striking affect: *intriguing*, *amazing*, *interesting*, *strange*, *striking*, *astonishing*, *startling*, *incredible*, *odd*, *surprising*, *extraordinary*, *puzzling*

Other: *very*, *new*, *other*, *similar*, *quick*, *breathless*, *paradoxical*, *voracious*

Further investigation suggests that the sequence 'adjective + *fact*' has three uses.

Firstly, the discipline to which the fact belongs is named. In this category we might include *scientific fact*, which is then the most frequent combination in the category, even though 'science' is not a discipline but a way of looking at the world. This in turn encourages us to see the disciplinary labels not as denoting a particular academic community so much as indicating that a proposition is a fact in the context of a particular worldview. Looking at more co-text in the instances of *scientific fact*, it is noticeable that they often construe dispute, as in Examples 6.28 and 6.29:

Example 6.28

But these authors state the Turing thesis as if an established **scientific fact**, which we are dummies not to accept.

Example 6.29

Is this phenomenon poetic licence or **scientific fact**?

Arguably, *scientific* here is a focusing adjective that is in a sense redundant, as it adds no new information. A *scientific fact* might be said to be a 'real fact' or, as we shall see in the following, a 'fact of good character'.[3]

Secondly, the adjective expresses an affective reaction to the fact: it is pleasing, displeasing or surprising. A frequent phraseology is '*the* adjective *fact is that* . . .', where the expression of affect provides a rationale for introducing the proposition, as in Examples 6.30 and 6.31.

Example 6.30

Yet *the most intriguing* **fact** of all about neutron stars is that they have a solid crust.

Example 6.31

The *sad* **fact is that** TB, malaria and other diseases of poverty have always been neglected.

Finally, the adjective indicates the role of the 'fact' in the discourse: it is important, or its factuality is emphasised (e.g. *inescapable fact*). Other roles are identifiable too: for example, *very fact* is used in paradoxes, as in Example 6.32:

Example 6.32

With small talk and chitchat the meaning conveyed by our verbal language is . . . often subordinate to **the very fact that** we are making an effort to vocalise.

The hypothesis that the status of 'fact' can be negotiable may be tested by looking at the word *fact* following *as*, that is, in the pattern **V n *as* n**. The preposition *as* indicates the placing of the noun phrase following into a class or category, in contrast with other potential categories. There are ten examples in the *New Scientist* corpus, shown here as Figure 6.8.

Figure 6.8 Examples of **Vn *as* n**

[1] . . . an accomplished geochronologist who was retained as a witness by the State of California in 1980, defending against a civil suit brought by creationists who objected to **the teaching of evolution as fact** in public schools.

[2]. . . this popular book actually de-dramatises it by **treating it as a technical fact** causing no surprise.

[3] . . .of more reactionary influences within the Church and, in trying to moderate these, advised Galileo **to hold the heliocentric hypothesis as a theory and not as fact**.

[4] Your correspondent Jeremy Henty and your reviewer Colin Tudge both **seem willing to accept as fact** the oft-quoted observations of Eugene Marais on African termites.

[5] 'They don't say, "We think there's gravity". **They teach it as a fact**.'

[6] Yet, despite these introductory papers, The Cyborg Handbook is more interested in **the cyborg as science fiction than as science fact**. This book is a cultural studies text, not a scientific examination of the technological enhancement of the human body.

[7] With the exception of **disguising his semifictional account of the invention of PCR as fact**, Mullis has every right to write about any idea that occurs to him-in his dreams or half-awake.

[8] You can see that history in the language. What counts as a fact, what needs to be taken account of, a final reckoning, 'value' as a monetary rather than an ethical or religious term: all these add up to evidence for the prestige of numbers.

[9] I cannot supply the source of the urban myth, but it is not a recent misapprehension. **It was being touted as a 'scientists tell us' fact** in advertisements for self-improvement courses in the 1920s and was uncritically cited by Albert Einstein.

(continued)

Figure 6.8 (continued)

[10] This use of the past tense suggests that the problem is sorted and, accordingly, **the authors treat it as fact**. But the problem is not sorted and the authors concede that the definitive experiment has yet to be done.

In each example from Figure 6.8 except number 8 a situation is construed in which the factual status of a proposition is a point of contention, with the current writer challenging the status accorded by another writer. (Example number 8 is rather less explicit, but the phrase *counts as a fact* suggests dispute about the status of a number of propositions rather than just one.)

This finding may be set against a similar investigation using the noun *discovery*. In the same corpus there are only three examples of *discovery* following *as*. These are shown as Examples 6.33–6.35.

Example 6.33

The race to confirm the existence of cosmic ripples—**heralded by Stephen Hawking last year as the discovery of the century**—has been won by a team of astronomers from Princeton University and the Massachusetts Institute of Technology.

Example 6.34

. . . there was no convincing evidence that Gallo stole the French virus and **passed it off as his own discovery**.

Example 6.35

. . . an invention concerning biological material **is not classed as a discovery** merely because the material already existed, and so can be patented.

Only in Example 6.35 is the status of *discovery* actually at issue—in the other two, the discovery itself is not in doubt, only how important it is (Example 6.33) and whose it is (Example 6.34). In Example 6.35, what is at issue is not whether something is *actually* a discovery or not but whether it is classified as such by a bureaucratic process. In other words, whereas the status of something labelled *fact* is treated as negotiable, the status of something labelled *discovery* is not.

6.4.1 Semantic sequences with *the fact that*

Of 1,161 instances of *fact* in the *New Scientist* corpus (this figure excludes *in fact*), about two-thirds (772 instances) occur in the phrase *the fact that*. Phraseologies including that three-word phrase are:

by the fact that	67
to the fact that	55
despite the fact that	48
of the fact that	44
on the fact that	37
from the fact that	36
and the fact that	26
is the fact that	24
for the fact that	22
in the fact that	20
but the fact that	16
with the fact that	13
exploits the fact that	13
was the fact that	10
[take] into account the fact that	10
ignores the fact that	9
about the fact that	6
exploit the fact that	6
given/reflect/ignore the fact that	

It will be noticed that the most frequent of these (the top six phrases and eight of the top 12) include a preposition. Using the methodology described earlier in this chapter, the following recurring sequences can be identified (based on the 356 lines of *the fact that* preceded by a preposition).

1. FACT IS THE BASIS FOR A PRACTICAL OUTCOME OR REASONING

   [system/test/technique/method] *relies on the fact that*
   BE based on the fact that
   rely on the fact that
   due to the fact that
   comes from the fact that
   BE down to the fact that
   BE related to the fact that
   HAVE something to do with the fact that
   BE reflected in the fact that
   BE consistent with the fact that
   BE strengthened by the fact that
   [idea] *be supported by the fact that*
   [system/method] *TAKE advantage of the fact that*

2. FACT EXPLAINS SOMETHING

   [result/effect] *can be explained by the fact that*
   [people] *attribute this to the fact that*
   [reason] *lies in the fact that*

3. FACT IS THE CAUSE OF A PROBLEM OR ITS SOLUTION

   [problem/answer] *lies in the fact that*
   [problem/difficulty] *stems from the fact that*
   [problem] *is compounded by the fact that*
   [problem] *made worse by the fact that*
   be caused *by the fact that*
   [process] *BE complicated by the fact that*
   [task] *be made easier by the fact that*
   BE hampered by the fact that

4. SOMETHING USES OR ASSUMES A FACT (OR NOT)

   account for/take account of the fact that
   get around the fact that
   use can be made of the fact that
   make use of the fact that
   allow for the fact that
   This BE despite the fact that (all sentence initial)
   despite/in spite of the fact that
   apart from the fact that
   were it not for the fact that
   in recognition of the fact that
   [entity] *TAKE/DERIVE its name from the fact that*

5. BE AWARE OR UNAWARE OF A FACT

   [people] *face/wake/wise up to the fact that*
   blind us to the fact that
   [negative] *lose sight of the fact that*

6. PEOPLE TALK ABOUT A FACT

   [people] *MAKE much of the fact that*
   [people] *point to the fact that*
   [people] *draw attention to the fact that*
   [people] *refer to the fact that*

7. AFFECTIVE REACTION TO A FACT

   [people] *find/take/draw comfort from the fact that*
   [people] *BE intrigued by the fact that*

To make this more manageable, these sequences in turn can be divided into three broad (and overlapping) areas, for which I will borrow Groom's (2007) term 'motifs'. The phraseologies expressing these motifs can be augmented by including instances where *the fact that* is preceded by a verb rather than by a preposition. The three motifs identified are:

1. **The 'cause' motif.** Facts are the cause of things—outcomes, reasons, problems and solutions. The phraseologies identified as groups 1, 2 and 3 express this motif.
2. **The 'orientation' motif.** Things are oriented around facts—they are ignored or taken into account. This is expressed by group 4, and also by these verbs: *exploit*, *given*, *reflect*, *(not) change*, *(cannot) help*, *(not) stop*, *ignore*, *disguise*, *overlook*.
3. **The 'human response' motif.** People respond to a fact—talking about it, becoming aware of it or having a reaction to it. Groups 5, 6 and 7 express this, as do the verbs: *says*, *explain*, *mention*, *face*, *highlight*, *accept*, *underline*, *recognise*, *dispute*, *acknowledge*, *emphasise*, *advertise*, *bemoan*, *resent*, *know*, *like*, *take*, *believe*, *respect*.

In the study reported earlier, what is significant to *the fact that* are the verbs and prepositions that come before it. In 117 instances of the 772 instances of *the fact that*, however, the phrase is clause initial and what is significant is the verb group of which *the fact that* is the subject. These instances, too, can be related to the three motifs outlined here.

The 'cause' motif. *The fact that* . . . has a cause or explanation or, more frequently, constitutes a cause. See Examples 6.36 and 6.37.

Example 6.36

The fact that P. suturalis has two different forms of shell . . . needs a different kind of explanation.

Example 6.37

The fact that drug use is illegal in France leads to harassment near needle exchange sites and police presence in the case of an overdose.

The 'orientation' motif. Most instances of *The fact that* indicate that the 'fact' means, shows or implies something or that it offers corroboration of another idea. See Examples 6.38 and 6.39.

Example 6.38

The fact that it [the brain] can support psychological functions does not show that computers can.

Example 6.39

The fact that the ripples have the same brightness, no matter what their size, is predicted by the theory of inflation.

The 'human response' motif. Clause-initial *The fact that* is used also to indicate that a 'fact' is significant or, less frequently, that it is valid. See Examples 6.40 and 6.41.

Example 6.40

The fact that human and chimp DNA is more than 98 per cent alike is certainly important.

Example 6.41

The fact that room size can amplify the effect is true, but . . .

6.5 CONCLUSION

The argument in this chapter can be summarised as follows. Propositions can be evaluated in terms of their status, using resources associated with modality and with evidentiality. One set of such resources are 'status nouns'. These constitute a sub-set of the nouns that may be followed by appositive that-clauses. Evaluation of status reifies propositions, and status nouns are the resource by which this is most obviously done.

Status nouns can be investigated in context. Their comparative frequency can be calculated (giving useful data comparing disciplines) and their phraseology can be investigated. Looking at the phraseology gives an insight into what is commonly said (in given contexts) about propositions evaluated in a particular way. The phraseology includes specific word sequences and less uniform sequences of semantic units.

This chapter has investigated a number of status nouns, among which *fact* stands out as being both the most frequent and the most contentious. Much of that study was inspired by a research project carried out by Morgan and colleagues at the London School of Economics and Political Science with the title 'How Well Do Facts Travel?' The project undertook a number of detailed case-studies all of which involved knowledge from one place, age or discipline, being used in another place, age or discipline.[4]

One instance is the story of Eyam (Wallis 2005), in which facts about this Derbyshire village from the seventeenth century (the village fell victim to plague, most villagers stayed in their homes rather than escaping to other parts of the country, a majority of the villagers died) appear to have 'travelled' to the eighteenth and nineteenth centuries and become part of a narrative in which the villagers, led by their Christian minister, heroically adopted a voluntary quarantine in an attempt to prevent the spread of infection to other villages. All these facts have travelled again to the twenty-first century and form part of Wallis's reinterpretation, which sees the quarantine as imposed from outside rather than voluntary, and as partial, affecting poor villagers but not the wealthy.

Morgan's (2007) concern with travelling facts coincides well with a concern for averred status. Morgan resists attempts to impose a choice between 'facts are discovered' and 'facts are socially constructed' positions. She is interested in how facts are used, not in how they are produced. This is somewhat similar to the concept of *averred* alignment of word and world, which is deliberately agnostic on what is real. If a writer calls something a theory or treats something as a fact, then for that writer it is just that.

Summarising the research that asks, 'why do some facts "travel", or get taken up and reinterpreted in other times and by other disciplines?', Morgan argues that:

- Facts travel well when they travel with good companions: 'Travelling well depends on a set of contingent circumstances and multiple actions that create good companions.'
- Facts travel well when they carry 'character', that is, characteristics, attributes or functions.

Interpreting the metaphor here is not entirely straightforward, but this is how I understand it:

- For a fact to travel from one context to another, the people in the new context have to accept it as true.
- They will do this if they see it as being supported by evidence, in particular evidence from a variety of different sources. Eventually a 'fact' will be taken for granted and will no longer be open to scrutiny, although it is also true that nothing remains taken for granted forever; changes in belief systems will open old facts to new scrutiny. An example is Wallis's study of Eyam, which shows the establishment and then de-establishment of the story.
- They will also do this if they see facts as having explanatory value, that is, as being useful to them. For example, people who believed that the people of Eyam were heroically selfless were at least partly motivated by a desire to see positive effects of Christian conviction.
- They will also do this if they see the fact as independent of its creators, that is, not being contingent upon the processes of its construction.

It should follow that when people label things as facts they might show awareness of each of the preceding. Let us now summarise what the corpus study of *fact* (in a particular corpus) has told us:

- Some of the phraseology of *fact* indicates negotiability. This can be observed by looking at the *as* construction with *fact* as its object, also by looking at *fact* co-ordinated with *and* and *or*. Here, common equivalent words are: *detail*, *figures*, *formulae*, *information*, *logic*, *observation*. Common non-equivalent words are: *artifice*, *fiction*,

interpretation, *prejudices*, *fantasy*, *imagination*. There is an apparent acknowledgement that (i) factuality as a category of 'epistemic objects' (Mattila 2005) can be seen to exist and to be definable but that (ii) each individual fact needs to be worked at and negotiated.

- A great deal of the phraseology indicates non-negotiability. Things labelled as *facts* are not created by *current* discourse but brought in from other discourses. The phrase *the fact that* is the most frequently occurring vehicle of travel.
- Establishing the phraseology of *the fact that* involves examining in detail the preceding prepositions and verbs, where *the fact that* is in Rheme and the predicates of *the fact that* where the phrase is in Theme, to use the terminology of Systemic-Functional Grammar.[5]
- Things commonly said about facts include:
    - Facts are the cause: they lead to outcomes, problems, solutions, conclusions; they mean/imply things, or are commented on because contrary to expectation they don't mean/imply things.
    - Facts are the immoveable objects around which other epistemic objects orientate themselves: they are ignored or taken into account, or they corroborate ideas.
    - Facts are important.
    - People also orientate themselves around facts: they talk about them and react affectively to them, either positively or negatively.

In a sense, once a proposition has been labelled *fact* in the *New Scientist* it is, in Morgan's terms, successful. It has travelled out of its original domain and into the area of popular scientific discourse. We can see that discoursally it has become situated at the heart of things; it is the proposition to which people react and around which other epistemic objects orientate. For the period of its lifetime it has consequences. The shadow of negotiability constantly hangs over it, however, and the possibility of being 'de-facted' is always present.

7 Grammar Patterns, Local Grammars and Evaluation

7.1 INTRODUCTION

This chapter reviews some possible answers to the question raised in Chapter 1 and again in Chapter 4: to what extent is it possible to use patterning in language to identify (and therefore quantify) evaluative meaning? The chapter starts from the work on pattern grammar carried out in the 1990s (Hunston and Francis 1999), but considers also more recent approaches, in particular the FrameNet project (e.g. Fillmore et al. 2003) and the local grammar of Affect proposed by Bednarek (2008). The chapter does not dwell on the Sentiment Analysis approach that was covered in Chapter 4, but in a sense it offers a corpus linguistics alternative to that Computational Linguistics approach. What lies behind all of this work is the question of the relationship between form, in particular phraseology, and meaning.

There are in fact many ways of expressing the relationship between co-text and meaning. As Sinclair (1991: 104–105) has noted, although ambiguity is common when words are considered in isolation, it is a rare phenomenon when ordinary running text is encountered as co-text typically disambiguates. The other side of the coin is that words which regularly occur with similar co-texts (*It is possible/likely/doubtful that*) tend to share aspects of meaning (Hunston and Francis 1999), to the extent that co-occurrence can be used as a search principle for identifying synonyms and antonyms (Renouf 2009). When words are considered to belong to semantic sets, recurring patterns in their co-texts can be parsed to identify common semantic elements (Fillmore et al. 2003). Sinclair (2003) prioritises meaning in context and argues for the concept of 'semantic reversal', where co-text imposes a meaning on a word, as in *Matt was usually the perpetrator of the best jokes*, where *perpetrator* imposes a negative interpretation of *best jokes*. Goldberg (2006: 6) similarly argues that it is constructions that impose semantic classifications upon verbs rather than vice versa. All such arguments tend to assume that two phenomena—word-meaning and co-text—are independent of each other, or at least can be spoken of as if they were. Teubert (2004) goes further and argues that meaning is to be identified only in text, and that it often makes no sense to talk of the meaning of a word independent of its co-text.

There are, then, at least three ways to talk about the relationship between the meaning of a word and the co-text of that word. One is to suggest that co-text can be used as a diagnostic, to *identify* words that share meaning (as Renouf 2009 mostly does). Another is to argue that co-text *selects* words that share meaning (as Renouf 2009 partly does and as Hunston and Francis 1999 mostly do). The third is that co-text *construes* meaning, so that all words occurring in a given co-text share meaning, but only in that environment (as Teubert 2004 does and as Hunston and Francis 1999 sometimes do). The difference may be due as much to research context as to conceptual disagreements. Dictionary compilers, for example, find it inconvenient not to assume that words have meanings.

This chapter explores the relationship between recurring patterns and evaluative meaning. In particular, it discusses grammar patterns and local grammars of evaluation, and asks whether grammar patterns can be used as a diagnostic to distinguish between types of evaluative meaning or to parse evaluative statements in running text.

The chapter takes as its starting point the grammar patterns identified by Francis et al. (1996, 1998; see also Hunston and Francis 1998, 1999) and the concept of local grammar developed by Barnbrook and Sinclair (Barnbrook 2002; Barnbrook and Sinclair 1995) and sometimes applied to evaluative language (Hunston and Sinclair 2000; Hunston 2003a; Bednarek 2008). However, the grammar pattern project was not envisaged as something that related to evaluation. The primary aim was to provide a transparent and flexible coding system for entries in the Collins COBUILD English Dictionary (Sinclair et al. 1995), and then to present the patterns independently of the dictionary in a way that made apparent the association between pattern and meaning, form and function (Francis et al. 1996, 1998; see also Sinclair 1991).

Inevitably, however, the patterns that attracted most attention were those that in some way conveyed evaluative meaning, and in particular those in which 'semantic reversal' (Sinclair 2003) seemed to take place. Work on patterns such as **V n *into*–ing** (e.g. *had tricked him into kidnapping her*), **V *way* prep/adv** (e.g. *elbowed their way through the crowd*) and ***there* v-link *something* ADJ *about* n** (e.g. *There's something strange about her*) led to the observation that these patterns imposed an interpretation of evaluative meaning on words that in other contexts would not be considered evaluative (Hunston and Francis 1999: 105).

When words that shared both pattern and some semantic similarity were grouped together (in Francis et al. 1996, 1998), it became apparent that functional roles could be mapped on to pattern elements with some consistency. Some of this was 'old news'. For example, among the verbs that occur in the pattern **V n n** (verb followed by two noun phrases) there is a group consisting of examples such as: *the man who had caused her all that pain*; *this defiant stand against Europe . . . finally cost her the premiership; it saved him the bother of getting washed and dressed for school*; *to spare him further embarrassment* (Francis et al. 1996: 275). These examples can all be parsed using similar terminology, as in Table 7.1.

Table 7.1 Parsing the **V n n** Pattern

AGENT	*PROCESS*	*PERSON AFFECTED*	*COST OR BENEFIT*
The man	caused	her	all that pain
This defiant stand against Europe	cost	her	the premiership
It	saved	him	the bother of getting washed and dressed for school
	spare	him	further embarrassment

Other examples are less traditional, although, as noted below, they do share much in common with the semantic roles proposed by Fillmore (e.g. Fillmore and Atkins 1992). One example is shown in Table 7.2, from the pattern **V n *from* n**:

This regularity and the possibility of consistency in mapping led to an attempt to develop a local grammar of evaluation (Hunston and Sinclair 2000; and see below).

There seems good reason to suppose, then, that grammar patterns and local grammar, and indeed the other similar ways of systematising corpus-based observations of form and meaning, will have a good deal of relevance to the study of evaluative language. This chapter explores the extent of that relevance.

Table 7.2 Parsing the **V n *from* n** Pattern

AGENT: PERSON, ACTIVITY OR THING	*PROCESS: PROTECT*	*BENEFICIARY: PERSON OR THING*		*UNDESIRABLE STATE*
Bingeing	can cocoon	a person	from	the pressures and challenges of everyday living
The suspension	is designed to cushion	passengers	from	the effects of riding over rough roads
They	insulate	us	from	despair
ozone	protects	us	from	the harmful radiation from the sun
that	would safeguard	them	from	future errors
	Shade	tender seedlings	from	bursts of scorching sunlight
Their charitable status	shields	them	from	the tough realities of trade publishing

7.2 PATTERN GRAMMAR AND ITS CLOSE RELATIONS

The concept of pattern is introduced in Francis (1993) and discussed at length in Hunston and Francis (1999) and subsequently in other papers (e.g. Hunston 2004b, 2007a). Pattern annotation of word senses can be found in the Collins COBUILD English Dictionary (Sinclair et al. 1995, and subsequent editions) and patterns are listed together with the words that occur as their core in Francis et al. (1996, 1998). The notation used in those publications expresses the pattern as a series of elements, which may be a specific word (usually but not always a preposition) or a group or clause type (e.g. noun phrase, that-clause). The pattern includes only those elements which are distinctive of the core word and not others. It does not analyse elements to express their clausal function (such as Object or Adjunct), although such an analysis can be undertaken, as is done in Francis et al. (1996). Hunston and Francis (1999: 152) argue that such functional analyses are inherently unsatisfactory, though others, such as Teubert (2008) and Reichardt (2008), suggest that the grammar pattern agenda might be enhanced by linking it to the concept of valency. Herbst et al. (2004) explore pattern from a valency point of view.

To give a brief example of grammar patterns, here is the verb *FORGET*, as coded in CCED (some of the codings only are given here):

V n (verb followed by a noun phrase: the Subject of the verb is not coded and neither are any accompanying adverbs or other elements; the fact that the noun phrase is an Object is not coded) e.g. *Sometimes I change the words because I forget them.*

V wh (verb followed by a finite wh-clause) e.g. *She forgot where she left the car.*

V to-inf (verb followed by to-infinitive clause) e.g. *She forgot to lock her door.*

V–ing (verb followed by an -ing clause) e.g. *I'll never forget going to Sunday School as a kid.*

V *about* n (verb followed by a prepositional phrase beginning with *about*) e.g. *She forgot about everything . . .*

V pron-refl (verb followed by reflexive pronoun: although with many verbs the noun phrase in a **V n** pattern may sometimes be a reflexive pronoun, the specific coding is given here because the meaning of 'forget oneself' is quite different from instances where *forget* is followed by any other noun or pronoun) e.g. *He forgot himself . . .*

As noted earlier, in two publications (Francis et al. 1996, 1998), lists are given of all the patterns for verbs, nouns and adjectives used in CCED; further lists are given for all the words (verbs, nouns, adjectives) that are the core words of those patterns. In the case of Francis et al. (1996), which deals with verb patterns, the pattern elements are mapped on to functional elements such as Object, Complement and so on. The words within each pattern are grouped into ad hoc 'meaning groups' which serve to highlight perceived semantic similarities.

For example, Francis et al. (1996: 145–153) collect together all the verbs in CCED that have the coding **V *about* n** and group them to highlight semantic congruence. Six such groups are given for this pattern. One group, for example, lists 31 verbs that indicate mental processes such as thinking or feeling, including *forget* and other verbs such as *agonise*, *agree*, *bother*, *brood*, *care*, *cogitate*, *daydream* and *deliberate*.

Pattern grammar arises from Sinclair's concept of the idiom principle as a descriptor of language. Like collocation, patterns represent an instance of the idiom principle in operation. (And, like collocation, they are often best seen as coming about because of a more pervasive phraseology than is represented by the pattern itself.) Each grammar pattern identified is a simple description of the recurrent behaviour of a given word. Behind the concept of pattern grammar lies an assumption that lexis and grammar are inseparable, and that any system of grammar is the outcome of the accumulation of each instance of each word in its context. It does not, however, take a theoretical stance on the mental processing of grammar, and, although some sense of 'similar meaning' between individual words is taken for granted, the existence of semantic classes in the mind is not presupposed. On the contrary, as shall be discussed in the following, it is assumed that semantic congruence is an ad hoc and mutable concept, and that meaning is invested in phrases rather than in words. In these ways pattern grammar is distinguished from other concepts to which it otherwise appears similar, such as construction grammar or frames (see below).

One of the issues in pattern grammar research is the relationship between pattern and semantic classification. Francis et al. (1996, 1998) somewhat naively group words into ad hoc 'meaning groups'. This has led to criticism (e.g. Teubert 2008) that the groupings are misleading because they appear to propagate the mistaken assumption that meaning 'belongs to' single words and that groups of words can 'mean the same thing', as well as ignoring aspects of semantic and grammatical theory that might more usefully inform the description.

The starting point for pattern grammar is the individual word, with 'word' used in its dictionary sense of a lemma (though see Chapter 5 for a discussion of pattern and lemma). To illustrate this, Figure 7.1 shows concordance lines for the lemma *RECOVER*. Each set of lines within the figure illustrates a different co-text.

Figure 7.1(a) Co-text: *RECOVER* is followed by a noun phrase (**V n**); also included here is the passive equivalent (***be* V-ed**).

do it myself, come to that." Gwen recovered a little of her commonsense at
disillusionment; maybe the Left is recovering an awareness once common among
should be given much wider powers to recover costs. <xr> 9141 </xr> <hl> Bricks
data retrieval service-and they recovered everything Alison had lost! Vogon
through, time, in particular, to recover his health and heal the horrendous
teeters for a moment on its bow, recovers his balance, and then walks the
wishes to sue the manufacturer to recover loss caused by a serious defect in
has been a way of exploring and recovering part of myself, of somehow
drive offers an opportunity to recover some of the deficit by virtue of
confidence, go back to his roots, recover who he was. <p> NAFTA was the
be changed. From now on, you quicklyrecover your sense of humour and your

If he waited until damages were recovered as they almost inevitably would
important if your stolen bird is recovered by the police and positive
sank off the Isle of Man could be recovered within a week if they are still

Figure 7.1(b) Co-text: *RECOVER* is followed by a prepositional phrase beginning with *from* (**V *from* n**).

job than his predecessor, who recovered from a poor start to clinch the
to provide cover while Hartson recovers from a knee injury and Gayle a
graver complications and almost recovered from bronchitis. He rested at
coming back in good numbers after recovering from disease. <p> There are also
from hospital, where he is recovering from a broken cheekbone and a
in two tie-break sets and Mecir recovered from a set down against Derek
last month after a long break to recover from a shoulder operation in
on Staff: Two out of three are recovering from addiction. Third is in AL-
been reversed. Scott's has never recovered from Roland Huntford's savage
from New York wrote, after recovering from an operation in China where

Figure 7.1(c) Co-text: *RECOVER* is followed by a noun phrase and a prepositional phrase beginning with *from* (**V n *from* n**).

go home." <p> Rescue workers have recovered 180 bodies from beneath the
<p> On 2 April, the lifeboat recovered a body from the sea which was
strategy by Mr Al Fayed's lawyers to recover costs from Mr Hamilton. The focus
such as estimating motion and recovering depth information from
percent certain of its ability to recover payment from your insurance
for other uses. Fishermen have recovered some from the sea, including a
Insurance were given fresh hope of recovering substantial sums from Michael
needs an exact copy of the key to recover the message from the cryptogram.
also in hand, he said, to physically recover the oil from the sea. <h> POPE IN
she will have great pleasure in recovering those expenses from her
simply being a miner, let alone recovering uranium from its Olympic Dam
is as foolish as saying that in recovering your property from the man who

Figure 7.1(d) Co-text: *RECOVER* occurs at the end of a clause or is followed by a non-dependent prepositional phrase or adverb (**V**).

shows signs—however faint—of recovering. <p> A small but growing minority
sound and I am confident we will recover and prosper as we have in the
in particular—had not entirely recovered at Monday's training session.
is sure her daughter will fully recover. `I'm very confident—she's a
Taylor Hawkins was last night recovering in hospital after the weekend's
and an eight-year-old girl is now recovering in intensive care. <p> Croydon
2000 </dt> A boy aged 11 was recovering last night after his grandfather
more hope we had she was going to recover. On occasion we felt she knew we
unless the world economy starts to recover soon. Financial crises and risks,
minutes against Everton, but recovered to keep his side in the game with

A few observations might be made about these examples. In Figure 7.1(a), the noun phrase following *RECOVER* (or the subject in the passive examples) may indicate either a positive cognitive, physical or emotional state (*her sense of humour*, *who he was*, *his health*) or a missing item (*some of the damages*,[1] *your stolen bird*). In 7.1(b) the noun phrase following *from* indicates an illness or other undesirable situation. The examples in 7.1(c) all deal with getting something from somewhere, while those in 7.1(d) are about regaining health, fitness or positive emotional state, that is, returning to 'normality'. In other words, if we distinguish two possible paraphrases of RECOVER: 'become physically or emotionally healthy after a period of reduced well-being' and 'get something from somewhere', then concordances 7.1(b) and 7.1(d) belong to the first and concordances 7.1(c) belong to the second. Concordances 7.1(a) show instances compatible with each paraphrase.

Other verbs that parallel *RECOVER* may also be identified. For example, verbs that also occur with the co-text of *from* followed by a noun phrase (the pattern **V *from* n**) indicating an illness or other difficulty include near synonyms of *recover* in the 'become healthy again' sense (*convalesce*, *recuperate* and *bounce back*) and verbs indicating suffering rather than recovery (*die*, *reel*, *smart*, *suffer*). Verbs followed by a noun phrase and then a prepositional phrase with *from* (**V n *from* n**) include several indicating a process of 'getting', such as *acquire*, *extract*, *get*, *steal* and *take* (information from Francis et al. 1996: 189, 375). We might conclude that each pattern makes salient one of the aspects of meaning of *RECOVER*. This concept of semantic grouping will be returned to later.

There are a number of assumptions and limitations involved in presenting the concordance lines and their patterns in this way. The most significant assumption is that only those items in the co-text that constitute defining characteristics of this particular verb need to be mentioned. For this reason, the co-text labels do not mention that the verb *RECOVER* has a subject in each case (as all verbs have subjects); they mention only what occurs after the verb. More controversially, it is assumed that some

items occurring after the verb are a defining feature of it, and so should be mentioned as significant co-text, while others are not. For example, in Figure 7.1(d) there are examples of *RECOVER* followed by an adverb or prepositional phrase indicating time (*at Monday's training session*, *last night*, *soon*); there are also examples of a more specific pattern—progressive *recovering* followed by *in hospital* or *in intensive care*. An examination of more concordance lines than can be shown here demonstrates that both are frequent. For example, a corpus of issues of the *Sun* and *News of the World* newspapers contains almost 27 instances of *recovering* per million words, of which over three instances per million words are *recovering in* followed by *hospital*, or the name of a specific hospital or clinic or a similar noun phrase such as *intensive care*. The exact phrase *recovering in hospital* occurs over 100 times in this 44 million word corpus. Yet, from the point of view of pattern, this is claimed to be irrelevant. A sharp distinction, therefore, is made between those prepositional phrases beginning with *from* and those beginning with *in*. Although a case for this can be made on frequency grounds (*RECOVER from* occurs about eight times as frequently as *RECOVER in* does), it is not a robust one. Rather, the argument is that whereas prepositional phrases beginning with *in* and indicating place are found following many verbs, those beginning with *from* and indicating a particular semantic role are found following only some verbs. To use Willis's (2003) terminology, *from* identifies a class of verbs whereas *in* does not. To use Fillmore's (http://framenet.icsi.berkeley.edu) terminology, *from* introduces an element which plays a core semantic role in the 'recovery' frame whereas *in* does not. Such distinctions are problematic for the automatic identification of patterns.

One observation made earlier about the formal co-text is that certain positions in it are consistently filled by specific semantic roles. For example, the agent of *RECOVER* is usually an intelligent being, though it is sometimes an abstraction from human activity (e.g. *the economy*, *the Left*) or a metonym (*the lifeboat*). In Figure 7.1(c) the object of *RECOVER* indicates either a corpse or a sum of money or a natural resource (*oil*, *uranium*). In Figure 7.1(b), the noun phrase following *from* indicates an undesirable circumstance such as an illness. In most cases this categorisation is so obvious as to be banal: the examples in Figure 7.1(b) include *a poor start*, *a knee injury*, *bronchitis*, *disease* and *an operation*. It should be noted, however, that in some instances a situation can be so classified, not because of its out-of-context meaning but because of its position following *recover from*. Here are some examples from a wider search than the examples shown in Figure 7.1:

Example 7.1

Meanwhile, with only a few weeks to go to the release of the latest Star Wars in Britain, The Sun has discovered some film-goers who *are still recovering from previous parts of the series*.

Example 7.2

Environmentalists say attempts to repair the airstrip will further disrupt birdlife which *is only just recovering from the initial building work*.

Example 7.3

When Mama *recovered from talking to Chrissie* she ran into Bob's room . . .

Here 'previous parts of the series', 'the initial building work' and 'talking to Chrissie' are not inherently negative, but are so in the context of '*RECOVER from*'. Here the pattern could be said to 'impose' meaning by the process of semantic reversal mentioned earlier, and that this is an example of benign circularity: *recover from* is followed by phrases indicating disruptive or unpleasant events, and those events are so classified because they follow *recover from*.

Attempts are now being made (cf. Hanks 2008) to write pattern descriptions which consistently capture semantic as well as formal information, but this project is in its early stages. The concept of 'local grammar', to be described in more detail later, also aims to map meaning elements on to pattern. A rather similar, and much more complete, project is the FrameNet project (www.framenet.icsi.berkeley.edu), which starts from specific semantic frames and progresses through the identification of words realising those frames to the specification of semantic roles within the sequences in which the core words frequently appear. A key advantage of this approach is that words that happen to belong to different classes are treated the same (e.g. *recover* and *recovery* belong to the same frame and have the same core semantic roles). The FrameNet project identifies a frame 'recovery', with the core semantic roles 'patient', 'affliction' and 'body part'. Examples such as *Pat is recovering from scarlet fever* illustrate the frame verb *recover* along with the patient *Pat* and the affliction *scarlet fever*. Verbs in this frame are: *convalesce*, *heal*, *perk up*, *recover* and *recuperate*. Nouns are: *convalescence*, *recovery* and *recuperation*. Some of these verbs share grammar patterns with *RECOVER* (and some of the nouns share patterns with *recovery*), others do not. Verbs such as *suffer* are treated quite differently as belonging to the 'Catastrophe' frame with semantic roles 'undergoer' and 'undesirable event' (rather than 'patient' and 'affliction'). Fillmore (2008) argues forcefully that such an approach is more informative than a valency approach alone; and although he does not mention grammar patterns as such, his argument would apply to those too.

In other words, what we have here are two approaches, each of which, in the end, identifies a set of verbs (among other words), a set of patterned co-texts (e.g. prepositional phrase with *from*) identified from a corpus, a general semantic congruence between the verbs and the semantic roles that go with this, mapped on to the pattern. What is interesting is that starting from either end of the process, that is, from the pattern or from the semantic frame,

leads to results that overlap to some extent but that are far from identical. *RECOVER* and *HEAL* are grouped together by FrameNet but treated quite differently in pattern terms. *RECOVER from* and *SUFFER from* are treated as similar by pattern grammar and as quite different by FrameNet. The difference is that whereas FrameNet assumes word-meaning that is independent of pattern (e.g. *suffer from* is a variant of *suffer*), pattern grammar assumes that meaning belongs more to pattern than to word (e.g. *suffer from* is more similar to *recover from* than it is to *suffer*). What pattern grammar assumes most of all, though, (and FrameNet does not) is the mutability of semantic categories. The placing together of *recover* with *recuperate* and *bounce back*, let alone with *suffer*, *reel* and *die* can only be an ad hoc suggestion based on an intuitively perceived congruence, and that perception is, of course, in turn based on the formal similarity of *recover from*, *bounce back from*, *reel from* and *suffer from*.

7.3 PATTERNS AND EVALUATIVE LANGUAGE

As noted earlier, the original concept of grammar patterns was not particularly connected with evaluation, but the association of pattern and meaning inevitably raises questions as to whether patterns can assist in identifying and characterising evaluative language. To put the argument very simply: a lot of words in English (verbs, nouns and adjectives) do not occur with anything other than the most basic patterns. Here 'basic' might be defined as 'that which defines the word class'. A noun is that which follows a determiner or an adjective. An adjective occurs before a noun or following a copular verb. A verb occurs without complementation or is followed by a noun phrase. In CCED, this behaviour is noted regularly only in the case of verbs (which are coded 'V' or 'V n'); for the other word classes it is assumed that the word class label carries this pattern information and the pattern is noted only when the frequency is unusual (e.g. *the north* is coded '*the* N' and *electric* as in *electric fire* is coded 'ADJ n', but *friend* is only coded 'Noun' and *large* is only coded 'Adjective'). Very many words, however, occur sometimes, often or usually with more extensive patterning. In a large proportion of cases, this more extensive patterning is associated with some kind of evaluative meaning.

For this argument to appear reasonable it is necessary to consider that most complementation patterns involve either clauses (that-clauses, to-infinitive clauses, wh-clauses) or prepositions. That-clauses, to-infinitive clauses and wh-clauses entail a packaging of information ('realis' or 'irrealis') that makes that information available for subjective comment (Hyland and Tse 2005). I have argued elsewhere (Hunston, 2008a) that prepositions serve to classify information; it might be added here that some of that classification will interact with evaluative meaning. This is fairly apparent in prepositions such as *against*, which construes

opposition, or *about*, which construes a topic about which comment may be made, but the most productive preposition in this sense is probably *as*, which construes subjective classification. For this reason, patterns involving dependent clauses and prepositions are all candidates for a contribution to evaluation.

The importance of pattern to evaluative meaning is illustrated most clearly in the case of adjectives. Adjectives are in any case the word class most associated with evaluation or, to put it more properly, evaluative meaning is most canonically articulated using what Halliday and Matthiessen (2004: 219) call 'intensive' clauses (e.g. *Sara is clever*) where an Attribute is assigned to a Carrier. Again, most canonically, the word realising the Attribute is modifiable with an intensifier such as *very* (*Sara is very clever*) and may modify another word (*Sara is a clever girl*); in short, it is what is commonly termed a gradable adjective. It is, of course, possible for adjectives, especially classifiers, to construe non-evaluative meaning (*electric fire*, *Catholic church*, *a tall building*), but the patterns that complement adjectives almost always co-occur with evaluative meaning of some kind. Figure 7.2 shows examples (all from Francis et al. 1998):

Figure 7.2(a) Adjectives indicating emotion.

- *I'm not surprised the staff support you* (ADJ that)
- *I was uncertain what his next move should be* (ADJ wh)
- *It was exciting watching her grow in confidence* (*it* v-link ADJ–ing)
- *Ann's friends were less enthusiastic about her plans* (ADJ *about* n)
- *I . . . was worried as to how my death would affect them* (ADJ *as to* wh)
- *I stood utterly astonished by what I'd done* (ADJ *by* n)
- *He was utterly absorbed in his private game* (ADJ *in* n)
- *I was very fond of her* (ADJ *of* n)
- *Canada is seriously worried over the level of spending . . .* (ADJ *over* n)
- *I've always felt very affectionate towards Karen . . .* (ADJ *towards* n)
- *I thought I was angry with them* (ADJ *with* n)

Figure 7.2(b) Adjectives indicating human qualities.

- *He appears powerless against the corrupt politicians* (ADJ *against* n)
- *We left for New York feeling . . . inadequate as parents* (ADJ *as* n)
- *Her mother was clever at many things* (ADJ *at* n)
- *I fear I'm not cut out for detection* (ADJ *for* n)
- *. . . he was absent from work for 35 days* (ADJ *from* n)

Figure 7.2(c) Adjectives indicating qualities of things.

- *Children's homes are expensive to run and difficult to staff* (ADJ to-inf)
- *. . . blues and greens are easy on the eye* (ADJ *on* n)
- *. . . insects which are beneficial to birds* (ADJ *to* n)

Figure 7.2(d) Adjectives indicating attitudes.

- *It was fortunate for George that the cinema manager could vouch for him* (*it* v-link ADJ *for* n that)
- *It is understandable why they hate the sight of him* (*it v*-link ADJ wh)
- *It is quite possible to alter or extend a house* (*it* v-link ADJ to-inf)
- *It would be ludicrous for him to travel economy . . .* (*it* v-link AJD *for* n to-inf)
- *. . . the trainer thought it best that I should rest the knee* (v *it* ADJ that)
- *. . . I don't feel it appropriate to start now* (v *it* ADJ to-inf)
- *It is possible some dates may change . . .* (*it* v-link ADJ that)
- *It was obvious to me that Mary needed a friend* (*it* v-link ADJ *to* n that)

The examples in Figure 7.2(a) indicate an emotion or reaction, the object or target of which is construed in the clause or prepositional phrase following the adjective. In 7.2(b), the adjective expresses a quality possessed by a person, with the prepositional phrase defining the scope of that quality. The examples in 7.2(c) are similar except that the quality belongs to a non-human entity. In each of the examples in 7.2(d), the clause following the adjective construes an event or situation to which the adjective expresses an attitude; in those examples where *it* is used, that attitude is construed as impersonal—a community rather than an individual reaction—except where a prepositional phrase (*to me*) reassigns the evaluation as personal.

In each case, then, the pattern elements (dependent clause or prepositional phrase) express either the object of the evaluation or the extent of its scope. In some cases there is both a prepositional phrase and a clause (*it was fortunate for George that . . .*; *it would be ludicrous for him to . . .*; *It was obvious to me that . . .*). In each case the prepositional phrase plays a slightly different role: the silent beneficiary of the positively appraised situation (silent in the sense that 'George's' own opinions are not known); the logical subject of the non-finite clause ('he travels economy and that is ludicrous'); the source of the evaluation ('I find it obvious that').

7.4 PATTERNS AND APPRAISAL

The grouping of the earlier examples is highly reminiscent of the three-part division of resources for evaluative meaning proposed by Martin and White (2005) as discussed in Chapter 2. Patterns appear to express 'what someone thinks about something' (Affect) or 'how good or bad an entity is' (Appreciation) or 'how good or bad what someone does is' (Judgement). This raises the prospect of using pattern to identify types of evaluation. It should be said at the outset that of course this will not work, because of the allusive quality of much evaluation; no 'test' can identify more than the most explicit or inscribed evaluation. However, we should note that Martin (2003: 173) draws on a concept of grammar pattern (which he

refers to as 'grammatical frame') to provide a simple way of distinguishing between Affect, Judgement and Appreciation. He argues for canonical expressions of each part of the Appraisal system thus (examples from Martin 2003: 173):

- Affect occurs canonically in the frame *I feel (very) x*, as in *I feel very happy/sad*.
- Judgement occurs canonically in the frame *it was x of him/her to do that*, as in *It was kind/cruel of him to do that*.
- Appreciation occurs canonically in the frame *I consider it x*, as in *I consider it innovative/unimaginative*.

This suggests that pattern, or frame, may be a useful heuristic in distinguishing between types of evaluative meaning, at least in respect of adjectives. Affect is associated with feeling, so adjectives construing Affect collocate with *FEEL*. The object or target of the emotion is not necessarily expressed, so the canonical pattern or frame is a simple Intensifier clause. Appreciation, on the other hand, is associated with a more cerebral response, and with an object or target, so adjectives construing Appreciation collocate with *CONSIDER* and modify a noun (or the pronoun *it* in Martin's example). Whereas Appreciation is associated with the evaluation of entities (construed by a noun phrase) rather than actions, Judgement is the reverse; thus, adjectives construing Judgement co-occur with to-infinitives while those construing Appreciation do not.

The frames suggested by Martin raise the possibility of using a preposition test to distinguish between adjectives expressing Appreciation and those expressing Judgement. If we consider two similar patterns, ***it* v-link ADJ *of* n to-inf** and ***it* v-link ADJ *for* n to-inf**, we might interpret Martin to mean that only instances expressed as or paraphrasable by the first of these would be genuine Judgement. Other instances paraphrasable with *for* would be Appreciation. In some cases, the distinction seems to work. For example, *brave* occurs with the *of* pattern but not with the *for* pattern (e.g. *It's very brave of her to give up after all these years* but not 'It is very brave *for* her to . . .'). *Difficult*, on the other hand, occurs with *for* but not *of* (*It's very difficult for people to have the time and energy to do that* but not 'It's very difficult *of* people to have the time . . .'). This suggests that an assessment of bravery is always Judgement whereas an assessment of difficulty is Appreciation, paraphrasable using the Appreciation frame: 'I consider having the time to do that difficult'. Not all instances are so amenable, however.

According to Francis et al. (1998: 501–502), three kinds of adjectives occur in the pattern ***it* v-link ADJ *of* n to-inf**: those evaluating an action positively (e.g. *brave*, *clever*, *decent*, *generous*, *kind*, *nice*, *right*, *sensible*, *wise*); those evaluating an action negatively (e.g. *absurd*, *cruel*, *foolish*, *gutless*, *improper*, *negligent*, *rude*, *selfish*, *silly*, *unfair*, *wrong*); and a smaller

number indicating that an action is typical or atypical behaviour for someone (*characteristic*, *typical*, *uncharacteristic*). Many more adjectives are listed under the pattern ***it* v-link ADJ *for* n to-inf**. This includes some groups that have no equivalent in the *of* pattern, including examples such as *it is difficult/easy/safe for him to . . .*; *it is important/essential/unnecessary for you to . . .*; *it is impossible/possible for her to . . .*; *it is enough/sufficient for us to. . . .* Other groups, and many individual adjectives, are found in both patterns e.g. *nice*, *right*, *sensible*, *wise*, *absurd*, *foolish*, *unfair*, *wrong*, *typical*. This suggests some similarities but also subtle differences. Table 7.3 shows some of the adjectives noted by Francis et al. (1998) as occurring in each pattern. Adjectives that occur in both lists are in bold.

Table 7.3 Adjectives in Two Patterns

Of pattern	*For* pattern
clever	*logical*
prudent	*rational*
responsible	*reasonable*
sensible	***sensible***
smart	***wise***
wise	
brave	
courageous	
big	*excellent*
decent	***fair***
fair	*fine*
generous	***good***
good	*great*
great	*helpful*
kind	***lovely***
lovely	***nice***
nice	*okay*
right	***right***
sweet	*tremendous*
thoughtful	*useful*

(continued)

Table 7.3 (continued)

Of pattern	*For* pattern
	gratifying
	interesting
	satisfying
	acceptable
	appropriate
	legal
	legitimate

There are a number of conclusions that might be drawn here. The first is that the *of* pattern is more straightforward, in that it is always the speaker who evaluates the actions of a person construed by the noun phrase following *of*. The *for* pattern is less straightforward. The person indicated by the noun phrase following *for* is sometimes construed as the evaluator of an action for which they are not directly responsible. This is certainly true of the group of adjectives (*awkward*, *dangerous*, *difficult*, *easy*, *hard*, *risky*, *safe*, *simple* and *tough*) which occur in the *for* pattern but not in the *of* pattern. An example such as *it was difficult for him to act* may be paraphrased as 'He acted. He considered his action difficult' and also 'He acted. I considered his action difficult'. Other examples include *It must have been gratifying for you to discover . . .*; *it'll be nice for you to have someone to play with*. In each case it is the speaker who imposes the evaluation; this is quite different from, for example, 'She considered having someone to play with nice'. In other cases, the noun phrase following *for* does construe the one with responsibility for the action, but the evaluator is the speaker alone (e.g. *it's okay for managers to throw cups around . . .* or 'I consider managers throwing cups okay'). Sometimes more context is needed to identify who the construed evaluator is: in *Sometimes it is helpful for a child to keep a record of anything to do with the tics* it is not clear without more context whether it is the child who is helped by this activity or another person (a medical professional, for example). A further feature to note is that the activity construed is typically 'irrealis': it is a potential activity rather than an actual one, whereas the *of* pattern typically construes a 'realis' or actual activity. Finally, it must be noted that the *of* pattern cannot be used with 'social' Judgement, only with 'moral' Judgement. Judgements of legality or appropriateness, as the final group in Table 7.3 indicates, are made using the *for* pattern. This suggests that both construe Judgement but of different kinds.

Some pairs of examples show a clear difference between the patterns, such as Examples 7.4 and 7.5.

Example 7.4

It was lovely of her to send the card (She sent the card; I judge the action to be lovely)

Example 7.5

It will be lovely for her to be part of it (She will be part of it; I judge that she will find the experience lovely)

Others are less clear-cut, as in Examples 7.6 and 7.7.

Example 7.6

It was wise of you to come, and alone (You came alone; I judge the action to be wise)

Example 7.7

It would be wise for you to refuse (You will refuse; I judge the action to be wise (and that you will benefit from it?))

The upshot of this discussion is that Martin was right to propose the *of* pattern as a diagnostic, except that what will be highlighted will be Judgement on moral rather than on social grounds. It is not entirely clear whether the *for* pattern is used for Judgement or Appreciation. The *for* pattern is more likely to be used for irrealis actions and in circumstances where the noun phrase following the preposition construes someone affected by, as well as doing, the action. The study also, incidentally, supports Teubert's (2008) criticism that some of the 'meaning groups' in Francis et al. (1998) are less finely delineated than they could be. The second group in the ***it* v-link ADJ *for* n to-inf** pattern, for example, is a mixture of items expressing different kinds of Judgement (*acceptable*, *legal* as well as *lovely*, *nice*) and some reported Affect (*gratifying*, *interesting*).

Most clearly, however, this study suggests that it is the pattern as much as the adjective itself that construes the evaluative meaning. As a result, where an adjective is used non-canonically the frame (or pattern) may alter the classification. This is true of Martin's other examples, too. For instance, Martin notes that *unimaginative* construes Appreciation rather than Judgement because it is more likely to co-occur with a noun phrase, and so appraise an entity, than with a to-infinitive clause, thus

appraising an action. Some typical examples from the BoE (Examples 7.8 and 7.9) support this:

Example 7.8

Haig . . . therefore proves himself to have been *unimaginative*, indifferent, obstinate and incompetent . . .

Example 7.9

The Warana parade was poorly organised, lacked any cohesion and was accompanied by boring repetitive music and *unimaginative* choreography.

Much less frequently, the adjective occurs in the frame or pattern noted by Martin:

Example 7.10

Most observers considered Ford *unimaginative*, certainly not brilliant.

When *unimaginative* is used to appraise an action, that action is usually nominalised:

Example 7.11

. . . once they had made the desperately *unimaginative* selection of Tim Wren . . . they had little option.

It is worth looking, however, at what happens when the word is used non-canonically.

In Example 7.12 does *unimaginative* construe Appreciation (of us as critics) or Judgement (of our action in saying something untrue)?

Example 7.12

Wordsworth is not saying, and *it is sentimental and unimaginative of us to say*, that he [Wordsworth] has become less a feeling man and less a poet. (Leader 1999: 30)

Similarly, in Example 7.13, is Mr. Lilley the object of Appreciation, or is this action of 'falling back on this old chestnut' the object of Judgement?

Example 7.13

It was in a way somewhat *unimaginative* of Mr Lilley to fall back on this old chestnut. (James Fenton, 'Do we want naked policemen on the beat?' The Independent 14.6.1993)

In a question that reminds us of the earlier discussion of pattern and meaning we might ask: is it the adjective *unimaginative*, construing Appreciation, or the pattern '*it is x of you to do that*', construing Judgement, that carries most weight? Without attempting to answer this conundrum, because the whole point of it is that it is unanswerable, I would argue that this apparent blurring of Martin's distinction provides a good test of it. The mismatch of pattern and adjective (*unimaginative* appears very rarely in the pattern 'it is adjective of person to do this') is congruent with the mismatch between what appears to be being evaluated in the examples and what is actually being evaluated. In Example 7.12, it is the utterance 'he has become less a feeling man and less a poet' that is evaluated as 'obvious but untrue' via the evaluation of the saying of it as the product of an unimaginative critic. In Example 7.13, what is evaluated as 'unimaginative' is 'Mr. Lilley—but only in the context of his current pronouncements'.

In more general terms, Martin's use of grammar frames suggests the possibility that other patterns might typically co-occur with one type of evaluative meaning than another. Patterns with that-clauses or wh-clauses, for example, tend to construe Affect, either expressed by the speaker/writer or reported by them. Example adjectives are: *afraid*, *angry*, *glad*, *horrified* and *surprised*. Similarly, adjectives followed by *about* typically construe Affect. Examples include: *angry*, *apologetic*, *cool*, *cynical*, *happy*, *passionate*, *philosophical*, *nervous*, *serious* and *unhappy*. As will be discussed in the following, some adjectives with *about* construe an appraisal of the reported Affect. Examples include: *analytical*, *arrogant*, *articulate*, *charitable*, *entertaining*, *fair*, *honest*, *reticent*, *right*, *scathing*. A small number construe evaluation of what someone has said, their state of mind or their actions e.g. *aware*, *careful*, *ignorant*, *marvellous*, *ruthless*, *specific*, *selfish*, *vague*. Of the adjectives followed by *in*, however, only those such as *confident* and *interested* express Affect. Many more express either Appreciation or Judgement, such as *awash*, *careful*, *fluent*, *helpful*, *important*, *lacking*, *lucky*, *right*, *slow*, *successful*. In fact, most patterns have exemplars within and without the Affect type, as Table 7.4 suggests (adjectives listed in the table are representative of groups cited in Francis et al. [1998] and appear in the order of the groups in that publication). A few, however, such as *against*, *as* and *towards*, favour one or other type of evaluation.

Table 7.4 Adjectives in Appraisal Types

Pattern	*Affect*	*Judgement/Appreciation*
ADJ *about* n	passionate, cool, happy, unhappy, philosophical, nervous, angry, cynical, serious, apologetic	arrogant, specific, vague, scathing, charitable, articulate, reticent, honest, entertaining, marvellous ruthless, careful, fair, selfish, analytical, aware, ignorant, right
ADJ *against* n		effective, defenceless, safe, successful
ADJ *as* n		marvellous, important
ADJ *at* n	nervous, angry	good
ADJ *by* n	fine, astonished	hidebound, available
ADJ *for* n	eager, afraid, happy, guilty	suitable, ready, vital, useful, good
ADJ *from* n		safe
ADJ *in* n	confident, interested	lacking, awash, fluent, slow, helpful, important, careful, successful, right, lucky
ADJ *of* n	fond, critical, afraid, tired, desirous	considerate, careless, empty, full, destructive, deserving, kind
ADJ *on* n	keen, optimistic	fair, rough, weak, strong
ADJ *over* n	angry, despondent, worried, enthusiastic, jealous	
ADJ *to* n	partial	beneficial, detrimental, generous, fascinating, infuriating, important
ADJ *towards* n	sympathetic, aggressive, ambivalent	
ADJ *with* n	angry, patient, breathless	lavish, excellent, okay, lucky

Attempts to support finer distinctions through pattern differentiation are less successful. For example, the pattern 'adjective + that-clause' occurs with most but not quite all of the sub-divisions of Affective meaning suggested by Martin and White (2005):

- FEAR: *afraid, anxious, apprehensive, fearful, nervous, pessimistic, petrified, scared, suspicious, terrified, wary, worried*
- DESIRE: *desperate, determined, eager, hopeful, keen, optimistic*

- MISERY: *heartbroken, sad, sorry, unhappy, bitter*
- ANTIPATHY: *critical*
- CHEER: *happy*
- AFFECTION: NONE
- DISQUIET: *concerned, disturbed, impatient, perturbed, worried, puzzled*
- SURPRISE: *amazed, astonished, astounded, bemused, incredulous, shocked, surprised*
- CONFIDENCE: *confident*
- TRUST: NONE
- ENNUI: NONE
- DISPLEASURE: *aghast, disappointed, disgusted, dismayed, distraught, frustrated, horrified, regretful, angry, annoyed, cross, furious, incensed, indignant, irate, irritated, livid, mad, outraged, resentful*
- INTEREST: *interested, fascinated*
- PLEASURE: *awestruck, chuffed, content, ecstatic, encouraged, flattered, glad, feel good, grateful, gratified, heartened, jubilant, pleased, proud, thankful, thrilled*

This list supports the fairly obvious observation that where Affect can be expressed towards a proposition (as in *I secretly felt quite pleased that they had missed me so much*) it can be realised by an 'adjective + that-clause' pattern, whereas in the few cases where Affect is directed only towards a person or thing (as is the case with 'Affection' or 'Trust'), no examples of the pattern are found.

Conversely, if a particular sub-division of Affect, such as Affection, is considered, the realisations are not restricted to any particular form: examples include *be passionate about*, *be fond of*, *be keen on*, *be enthusiastic over*, *be partial to* and *be sympathetic towards*.

It seems, then, that adjective patterns in general can be of some use in distinguishing between types of evaluative meaning, most decisively identifying Affect. That use is, however, fairly limited and is not helpful in making finer distinctions. What a 'pattern grammar' is useful for, however, is providing evidence for the wide range of adjective-in-pattern exemplars that are available for consideration. Lists of such patterns and adjectives based on corpus investigation are, arguably, a useful supplement to and confirmation of intuition. If the priorities are reversed, however, and patterns themselves are taken as a starting point, other distinctions emerge. These will be considered in the next section.

7.5 PATTERNS AND EVALUATION

As noted earlier, patterns make some contribution towards distinguishing between the types of evaluative meaning identified by the Appraisal model.

Generalisations from patterns, however, may make other distinctions. In other words, instead of asking 'what kinds of evaluative meaning are there and do patterns contribute to these distinctions?', we might ask 'what, in relation to evaluative meaning, do patterns do?' Or, if we take patterns as a heuristic to investigate evaluative language, what are the phenomena that emerge? I suggest that from a study of each of the meaning groups in each pattern set two distinctions can be observed.

Firstly, there is a distinction between patterns that perform the function of evaluating and those that report evaluation. The 'performative' patterns are primarily the '*it*' patterns and the '*there*' patterns. In each case, unless there is an explicit attribution ('Smith says that' or 'According to Smith' or 'Smith finds it adjective that' or 'it is adjective to Smith that'), it is the writer/speaker who is the source of the evaluation. Here are two examples:

***it* v-link ADJ that**

- Assesses likelihood: *arguable, axiomatic, believable, certain, conceivable, correct, definite, demonstrable etc.*
- Assesses obviousness: *apparent, clear, evident, manifest, obvious, plain, transparent*
- Judges the proposition as good or bad, including assessments of likely affective response: *acceptable, admirable, advisable, absurd, annoying, appalling, awful, awkward, bad etc.*
- Assesses the importance of the proposition: *compulsory, critical, crucial, essential etc.*
- Assesses the ordinariness or otherwise of the proposition, including assessments of likely affective response: *amazing, anomalous, astonishing, astounding, baffling etc.*
- Assesses the relevance of the proposition: *incidental, irrelevant, pertinent, relevant, significant*

***there* v-link *something* ADJ *about* n/-ing**

- Judges or appraises the entity: *admirable, attractive, beautiful, cosy, elegant etc.; absurd, artificial, awful, bad, bogus etc.; abstract, concrete, fanciful etc.*
- Assesses likely affective reaction to the entity: *alluring, appealing, endearing, exciting, fascinating etc.; boring, depressing, disconcerting etc.; addictive, compelling, compulsive, hypnotic etc.; amusing, comic, funny etc.*
- Assesses the degree of normality of the entity: *abnormal, average, common, curious, different etc.*
- Picks out a typical characteristic and makes this evaluative: *Chaplinesque, Dickensian, English, exotic etc.; adult, boyish, childish etc.; military, political, scientific, theatrical etc.*

- Indicates degree of openness: *anonymous, covert, secret etc.*
- Indicates degree of fixedness: *controversial, equivocal, final etc.*

Other patterns may perform evaluation, but only if the source of the evaluation is explicitly marked as the speaker (as in *I find it strange that a lower court can turn around a decision of a higher court*); otherwise, the evaluation is reported rather than performed (as in *People find it strange that people on a high income end up getting the same child benefit as everyone else*). The distinction between 'performed' and 'reported' is important for the implementation of local grammar (see the following section).

The second distinction is less clear; it relates to the fact that whereas evaluation in some instances appears to be straightforward, in other cases it is multilayered. As illustration, consider Figure 7.3, which shows examples of the **ADJ *about* n** pattern:

Figure 7.3 Examples of **ADJ *about* n**.

1. *Everybody is happy about Cindy and Jerry finally getting married.*
2. *People were anxious about the future.*
3. *Most people are curious about a murderer.*

4. *They're fanatical about what they eat.*
5. *He would sit in his chair . . . getting grumpy about the temperature of the milk on his cornflakes.*
6. *Americans have become complacent about immunisation.*

The first three examples in Figure 7.3 report Affect, attributing a reaction to events to the people construed by the subject of the clause. In examples 4–6, however, a reaction is attributed but it is also judged as excessive or inappropriate (*fanatical*—'too concerned'; *grumpy*—'unreasonably displeased'; *complacent*—'too relaxed'). In these examples there are two layers of evaluation. In example 6, for instance, 'Americans' are the source of Affect in response to immunisation, but the writer/speaker is the source of Judgement regarding the Americans' attitude.

In the 'ADJ *about* n' examples, the multilayering is dependent on the choice of adjective itself rather than on the pattern. For some patterns, however, multilayering is built in. Four patterns of this kind are: **V n *as* n; V n *as* adj; V n n; V n adj.** Examples of these patterns are:

Example 7.14

[The history of Irish music] sought to **represent the story as a steady progression from the dark ages of the 1960s towards the bright lights of the boy-band era.**

Example 7.15

Colin was amazed that most people **described him as aloof**.

Example 7.16

I have always **considered us a happy family**.

Example 7.17

. . . firing those who **thought him insane** and replacing them with those who **pronounced him sane**.

These may be parsed as shown in Table 7.5.

Table 7.5 Parsing the **V n *as* adj, V n n** and **V n adj** Patterns

	verb	*noun*	*as*	*noun*
The history of Irish music	sought to represent	the story	as	a steady progression from the dark ages of the 1960s towards the bright lights of the boy-band era.

	verb	*noun*	*as*	*adjective*
most people	described	him	as	aloof.

	verb	*noun*	*noun*
I	have considered	us	a happy family.

	verb	*noun*	*adjective*
[they]	thought	him	insane
[they]	pronounced	him	sane

In each of these examples an evaluation of an entity is attributed to someone. The evaluations are: 'the story of Irish music is a steady progression . . .'; 'Colin is aloof'; 'we are a happy family'; 'he is insane/sane'. Where attributions using these patterns differ from those using other patterns such as **V that** ('most people said that he was aloof', 'I think that we are a happy family' and so on) is that with this set of patterns (**V n *as* n** etc.) the mediation of the attributor is made apparent. Whereas in 'most people said that he was aloof' there is an apparent 'original' utterance—'he is aloof' (though this is only apparent, of course)—in *most people described him as aloof* such an utterance is less recoverable. It is more obvious that the opinions being attributed could have been expressed in very different terms or not expressed at all. In the first example, it is far from clear that the TV programme makers would recognise their own narrative as charting *a steady progression from the dark ages . . . towards the bright lights. . . .* The 'steady progression from darkness to light' might be read as a pastiche of the programme's message and therefore as an implicit comment on the quality of the programme itself. This pattern allows or even encourages the expression of attribution as evaluative interpretation rather than as report.

Moreover, many of the verbs that occur with these patterns themselves indicate a degree of subjective interpretation of the attribution. For example, verbs used with *as* and either a noun group or an adjective group include: *attack*, *brand*, *condemn*, *decry*, *denounce*, *dismiss*, *misrepresent*, *scorn*, *stereotype*, *pass off* and *write off* among others. Verbs used with the **V n n** pattern include *brand* and *label* and others, while those with **V n adj** include *believe*, *brand*, *judge* and *label* among others.

The distinctions drawn in this section, between performed and reported evaluation, and between single and multilayered evaluation, are important for the establishment of a local grammar or evaluation.

7.6 AN INTRODUCTION TO LOCAL GRAMMAR AND EVALUATION (AND A RETURN TO FRAMENET)

The chapter 'A Local Grammar of Evaluation' (Hunston and Sinclair 2000) was an initial attempt to bring together Sinclair's concept of a truly functional grammar and the concept of evaluation. By 'functional grammar', Sinclair meant a grammar that would label each element of an analysed unit in terms that related directly to its discourse function. The clearest example of this is the local grammar of dictionary definitions developed by Barnbrook (2002), in which each element of the definition is identified in terms that give a precise indication of how it contributes to the function of definition. The advantages of definitions as a test bed for the concept are that they are unambiguously identifiable (they occur in particular slots in a dictionary database) and their function as definitions (or explanations of word-meaning) is equally unambiguous. Table 7.6 is an example of a definition as parsed by Barnbrook (2002: 172).

Table 7.6 Parsing Definitions

Hinge	*Left-hand side*			*Right-hand side*		
	Co-text	Definien-dum	Co-text	Matching element	Definiens	Matching element
If	someone or some-thing	is geared	to a particular purpose	they	are organised or designed to be suitable	for it

Applying the concept of local grammar to evaluation is less straightforward. Leaving aside for a moment the (very large) issue of implicit evaluation, it is still not easy to identify clauses whose sole function is to evaluate. It is a premise of local grammars, as defined by Barnbrook (2002) and by Sinclair (Hunston and Sinclair 2000), that they parse the function of a sentence. That is, they treat the sentence as performing an action. This is wholly appropriate in the case of dictionary definitions; such definitions perform the function of defining. One of the intriguing aspects of grammar patterns, however, is that for the most part they are used to report a function rather than to perform it (see Hunston and Francis 1999: 121–122). The previous section illustrated the fact that while a few patterns with adjectives might be said to perform evaluation, most report evaluation and some do so in a multilayered way. Furthermore, as is obvious, no ready-made corpus of evaluative sentences exists (although there are data sets of texts that are likely to be heavily evaluative, as described in the Sentiment Analysis literature).

In Hunston and Sinclair (2000) no attempt is made to identify a corpus of evaluative sentences (in the way that Barnbrook uses a corpus of definitions). Instead, some patterns are taken as a starting point, and also single words are selected which are known to be evaluative (*nuisance* and *difficult*). Concordance lines for each word are obtained from the BoE corpus, and the various patterns of occurrence for each word are parsed using terminology such as 'Evaluated Entity' and 'Evaluator'. The patterns of occurrence include both those associated with the pattern grammar of Francis et al. (e.g. *It is difficult to imagine*) and those that are simply associated with the word class (e.g. *It was a difficult task*). The aim was to make advances towards establishing a local grammar (i.e. one that identified all the elements needed to parse instances of evaluation) that could be used to parse automatically a corpus of evaluative clauses. It did not go very far towards achieving that aim, at least partly because it failed to differentiate between performed and reported evaluation and between evaluation-as-emotive reaction (Affect) and evaluation- as-social judgement (Judgement or Appreciation).

Before continuing with the notion of local grammar, I shall return to the approach of FrameNet. As mentioned earlier, FrameNet takes as its starting point the notion of 'semantic frame'. Words which belong to each frame are studied in a corpus, typical uses are extracted—these may be

grammar patterns (termed 'valency patterns') or frequent collocations—and the semantic frame elements are mapped on to parsed examples. A great deal, then, depends on the initial identification of semantic frame. As noted earlier, in the discussion of *RECOVER* and *SUFFER*, this is the point at which an approach that prioritises pattern and one that locates meaning within individual words might most differ.

I shall here take an example, using the word *difficult*, which Hunston and Sinclair treated as expressing evaluation and which according to Martin's (2003) test most probably construes Appreciation. FrameNet allocates the word to the 'difficulty' frame. This frame identifies a scale of ease and difficulty, not just 'difficulty'. Other words in the frame include *a cinch*, *a doddle*, *a pain*, *child's play*, *challenging*, *tough*, *easy* and *trivial*. There is, therefore, a considerable difference in the degree of generalisation being employed here, with Hunston and Sinclair treating the concept of 'difficulty' as an instance of evaluation and FrameNet treating it as a distinct category. FrameNet identifies two core elements for this frame: 'activity' and 'experiencer'. The analysis identifies a number of valency patterns, which are very like grammar patterns except that some valency patterns conflate two grammar patterns. Leaving this difference aside, the mapping of semantic element on to pattern element is quite straightforward and is identical in the two approaches. The valency patterns are:

- vping (in pattern grammar terms ***it* v-link ADJ–ing**), as in *it was difficult getting motivated on a night like this*, where the -ing clause is the 'activity'.
- vpto (in pattern grammar terms this includes both ***it* v-link ADJ to-inf** and **ADJ to-inf**). Examples are *It was difficult to settle on the right key* and *this second distinction will be much more difficult to make*. The activity is either the to-infinitive clause alone (in the *it* pattern examples) or the subject of the clause and the to-infinitive (e.g. *this second distinction . . . to make*) in the others.
- ppfor (in pattern grammar terms both ***it* v-link ADJ *for* n to-inf** and **ADJ *for* n to-inf**). This includes the experiencer element in the prepositional phrase, as in *It is so difficult for some people to accept friendship* or *Max was difficult for her to deal with*.

Some of the other annotations note degree, identifying *quite*, *extremely*, *very* and *so . . . that*, and parameter, noting the adverb *technically*. Others note verbs which tend to govern the clause in which the difficulty is expressed, namely *find* and *prove*. More patterns are thereby added: **v n ADJ** (*find something difficult*), **v *it* ADJ to-inf** (*find it difficult to do something*), **v n ADJ to-inf** (*find something difficult to do*), **v-link ADJ** (*something proves difficult*), **v-link ADJ to-inf** (*something proves difficult to do*), **ADJ n** (*proves a difficult thing*). The other collocates (*choice*, *issue*, *question*, *road*, *situation* and *market*) add only one other pattern, **ADJ n to-inf** (*a difficult question to answer*). That is, 12 patterns.

The word *difficult* is also analysed by Hunston and Francis (1999: 130–136). They identify 21 'behaviours', of which some are associated with comparative and superlative or with pseudo-cleft with 'what' and 'how' or with 'as . . . as'. Omitting these there are 15 left. The difference in quantity between FrameNet and Hunston and Francis can be attributed to the fact that the latter include the identification of patterns with *make*.

Thus far, then, there is agreement as to how this word behaves and how the behaviour might be analysed. This continues. Where FrameNet identifies two core elements: 'Activity' and 'Experiencer', Hunston and Francis suggest 'Evaluated Entity' and 'Affected Entity'. Analyses of the patterns with *it* are identical except for nomenclature; two examples are shown in Tables 7.7 and 7.8.

Table 7.7 ***It*** **v-link ADJ–ing**

			Activity/Evaluated Entity
It	was	difficult	getting motivated

Table 7.8 ***It*** **v-link ADJ** ***for*** **n to-inf**

				Experiencer/ Affected Entity	*Activity/ Evaluated Entity*
It	is	difficult	for	some people	to accept friendship

Hunston and Francis, however, suggest that the Affected Entity is involved also in the patterns with *find*, as in Table 7.9.

Table 7.9 **V** ***it*** **ADJ to-inf** (Example from FrameNet)

F'net					*Activity*
H&F	*Affected Entity*				*Evaluated Entity*
	She	found	it	difficult	to wipe the thought from her mind.

There is more difference in the coding of to-infinitive clauses without *it*. FrameNet, quite sensibly, includes both the subject of the clause and the to-infinitive clause in the 'Activity' element, as in Table 7.10.

Table 7.10 **V** ***for*** **n to-inf**

Activity . . .			*Experiencer*	*. . . Activity*
Max	was	difficult	for her	to deal with

This makes the clause the equivalent of 'Dealing with Max was difficult for her'. Hunston and Francis opt for a solution that is less satisfactory, using another element label, 'Evaluation Limiter'; the equivalent coding for the example from FrameNet would be as in Table 7.11.

Table 7.11 **V *for* n to-inf**

Evaluated Entity			*Affected Entity*	*Evaluation Limiter*
Max	was	difficult	for her	to deal with

The rationale (that Max is not inherently difficult, only difficult to deal with) seems to be splitting hairs, and has justly been criticised (Warren, personal communication). However, other uses of 'Evaluation Limiter' appear more justified. One example from Hunston and Francis (1999: 134) is shown in Table 7.12.

Table 7.12 Using 'Evaluation Limiter'

Evaluated Entity			*Evaluation Limiter*
City governments in Poland	can be	difficult	business partners

The rationale here, that the 'city governments', when acting as business partners, are difficult (but in other respects are sweet reasonableness itself), seems more justified. The other advantage of this analysis is that the 'Evaluated Entity' is identified as the 'city governments'; these are, surely, the object of the evaluation. (Note that it would be difficult to find an 'activity' here.) A similar example from FrameNet, *International trade unionism was a difficult road to travel* identifies 'road' and 'to travel' as together comprising the Activity but leaves 'international trade unionism' unlabelled. Another use of Evaluation Limiter occurs in the Hunston and Francis example shown in Table 7.13.

Table 7.13 Using 'Evaluation Limiter'

Causer		*Evaluated Entity*		*Evaluation Limiter*
Modifiers	make	a sentence	difficult	to understand

(FrameNet does not include examples with *make* in its treatment of *difficult* and has no 'causer' equivalent.) A good argument could be made that this is a paraphrase of 'When modifiers are used, understanding a sentence is difficult', suggesting that 'understand a sentence' is the Activity or Evaluated Entity. Again, this would seem to be the more rational analysis. What the analysis shown here does, however, is to focus on 'a sentence' as the object of evaluation rather than the activity of understanding. This in turn suggests a consistency between this analysis and a categorisation in Martin and White's terms as 'Appreciation'.

FrameNet does, however, identify elements which Hunston and Francis do not account for. These are all either 'peripheral' or 'extra-thematic'. One is 'degree', which takes account of modifiers such as *extremely*, *so . . . that* or *as . . . as*. Another is 'parameter', which is to a large extent similar to 'evaluation limiter'; it relates either to adverbs such as *technically* and *economically* or to prepositional phrases with *in*. Examples (with the 'parameter' in italics) are:

Example 7.18

International trade unionism was a difficult road to travel, *both in principle and in practice*.

Example 7.19

The principle is . . . difficult to apply *in the many practical situations which arise*.

Finally, 'circumstance' suggests a cause of difficulty and occurs in a prepositional phrase with *with* or in a wh-clause; this element is in italics in Examples 7.20 and 7.21.

Example 7.20

Once the activities of an organisation have been divided up, they need to be coordinated and this becomes more difficult *with the complexity of the organisation's activities*.

Example 7.21

But identification is more difficult *when relatively few portraits in the round have survived*.

In spite of this agreement, there is considerable disparity between FrameNet and a local grammar approach because the starting point for a local grammar is the pattern itself. The example that will be taken here is the pattern **ADJ to-inf**. It is well known that this pattern is susceptible to contrasting analyses, depending on the choice of adjective, because of Chomsky's famous distinction between 'easy to please' and 'eager to please'. Hunston

and Francis (1998: 405) identify 16 main meaning groups, divided into two 'types'. In one type, the understood subject of the to-infinitive clause is different from the subject of the main clause, as in *Fish can be fiddly to cook* (ibid.: 405; cf. 'John is easy to please'). In the other type, the two subjects are the same, as in *We would be foolish to ignore them* (ibid.: 405; cf. 'John is eager to please'). In terms of evaluation, however, there are further distinctions. All examples following are from Francis et al. (1998: 404–411).

For the examples in the first five groups, the competing analyses discussed earlier are possible. That which is evaluated may be contained in the clause subject, with the to-infinitive clause indicating a limiting specification or context. Alternatively, it may be that the entity indicated in the subject and the action indicated in the to-infinitive clause together (as Verb and Object) make up that which is evaluated. In Table 7.14, both analyses are shown (in the first and last rows) with the examples placed closer to each analysis depending on which seems most reasonable. It might be noted that the first analysis brings the analysis closer to Appreciation (of a person or thing) and the final one brings it closer to Judgement (of an action).

Table 7.14 Parsing **v-link ADJ to-inf**

Entity		**Evaluation**	**Limiter/Specification**
Horses	are	pretty	to look at
. . . the place	is	fit	to live in
The printing	is	easy	to read
Such matches	are	boring	to watch
Children's homes	are	difficult	to staff
Action . . .		**Evaluation**	**. . . Action**

The next five groups express Affect; in each case an Evaluator (or Experiencer) is reported as feeling an emotion about or reaction to a situation. In the final two examples shown in Table 7.15, the reaction is one of willingness or unwillingness to perform an action.

Table 7.15 Parsing **v-link ADJ to-inf**

Evaluator		**Reaction**	**Situation**
They	were	puzzled	to find the kitchen door locked
She	was	very angry	to find him still with the circus
You	've got to be	very thankful	to win once
A spokesman	was	reluctant	to reveal the actual figures
He	is	most anxious	to avoid appearing weak

Table 7.16 Parsing **v-link ADJ to-inf**

Action . . .		**Evaluation**	**. . . Action**
People	are	slow	to learn
The Labour Party	looks	increasingly certain	to win the next election
The lion	had not been	able	to escape

In the next three groups (one other group consists only of 'be sure/careful to' and is left unanalysed here), an action or situation is expressed by both the subject of the main clause and the to-infinitive clause, with the evaluation occurring between them. The evaluation is either Judgement or an assessment of probability or ability, as shown in Table 7.16.

For the final two groups, we are again caught between competing analyses, one of which favours Appreciation (a person is evaluated in respect of an action they have done); the other favours Judgement (an action performed by a person is evaluated). As in the first set of groups, both analyses are shown in Table 7.17.

Table 7.17 Parsing **v-link ADJ to-inf**

Entity		**Evaluation**	**Limiter/Specification**
You	are	right	to say we are dealing with people who are feeling emotional
He	was	lucky	to escape with his life
Action . . .		**Evaluation**	**. . . Action**

The upshot of this is: whether we start with a semantic frame or pattern there is considerable consistency in mapping semantic and pattern elements. From the analysis of **ADJ to-inf**, and the earlier analysis of **ADJ *about* n**, however, it appears unlikely that any pattern would have a one-to-one mapping on to local grammar elements. Regrettably, there appears to be no 'quick fix' here. The FrameNet approach would appear to work better, but the early decision concerning semantic frame limits its scope. One of the advantages of pattern is that it is possible to look wider and include other kinds of evaluation in the same frame. It is becoming clear that the main challenge to the concept of either local grammar or semantic frame is the level of generality of the frame adopted.

An illustration of a level of generality that seems to work is the local grammar of Affect (Hunston 2003a; Bednarek 2008: 65–99). This is partly because Affect itself is relatively self-contained. Bednarek's version is probably the most successful. It builds on and considerably extends work by Hunston (2003a) and shares common ground with FrameNet by borrowing some terminology from it, notably 'Emoter', 'Emotion' and 'Trigger', as the

core semantic elements associated with Affect. Another point of commonality with FrameNet is that Bednarek considers all word classes, to considerable advantage. Some of Bednarek's examples, with their parsing, are:

- *Everyone / loves / compliments* Emoter / Emotion / Trigger (example from FrameNet)
- *I / 'm happy / for him* Emoter / Emotion / Empathy target (example from Hunston 2003a)
- *He / has impressed / as stand-in for the injured Tommy Wright* Trigger / Emotion / Trigger (Example from the BNC)

Bednarek (2008: 95) notes that analyses depend on the presence or absence of either the Emoter (*He hates days like that* vs. *He has impressed as a stand-in*), the Trigger (*He still hankers after office* vs. *I don't mind*) or an overt expression of Emotion (*I yearned for something new* vs. *It came as a surprise*).

7.7 CONCLUSION

This chapter has considered pattern grammar in relation to concepts of evaluation, in particular the Appraisal system (see Chapter 2). It was found that patterns are of some but limited use as tests for different kinds of Appraisal. On the other hand, patterns are a very useful heuristic, identifying distinctions in evaluative meaning that the Appraisal system may overlook.

The chapter also examined local grammars alongside other attempts to map meaning on to form, most notably FrameNet. Considerable similarities were uncovered but no one-to-one match.

The chapter has considered the challenge posed by any approach that maps formal and functional features: to derive a system that would reliably identify functional roles in unannotated text. A number of possible Local Grammars have been suggested and it is argued that lessons can be learned from the difficulty in implementing them. One is that pattern alone does not reliably identify functional roles. Rather, it is the word and the pattern together that does that, and that causes problems for the grammar writer in that the grammar becomes very detailed indeed, with a single pattern having two, three or four potential analyses depending on the node word. The alternative, to work from the word to the grammar, is promising, but runs into difficulties in identifying the optimal level of generality at which the grammar might aim.

Most significantly, perhaps, the chapter has argued for the primacy of pattern in identifying meaning and for the mutability of semantic categorisation. This alone argues for the desirability of taking pattern rather than word as the starting point for a local grammar.

8 Phraseology, Intensity and Density

8.1 INTRODUCTION

The purpose of this chapter is to examine the role of phraseology in expressing strength of evaluation. The chapter begins with a reconsideration of one of the examples presented in Chapter 1 (Ali 2008) and repeated here as Example 8.1.

> Example 8.1
>
> As I write this, Professor Smith, now a distinguished scholar, has her job under threat from the ghastly, grey accountants who run the University of Biggin-on-Sea. We are now in an epoch of production-line universities with celebrities paid fortunes to teach eight hours a week and genuine scholars dumped in the bin.

The extract makes two main evaluations: disapproval of the University of 'Biggin-on-Sea', which is planning to discontinue Professor Smith's employment, and, more generally, disapproval of all UK universities who prioritise financial interest over scholarship. In terms of resources used to make these evaluations, the extract is both implicit and explicit. It is implicit because it relies on shared assumptions about the social unacceptability of: 'universities are run by accountants', 'celebrities teach instead of scholars' and 'celebrities are paid a lot of money to teach'. There is also, however, explicitly evaluative lexis: adjectives *distinguished*, *genuine*, *ghastly*, *grey*; and phrases *under threat*, *dumped in the bin*. There are also phrases that flag or suggest evaluation without being wholly explicit: *production-line*, *paid fortunes*. It is also notable that every entity and action in the extract is the subject of evaluation. The human participants are a *distinguished scholar*, *ghastly accountants* and *celebrities*. The universities are *production-line*, run by accountants, with skewed priorities. The action of discontinuing someone's employment is described with the phrases *under threat* and *dumped in the bin* and the practice of employing non-academics as teachers is evaluated negatively by contrasting *celebrities* with *genuine scholars*.

In short, almost every word in the extract contributes to the evaluative meaning. As a result it might be said that the extract is 'saturated' with subjectivity. In addition, the words and phrases chosen are stronger or more intense in evaluative meaning than alternative paraphrases might be: *ghastly* is stronger than 'unsympathetic'; *has her job under threat* is stronger than 'may lose her job'; *production-line* is stronger than 'efficiency-oriented', suggesting an inhuman process; *under threat from . . . accountants* is stronger than 'for financial reasons'; *in an epoch of* suggests a depth of historical significance which the alternative 'nowadays' does not. Thus the phraseology of the cited extract makes the text 'more subjective' and 'more evaluative' than a paraphrased alternative would be.

This chapter investigates further the role of phraseology in increasing the evaluation in texts. It interprets 'more evaluative' in terms of both intensity (how strongly evaluative) and density/saturation (how extensively evaluative), as explained in the following. It is argued in this chapter that the sheer quantity of evaluative language (the density) adds to the strength of the evaluation (the intensity), and that aspects of phraseology play a key role in both.

8.2 INTENSITY, DENSITY AND REDUNDANCY

The concepts of intensity and density/saturation are taken from Martin and White (2005: 19–20). These writers note that evaluative meaning is prosodic, to use Halliday's (1979: 66–67) term (taken in turn from Pike 1982: 12–13). The term 'prosodic' appears to mean both 'scattered throughout a text, not restricted to one element of structure' and 'capable of gradation from strong to weak'. Martin and White refer to the first of these as 'saturation' and the second as 'intensity', though White later (personal communication) uses the word 'density' in preference to 'saturation'. As an example of saturation they suggest *I suppose he might possibly have, mightn't he?* where the modality is realised in four places ('I suppose', 'might', 'possibly' and 'mightn't'). They exemplify intensity with *It's a dirty rotten stinking lousy bloody low filthy two-faced lie*. Intensity forms one of the systems realising Force, which in turn is part of the system of Graduation (Martin and White 2005: 141ff.). Evaluations can be made more or less intense by a number of mechanisms, including the addition of an item such as a modifying adverb (*very/fairly miserable*), using a more intense lexical item (*brilliantly* as opposed to *competently*) and repetition (*we laughed and laughed and laughed*). Although Martin and White treat density/saturation and intensity separately, it is arguable that examples of the two concepts work together to make evaluation more or less strong in a text. In Example 8.1, for instance, it could be said that the density of the evaluation—the fact that almost every word contributes to evaluative meaning—increases the intensity. Each of Martin and White's examples, too, might be said to be

more intense because of its density. The first example sounds more tentative than the less dense alternative *He might have* does. The second gains intensity not just from the relative strength of the adjectives but from the fact that the clause is saturated with them.

Two further points might be made. The first is that consistency of evaluative phraseology adds to the coherence of a text (this point is adapted from Stubbs 2001: 108–109; 2009: 131). The second is that texts that are both intense and saturated in terms of evaluative meaning tend also to display a high level of redundancy. This means that there is, as it were, more language than is strictly necessary (as shown in the discussion of paraphrases of Example 8.1) and also that a reader is able to predict with increasing accuracy the evaluative meaning to be found in the text. Because the evaluative meaning is coherent, each new evaluative item confirms rather than initiates that interpretation.

To continue this discussion further, I return to texts first presented in Chapter 3. Below are three short text extracts which are similar in terms of topic but very different in terms of the intensity and density of their evaluation. All three texts were written at the time of the 'bird flu' epidemic of 2006. Example 8.2 comes from a website maintained by the United Kingdom Foreign and Commonwealth Office; the website offers advice to UK citizens intending to live or travel abroad—this includes advice on health issues, and in 2006 part of that advice related to the threat of bird flu. Example 8.3 is from a site called 'What Doctors Don't Tell You'; the site gives what might be called 'alternative' medical advice, advocating holistic and complementary treatments and challenging information given by the orthodox medical community. Example 8.4 comes from a weekly column in the *Guardian* newspaper entitled 'Bad Science'. The columnist attacks what he considers to be misleading reports of scientific, often medical, developments in the popular press. Example 8.4 itself constitutes an attack on the approach propagated in Example 8.3. It might be said that the text from which Example 8.2 is extracted is consensual, representing knowledge about bird flu as a unified entity, while the other two are conflict-oriented, representing knowledge as the site of dispute. Intuition also suggests that the texts are ranked in terms of their evaluative strength, with Example 8.2 being the 'weakest' and Example 8.4 the 'strongest'. This can be discussed in terms of both intensity and density. (Underlined phrases are the focus of further discussion later.)

Example 8.2[1]

There is at present no vaccine against any future pandemic flu strain. The normal seasonal flu vaccination *protects against* currently circulating human influenza strains, but *is unlikely to offer any protection against* avian flu strains or against a new pandemic flu strain. The use of seasonal flu vaccination, by minimising numbers of cases of seasonal flu, would *reduce opportunities* for

avian strains to mix with human strains, and may *allow* people with avian or pandemic flu to be *more easily* identified.

Example 8.3[2]

Scientists made similar *warnings* over the SARS . . . virus in 2003, which was also expected to become a major pandemic. Like H5N1, it started in Asia before spreading to the West, where it seemed set to reach Canada. The Canadian health authorities—*so concerned by the threat—allowed* the United Nations and the WHO *to take over* the day-to-day responsibility for controlling the outbreak. *In the event*, SARS proved to be *a damp squib*, killing *fewer than a thousand people* worldwide.

Example 8.4

It [the WDDTY website] *goes on for pages and pages*, *rehashing* the Tamiflu information leaflet's safety data, in *the most scaremongerish tones they can muster*, quoting scientific journal articles . . . all to make themselves look *as authoritative as is humanly possible*, all while *cursing* and *mocking* the medical profession. And all the while they are *blissfully, beautifully, wonderfully ignorant* of the fact *they have got the most important thing, at the heart of the matter, completely and utterly, pathetically, stupidly, obviously wrong.* This isn't *bad science*. It's *performance art*. (Goldacre 2006)

Example 8.2 contains low intensity verbs and nouns: *protect* and *protection*, *reduce opportunities* and *allow . . . more easily*. An interesting additional phrase, however, is *unlikely to offer any protection against*. This shows evidence of redundancy, in that it could be paraphrased rather more briefly as *will/may not protect against* (though this would entail repeating the verb *protect*). The longer phrase may be said to intensify the negative evaluation and to draw attention to it ('this warning is important') at the same time as apparently hedging the advice through the use of *unlikely*.

In Example 8.3 the lexis is more intense, including *warnings*, *threat* and the figurative *a damp squib*. There are also two instances of phrases flagging an attitudinal response. The first is *allowed . . . to take over*. Although the evaluation is not explicit here, one point of view would be that it is unwise for a national agency to give up responsibility to an international one. In the Bank of English all instances of 'allow x to take over y' are in the context where there is other evidence that the decision involved is unwise. The second instance is *in the event*, which can also be shown to co-occur with an expectation that is proved wrong. In each case the phrase is longer than possible paraphrases: 'The UN took over' and 'However'.

Example 8.4 is the most intense and the most dense of the three in terms of evaluative language. It contains intensifying adverbs (e.g. *completely and*

utterly, *pathetically*, *stupidly*, *obviously*) and other strongly loaded lexis such as *rehashing*, *scaremongerish*, *mocking*. In addition there are phrases that make the evaluation more dense as well as more intense: *goes on for pages and pages*; *in the most scaremongerish tones they can muster*; *as . . . as is humanly possible*. Each of these phrases adds redundancy and could be rephrased as 'is lengthy', 'in scaremongerish tones' and 'as . . . as possible' respectively. In other words, the writer has chosen a longer expression over a shorter possible one. The phrases *as . . . as is humanly possible* and *the most . . . they can muster* are open to corpus-based scrutiny to test the hypothesis that their usual usage patterns add evaluative weight to Example 8.4

The first investigation is of '*as* Adjective/Adverb *as is humanly possible*'. It is not possible to 'prove' that this phrase is more intense than the alternative 'as . . . as possible', but all the corpus examples imply a sense of striving, of attempting to achieve something that is difficult. Here are some examples to illustrate:

Example 8.5

To approach optimal experience **as closely as is humanly possible**, a last step in the control of consciousness is necessary.

Example 8.6

(headline) The Bosnian burden: **As long as is humanly possible**, the aid must get through

Example 8.7

It has to be said that sex is shoehorned into the plot **as often as is humanly possible . . .**

Example 8.8

If any of us should have to dial 911 for a medical emergency, we'd like to think that help would be arriving **as quickly as is humanly possible**.

Example 8.9

Since adoptive parents are unsupervised once an adoption order is made, approving authorities have to be very careful that, **as far as is humanly possible**, the adoption applicants will be able to cope with the child or children . . .

In Example 8.4, then, *as authoritative as is humanly possible* implies a striving after the appearance of scientific integrity, which is coherent with the more explicit message of the extract.

Investigation of the phrase *the most . . . they can muster* leads to surprising results. The use in Example 8.4 ('superlative adjective . . . *they can muster*') is not typical. Most instances are '*as much* noun *as he could muster*' or '*all the* noun *he can muster*'. Where a superlative adjective is used it is usually *best*. Again, however, there is a commonality of pragmatic meaning. The phrase is used when 'all that someone can muster' is not very much, and when a kind of pretense is taking place. Examples 8.10–8.12 illustrate this:

Example 8.10

With **all the gentle, confident command he could muster**, Bunner said, 'Ken, show me,' and he held his hands open and out to Ken Nevins. Ken hesitated, and Bunner knew he'd played out all his line. If Ken wouldn't do it, he couldn't ask him again. [Bunner does not feel confident or in command]

Example 8.11

I made my exit with difficulty, shrill abuse and threats following me from the entire family as I moved, with **whatever dignity I could muster** as a cold sweat trickled between my shoulder blades, through the gate and to freedom. [The speaker leaves with little dignity]

Example 8.12

If Jacques Arnold is **the best recruit he can muster** he must be very disappointed. [JA is not very good]

In Example 8.4, then, the implication is that the *scaremongerish tones* are not particularly frightening and that the writers of WDDTY are failing to achieve something they have attempted.

To summarise the argument from these short discussions of individual texts: the strength of evaluation in each text is a product of both intensity and density. These in turn work together: intense expressions of evaluation are likely to add to the density of evaluation in the text, and intensity can be achieved by density of expression. Expressions which add to the density of expression are likely also to have a level of redundancy.

What are of particular interest are phrases where the evaluative meaning is not particularly obvious, but where a search of a large collection of texts will reveal a commonality of usage. Examples include *in an epoch of*, *in the event* or *as is humanly possible*.

8.3 INTENSIFYING PHRASES

The previous section examined individual text extracts and noted how the choice of phraseology, as opposed to possible paraphrases, adds to the

intensity and density of evaluation in that text. Such an exercise may or may not be convincing, but it has the disadvantage of being restricted in application to the texts under discussion, for the most part. Apart from the general observation that phraseology matters, there is little that can be generalised from the discussion. It has become apparent, however, that some individual phrases do have a more general function in intensifying evaluation in whatever text they occur. This section continues this theme.

For the discussion in this section I draw on another short text extract, with one sentence that I find particularly interesting shown in bold:

Example 8.13

The intention of this article is not to offer 'new ideas' for pedagogical application, but to share a very real concern about our survival both as teachers and as human beings. If this seems unduly dramatic, then so be it. **Our situation is already dramatic almost to the point of tragedy.** We are at risk—from the pressures of consumerism, the media, technology and rampant trivialisation. Our lives are lived at an increasingly accelerating pace, leaving less and less time for mature reflection and the exercise of independent choice.

This extract was discussed in Hunston (2004a); one of the points made there was that *to the point of* can be shown to occur consistently in the sequence:

'less intense negative evaluation' + *to the point of* + 'more intense negative evaluation'

This suggests that *to the point of*, although it is not in itself obviously evaluative, does play a significant role in the expression of evaluation. It adds to the redundancy inherent in the context in which it is found, because a reader can anticipate that what follows *to the point of* will be negatively, and intensely, evaluative. In addition, the phrase adds to the density of evaluation in the sentence. In effect, the reader is told three times that 'things are bad': once in the word *dramatic*, once in *to the point of* and finally in *tragedy*. Both the redundancy and the density contribute to the intensity of the evaluation, in that *dramatic to the point of tragedy* is more intensely negative than *tragic* alone would be. As will be argued in the following, there is a paradox here, in that there is also apparent hedging, with *to the point of tragedy* meaning 'almost but not quite tragic'. This is similar to the argument around Example 8.2, where it was suggested that *unlikely to offer protection against* both hedged the unavailability of protection whilst at the same time intensifying the message.

The phrase *to the point of* is one of a set that perform a similar function. The phrase *bordering on* was discussed by Sinclair (2003), and *almost* is a single word paraphrase that similarly links lower and higher intensity items. Figure 8.1 shows some examples of each, with the low and high intensity items in italics.

Figure 8.1 Intensifying sequences.

what is unconscious, *spontaneous*, "almost *brutish*." Manhood, by contrast, is
to see Michelle Collins in a *sweet*, almost *ladylike* role. DAVID EVANS Walsall,
as their recent talk had been *light*, almost *trifling*. Fox was so moved that he
quickly grown to be a protective, almost conformist, shield around the Pill
the eyes: they are deep and *dark*, almost *black*, always full of light and,
em!) when Marijne gives a *throaty*, almost *inaudible* growl of regret before
of German accent (and with a *formal*, almost *Prussian*, courtliness that always
years, a thinning of the *subtle*, almost *invisible* threads of
with flowers. In a very small, almost square format she took great pains
a cast performing *whisper-quiet*— almost *spectral*—interpretations of
<p> Julio's problem is *simple*, almost *trivial*, but it has been weighing
collars—transforms a *preconscious*, almost *vegetative* existence into a
generals, he is wiry and *donnish*, almost *ascetic*. But he has a reputation
border crossing was a *peaceful*, almost *idyllic*, backdrop to the sad,
s name with passion. It was a slow, almost painful start for him but he has

from the hip *negligent*, at least bordering on *criminal*. Haled, from the
and legs. Her manner was *clinical*, bordering on *cold*, and she had brought a
exfoliation, and found it *ticklish* bordering on *painful*; then a `vibrosaun",
younger colleagues as *overconfident* bordering on *foolhardy*. The study
these positions will be *difficult*, bordering on *impossible*. But neither Mr
East Germany. <p> She is *anxious*, bordering on *fearful*, as she addresses the
experience, expect the *ridiculous* bordering on the *insane*. Labor's Education
said Brian. His tone was colorless, bordering on hostile. <p> She's good," said
and true, tend to be *dogmatic*, bordering on the *dull*. Use some restraint,
in the interior to be `*filthy*, bordering on *beastly*," unredeemed for Faux
at the waist. He was very *thin*, bordering on *emaciated*. His face was the
game at large, would be *difficult*, bordering on *impossible*, to find. Two have
incident in India. IT is *difficult* bordering on *impossible* to avoid the
someone with this *extraordinary*— bordering on *obsessive*—commitment to
to come his way, Ingram is *upbeat*, bordering on *excitable*. Twice before he has

to do when arrested was *dismal* to the point of serious *depression*. I admired my
at Bright's Inn were *generous* to the point of *overwhelming* and Flodge was
realisation, can play it *cool* to the point of dazed *blankness*, but let me tell
backing that sounds *thin* to the point of *threadbare* (I'm on side two now,
dramatic heights and be *crisp* to the point of near *charredness* in places. It
language should not be *simple* to the point of *condescending*; rather the
Buddhist, is *positive* to the point of *evangelistic*. `I think of it
persons who are *demented* to the point of total *incomprehension* receive
metals are much calmer—*calm* to the point of chemical *indifference*. Tin, for
firm. Diligent, modest and *shy* to the point of near *invisibility*, he is stuck
solicitous staff, *sincere* to the point of *oleaginous*, shaking hands with
II, for instance -are *rash* to the point of *reckless*, and he supports his

(continued)

Figure 8.1 (continued)

and with back play *ponderous* to the point of *stationary*, it was a wonder that
it is incredibly green-to-the-point-of-transcendental and is made in a
with his money and is *modest* to the point of apparent *forgetfulness* about his

There are a number of observations that might be made about these sets of examples. In grammatical terms, the most typical sequence is 'adjective + phrase + adjective'. In the case of *to the point of*, however, the sequence 'adjective + phrase + noun' is also found. Turning to the evaluative meaning, although the pattern of 'less intense → more intense' is very frequent it is not universal, particularly in the case of *almost*. In sequences such as *protective almost conformist*, *small almost square* and *slow almost painful* the adjectives are not (necessarily) to be seen as points along a single trajectory. In most cases, however (e.g. *formal—Prussian*, *anxious—fearful*, *rash—reckless*), the argument of increased intensity holds good. Secondly, in the case of *almost* the adjective pairs indicate evaluations of both positive and negative polarity as well as judgements that are clearly subjective but not particularly strong in terms of positive and negative (*sweet—lady-like*, *light—trifling*). With the other two phrases, however, the evaluation is predominantly negative. A final point is that the two contextually linked adjectives are not necessarily ones that might be linked out of context. This point might be expanded by considering examples of 'adjective + *almost* + *clinical*' (*clinical* being one of the most frequent adjectives occurring in this pattern). The first adjective slot is filled by: *astringent*, *bright*, *cool*, *detached*, *fastidious*, *serene* and *simple*. These do not form a particularly strong lexical set, suggesting that each imposes a specific interpretation of *clinical* but also that the sequence itself construes a connection that would be missing otherwise. This is another example of Sinclair's concept of 'semantic reversal' (2004: 134–136). Sinclair discusses the example *The ambience borders on the holy*, which, he argues, presents holiness as something 'outside normality'. This is an instantial meaning, Sinclair argues, imposed by *borders on*.

We might also note, as mentioned earlier, that the examples with *almost*, *bordering on* and *to the point of* perform two apparently contradictory functions. In each there is apparent hedging, as illustrated by these paraphrases:

- *simple almost trivial* (not quite trivial)
- *anxious bordering on fearful* (not very fearful)
- *crisp to the point of near charredness* (not exactly charred)

In each, however, density and therefore intensity are introduced into the text. The examples are triply evaluative. Each contains not one but two evaluative adjectives or nouns, plus a sequence (adjective + *almost* / *bordering on* / *to the point of* + adjective/noun) that is closely associated with evaluation. The three expressions of evaluation add to the density. There is also considerable

redundancy, in that the force of the final adjective in the sequence is wholly predictable (e.g. *crisp to the point of—brittle / inedible / dangerously sharp*). The intensity of the phrase as a whole is greater than that of either of the two adjectives and might be said to be the sum of the three-point evaluation.

A final point of consideration is what the effect of these 'two adjectives' sequences is. Here are three comparable examples, each containing the phrase *clinical detachment*. In Example 8.14, *clinical* is linked to *cool* and modified by *almost*. In Example 8.15 it is modified but not linked to another adjective. In Example 8.16 the adjective is unmodified. Each example is from a text describing the consequences of instances of abuse and rape.

Example 8.14

Seated in a straight-backed chair, Hebert tells his life story with *cool, almost clinical detachment*. He seldom registers any emotion except occasional sadness for the little boy he once was.

Example 8.15

The girls look tearful when they recall how an unknown boy offered to buy them a coffee in a Leeds shopping mall, but recount their experience in Pristina with *almost clinical detachment*.

Example 8.16

She talks of the abuse with a *clinical detachment*, trying to diminish it in an effort to bolster the family reputation.

Interpreting the effect a text has on a reader must always be a subjective enterprise. To me, these examples are ordered in terms of the degree of surprise they express that someone in such a situation could tell their story in a 'detached' way. Example 8.14, with the 'two adjectives' sequence, sounds to me the most surprised, example 8.16 the least.

My argument is, then, that phrases such as *to the point of* play a specific role in intensifying evaluative meaning. The starting point has been individual texts, with a corpus being used as corroborating evidence for a phrase consistently having an intensifying function. The next section will consider how a corpus might be used in a more focused way to identify intensifying phrases.

8.4 SEARCHING FOR INTENSIFYING PHRASES

The question to be addressed in this section is how intensifying phrases can be identified using a method that is more systematic than simply commenting on sample texts. I will here suggest a method that should identify some

though not all such phrases. I begin once again with the phrase *dramatic almost to the point of tragedy*.

One of the implications of this phrase is that the combination of a preposition (*of* in this case) and a noun that expresses intense and stable evaluation (*tragedy* in this case) might be worth further investigation. The perhaps surprising contribution of prepositions towards evaluative meaning has been commented on in Chapter 5 and Chapter 7. Further evidence that *of tragedy* might be worth investigation comes from the fact that in the Bank of English, *of* is the third most frequent L1 collocate of *tragedy*.

The words occurring most frequently before *of tragedy* (with those mentioned in the following in bold) are:

> ***sense***, ***kind***, ***birth***, *sort*, ***face***, *lot*, *time*, ***out***, *type*, ***share***, *scale*, *years*, ***time***, *mask*, ***elements***, *element*, *one*, *moment*, *life*, *side*, ***air***, *nature*, *toll*, ***lighthouse***, *shades*, ***seeds***, *stories*, ***midst***

Each of these occurs three or more times in this context in this corpus. Not all of these words are likely candidates for intensifying phrases. *Birth of Tragedy* and *Lighthouse of Tragedy* are the titles of books, so *birth* and *lighthouse* are excluded from further consideration. Some of the words in the list form phrases that occur with a wide variety of nouns, such as *a sense of (humour, community, security, urgency, loss)*; *the kind of (people, society, behaviour)*; *born out of (wedlock, love, necessity, desperation)* or *key elements of (success, democracy, government)*. These uses suggest that the co-occurrence with *tragedy* is not significant. Other words in the list, however, do seem to be more reliably associated with negative evaluation. For example:

- *In the face of* occurs with *adversity*, *opposition*, *competition*, *pressure*, *death*, *criticism* and *odds*.
- *A time of* occurs with *crisis*, *war*, *change* and *tension*.
- *In the midst of* occurs with *crisis*, *war*, *recession*.

Yet others co-occur with both negative and positive evaluation:

- *An air of* occurs with *mystery*, *authority* and *confidence*, but also with *unreality*, *desperation* and *resignation*.
- *Fair share of* occurs with *luck*, *success* and *attention*, but also with *problems*, *stick*, *bad luck*, *bad press*, *ups and downs* and *injuries*.
- *The seeds of* occurs with *love*, *future success* and *future victory*, but also with *doubt*, *change*, *disaster*, *destruction*, *future instability*, *future conflict* and *future trouble*.

In all these cases, the phrase concerned might be said to add to the redundancy, density and therefore the intensity of the evaluation concerned.

The argument is strongest in those cases where the polarity of the evaluation is consistent.

The procedure can be repeated using the second most frequent preposition preceding *tragedy*, which is *in*. This preposition occurs most frequently following the verbs *END* and *RESULT*, that is, in phrases such as *end in tragedy* or *resulted in tragedy*. Repeating the earlier procedure, each instance of *END in* and *RESULT in* was investigated, and as expected the nouns following those verbs are consistently negative, with *tears*, *divorce*, *failure* and *disaster* being the most frequent.

In short, the word *tragedy* is being used as a probe to find phrases ending with a preposition that reliably co-occur with either evaluative meaning in general or with negative evaluation in particular. This can be repeated with other probes. The only criterion for a probe is that it should be, using White's (personal communication) term, stable in meaning, that is, it should consistently indicate a particular polarity of evaluation. There is no requirement to obtain a complete list of such items as they are only being used as probes. A possible source is the General Inquirer, a semantically annotated wordlist used by, among others, those working in Sentiment Analysis (Stone et al. 1966; Kelly and Stone 1975; and see http://www.wjh.harvard.edu/~inquirer/homecat.htm).

To carry this out, ten nouns from the beginning of the 'Negative' category in the General Inquirer database were selected and subjected to the same process as that described for *tragedy* earlier. The ten nouns are: *absurdity*, *adversity*, *affliction*, *aggression*, *agony*, *anarchy*, *anguish*, *animosity*, *apathy* and *arrogance*. The investigation leads to some interesting observations. For example, a potential intensifying phrase is *months of* which occurs before *agony*. Further investigation of *months of* indicates that it occurs before a wide range of nouns with no consistency in meaning. If a further restriction is added, so that it is specified that *months of* follows a verb, then sets of nouns do emerge. Table 8.1 shows both the frequent verbs and the frequent nouns before and after *months of*.

Some of the verbs have an evaluative meaning, most notably *ENDURE* and *SUFFER*. It is not surprising that the nouns that co-occur with those verbs are also negative. The nouns *criticism*, *isolation* and *delays*, for example, follow *endured months of*. In other cases the meaning is not linked to an evaluative polarity. The verb *INVOLVE*, for example, co-occurs with a range of nouns including the apparently neutral *planning*. Arguably, the effect of adding *months of* is different in each of these examples. In the phrase 'endured criticism', for example, the evaluative meaning is clear and is to a large extent dependent on *endured*. Extending the phrase to *endured months of criticism* intensifies the negativity, partly by adding the information that the situation continued for a lengthy time period, but also simply by adding more words to

Table 8.1 Verbs and Nouns with *months of*

Verb		*Noun*
FOLLOW	*months of*	*treatment*
END		*therapy*
TAKE		
FACE		*negotiation*
ENDURE		*talks*
REQUIRE		*argument*
INVOLVE		*bridge-building*
MEAN		*consultation*
SUFFER		
		rumours
		controversy
		protests
		criticism
		unrest
		turmoil
		torment
		misery
		isolation
		despair
		delay
		delays
		planning
		effort
		anticipation

the phrase. *Months of* is very nearly redundant, meaning that its main function is to intensify. In *involved months of planning*, on the other hand, conveying the arduous and time-consuming nature of the task, and therefore the negative polarity, rests on the phrase *months of*. In this case, *months of* is by no means redundant. It adds evaluation rather than intensifying it.

What this example alerts us to is the fact that the intensifying phrases may be quite precise. For example, *a dose of* does not reliably indicate evaluative meaning but '*a/an* adjective *dose*' does. This in turn is linked to what is usually considered a distinction between congruent and figurative meaning.

The intensifying phrases identified by this means are shown in Table 8.2. They have been grouped according to meaning and are annotated with (-) for negative polarity and (+/-) for both positive and negative.

Table 8.2 Instances of Intensifying Phrases

Phrases	*Comment*
To the point of (-) *Bordering on (-)*	Discussed earlier. A figurative indication of proximity.
The depths of (-) *As much as (-)*	Indication of quantity.
On the brink of (-) *On the edge of (-)* *On the verge of (-)*	A figurative indication of proximity.
ADJECTIVE dose of (-) *A hint of (+/-)* *A touch of (+/-)* *A degree of (+/-)* *A wave of (-)*	Indication of quantity.
In the face of (-) *ADJECTIVE act of (-)* *Be greeted with (-)* *A tendency to (-)*	A collection of other examples.

The comments in Table 8.2 suggest some perhaps surprising generalisations. Several of the phrases indicate proximity, including *to the point of*, *on the brink of* and *on the verge of*. Others indicate quantity, including *a hint of* (small quantity) and *a wave of* (large quantity). Many of the examples use what might be called figurative language. The proximity, for example, is symbolic rather than actual, and many of the indications of quantity are also figurative, as with *the depths of* or *a wave of*. Other phrases such as *in the face of* or *be greeted with* are also figurative.

To illustrate again how these phrases might operate in actual text, here are two further examples. The first is:

Example 8.17

I can't emphasise enough the depths of my ignorance (Independent 8 July 1995)

This can be paraphrased as 'I was very ignorant'. It includes the explicitly evaluative word *ignorance* and two intensifiers: *I can't emphasise enough* and *the depths of*. Both of these add to the density of evaluation in the sentence and to redundancy. Both can be shown to co-occur consistently with evaluative meaning. Out of the 31 instances of *can't emphasise enough* in the Bank of English, all but eight co-occur with evaluation. This is mostly evaluation of importance, but there are also instances of positive and negative polarity e.g. *the importance of . . .*, *how inappropriate it would be . . .*, *the friendliness of. . . .* It might be true to say, then, that *can't emphasise*

enough contributes to density but not to redundancy, because we cannot predict from it what the polarity of the evaluation will be. As noted earlier, *the depths of* co-occurs with negative evaluation.

The second example is:

Example 8.18

Going abroad with Raleigh International entails a fair dose of misery, what with primitive conditions, food, latrines and other niceties you encounter.

The first part of the sentence can be paraphrased as 'Going abroad with Raleigh International entails misery'. The phrase *a fair dose of* intensifies the evaluation and makes the sentence more saturated with evaluative meaning. It adds to redundancy in the sense that it is consistently associated with negative evaluation so that *misery*, or something like it, is predictable.

This very brief investigation of some target phrases suggests a possible way forward in adding to our knowledge of how the evaluative character of a text is achieved.

8.5 CONCLUSION

In this chapter I have drawn on the well-known observation that some phraseologies are frequently associated with evaluative meaning, to the extent that they are able to bring about semantic reversal and imply a meaning that would be otherwise missing from a text. Where this phraseology co-occurs with other evaluation the effect is of redundancy, also heightened intensity and saturation of evaluation. This can be illustrated through analysis of sample texts, though investigation of this kind is necessarily unsystematic and ad hoc. Systematising the search in a corpus is not easy and requires some ingenuity. Only the tip of the iceberg has been investigated here. There are some promising leads though, including a role for figurative language and a suggestion that sheer quantity of words may be significant.

9 Conclusion

9.1 EVALUATION, PHRASEOLOGY AND CORPUS LINGUISTICS

An act of evaluation or stance-taking minimally involves the evaluator and the person, utterance, object or situation evaluated. Given that evaluation is an inherently interpersonal act, however, it also involves a receiver of the information, who may be a participant in a conversation or the reader of a text. Other possible roles are the beneficiary in a positive situation, or the equivalent in a negative situation, the causer of an act of evaluation and the reporter of the act.

Under the broad heading of 'evaluative language' the following are often included:

- Words which indicate the act of evaluating an entity that is external to the text; these are especially but by no means exclusively adjectives. The research challenge is to recognise them and also to be able to map them on to instances of evaluation.
- Clauses and phrases that imply but do not state evaluation. It is not always possible to reach agreement on which items constitute this set.
- Evaluators of propositions within a text; these are often adverbials but a range of other features, including attribution, may perform this function also.

The objects of evaluation include people, objects (including text-objects), events and situations construed as external to the text or an internal to it.

Corpus investigation techniques play a number of roles in the study of evaluative language. Some of these are:

- They allow a researcher to establish that a given word or phrase has a typical evaluative use or polarity.
- They permit quantification of evaluative meaning in one set of texts over another, by counting the occurrences of given forms.
- They permit mapping of meaning elements on to form elements where these coincide consistently.

- They allow a researcher to observe consistency in co-text in meaning as well as in form.

More profoundly, perhaps, the insights into language gained from corpus linguistics studies inspire ways of looking at evaluative language. Most important of these is the search for and perception of pattern: consistency in the mapping of meaning on to form, and consistency in the items that co-occur with a node.

Phraseology has been interpreted in this book to mean:

- the identification of sequences of words i.e. MWUs that play a role in the evaluative act
- differentials in wordform frequency that can be used to establish likelihood of a kind of evaluation occurring
- consistency in how particular kinds of textual item are evaluated within a specialised corpus

9.2 CORPUS LINGUISTICS—THEORY, METHOD, TECHNIQUE

There are by now many ways of doing corpus linguistics, some involving very sophisticated quantification techniques (e.g. Biber 1988; Gries 2008b), others focusing on the detail of individual word use (e.g. Teubert 2008). Some see corpus linguistics as a method (e.g. Thompson and Hunston 2006: 8); others restrict it to a particular theory of language (Teubert 2005). In many ways, although corpus linguistics comprises a (growing) number of fairly standard techniques (finding continuous and discontinuous MWUs, identifying Keywords, finding statistically significant collocation), researchers in corpus linguistics frequently find themselves still searching for a method, especially when trying to answer questions of a discoursal nature. It is still the case that whereas corpus investigation is very fruitful when the object of enquiry is known and identifiable—for example, a particular grammatical form in a parsed corpus or a given set of words—a question still posed when embarking on corpus research is: having collected our corpus, what shall we look for in it?

In essential terms it remains impossible to 'analyse' a corpus, unless we are content with an analysis that lists and quantifies words or phrases or clause types. If our questions are something like 'what is said about . . .?' or 'how is . . . done?', the method of analysis remains obscure. Rather, ways have to be found to translate research questions prompted by discourse analysis into corpus interrogation questions. This is particularly demonstrable in the case of evaluation. The only way to analyse the evaluative language in a corpus would be to annotate each instance manually (as Martin and White 2005 do for individual texts). An approach of that kind would treat the corpus as no different from a collection of texts, each one

of which can be considered as an independent entity. Sinclair (forthcoming), however, argues that what distinguishes a corpus from a collection of texts is not so much its size or content as the way in which it is interrogated. Searching the corpus in a way that does not prioritise the integrity of each constituent text but instead makes quantitative generalisations across texts can answer specific questions about evaluative language (as Hyland and Tse 2005 or Biber 2006a do) but cannot identify or quantify all of it.

The techniques used in this book have been very simple. Apart from some discussions of relative frequency, and use of the concept of most-frequent co-occurring item, the bulk of the work has focused on the scrutiny of concordance lines. This is not accidental. The aim of the book is not to analyse a corpus, or even to compare corpora in the way that researchers using the terminology of stance do. Rather, the aim is to propose approaches to the study of evaluative language that make use of insights into phraseology that have themselves been inspired by the study of individual words in context, that is, in concordance lines. It is hoped that each of the approaches suggested in the book can be used in further research, and some suggestions are outlined in the following section.

Before turning to them, and at the risk of undermining much of what has been said, a brief caveat about concordance lines must be added. The example used here is the proposal for modal-like expressions explained in Chapter 5. The methodology employed in that investigation illustrated the peculiar view of language that results from examining a corpus via concordance lines. Concordance lines derived from corpora draw attention to the similarity of items occurring in proximity to a 'node' item, especially when the lines are selected and sorted in the ways demonstrated in this book. They might also, however, be said to *impose* a similarity through proximity. In other words, seeing many examples of a given wordform together facilitates the observation of similarity that leads to the identification of patterning. Out of context, in the somewhat unusual sense of 'out of the context of the concordance lines', a sequence such as *you are slowly learning how to distinguish fact from fiction* is unlikely to be recognised as sharing a modal-like function with *surely it's a question of being able to distinguish between facts and propaganda* and *. . . a refusal to believe that the average viewer can distinguish between fact and fiction*. Once they occur in the same space, and alongside *difficult/impossible/have to distinguish*, the modal function becomes clearer.

On the other hand, it could be argued that the concordance lines themselves construct 'artefacts': objects created by the investigative procedure itself. The observation of concordance lines places the node word or phrase both literally and metaphorically at the centre of investigation. The keyword-in-context layout predisposes the observer to see the node word as the dominant feature to which all other elements in the concordance line are secondary. Interpreting a range of features as 'modal-like' implies that their function is to modify the node word. This is at odds with the

grammatical structure of many of the sentences that appear in the concordance lines. There is no grammatical sense in which *you are learning how to*, for example, is a modifier of the verb *distinguish* in the sequence *you are learning how to distinguish*. The modification is discoursal and pragmatic rather than grammatical. A further point to be made is that it is certainly not true that sequences such as *learn how to*, or patterns such as **verb + noun + to-infinitive** always realise modal-like meaning. Only when they are drawn together by their co-occurrence with a common lexical item, *decide* or *distinguish*, does their similarity emerge. It could be said, however, that this similarity does not 'emerge' at all, but that it is imposed by the observer, who is seduced by the layout of the concordance lines into seeing similarity where none exists.

There is ultimately no proof of the existence of categories such as modal-like expressions. Each individual language user will look at the evidence and either see similarity or fail to see it. Accepting the category as a valid one involves accepting a view of language as lexically and discoursally rather than grammatically driven and as being composed of categories that are post-experience rather than structural.

A corpus is a collection of independent texts, each produced in its unique social situation. Each instance of a given word or phrase, each concordance line, has arisen from a particular communicative context in which a combination of ideas has been put together. What sets of concordance lines reveal is the outcome of the linguistic choices made in all those different contexts, taking a single word or phrase as the common factor. Some writers, such as Hoey (2005: 178), argue that each language user is influenced by their experience of particular lexical items in other contexts, so that they are likely to produce a phraseology that conforms with the norm. Others, such as Teubert (2003), argue that a corpus cannot provide evidence for cognitive processing, only for outputs. Whichever view is adopted, the concordance lines and list of collocations are derived from the outcome of many different communicative actions and represent the outcome of many different interactions between communicative need, grammatical structure and lexis. They suggest that to a remarkable extent diverse communicative situations have led to similar outcomes.

9.3 IMPLICATIONS AND APPLICATIONS OF THIS RESEARCH

Each of the main chapters of this book suggests possible applications and further research. In some cases some more general implications are suggested.

Chapter 5 offered new resources for quantifying the modal meaning in a corpus, beyond the identification of modal verbs. It also argued for the colligation of the **verb + wh-clause** pattern and modal meaning. Further research would be necessary to establish this more securely and

to investigate whether text-type or register was a factor here. More generally, the tendency of individual words to co-occur with a given structural or semantic feature such as modality has implications for the study of probability in register (e.g. Matthiessen 2006). From one end of the scale, it can be established that some registers show a high proportion of a given feature. From the other end, it can be shown that some words co-occur with that feature. What, then, is the relationship between the word and the register? For example, do verbs that are disproportionally frequently found in negative clauses also occur disproportionally frequently in registers that have a high incidence of negative clauses? Or are the two facts unconnected? How do lexical choice and register choice intersect?

Chapter 6 suggested that epistemic objects—facts, hypotheses, discoveries and so on—may attract particular forms of comment and evaluation. The corpus used to investigate this was a fairly general one, consisting of issues of the *New Scientist*. The same principles of investigation could be applied to corpora based on single academic disciplines. It might be expected that one of the differences between disciplines might be the prevalence of evaluation of these epistemic objects. In a more general way, Chapter 6 might be said to extend the range of what corpus linguistics does, possibly beyond linguistics. The semantic sequences that are identified are not really linguistic objects at all (and this distinguishes them from the 'units of meaning' identified by Sinclair, for example). They do not tell us what language is like, or even what the English used in the *New Scientist* is like. Rather, they tell us what things are often said, and that is a very different question. Increasingly, however, corpora are being interrogated for what their component texts say, as well as how they say it, and semantic sequences offer a way of expressing that information.

In Chapter 7 the feasibility of establishing local grammars was discussed. Taking this project further would have a number of potential applications. More rigorous and complete local grammars of evaluation could be an additional resource for sentiment analysis, potentially parsing texts to identify evaluator and object, for example. Moving beyond evaluative language, an 'intelligent pattern grammar' could offer a resource to language learners that would approximate to a thesaurus of functions.

Chapter 8 sketched out very briefly some suggestions for investigating phrases that might reliably intensify or accompany evaluative language. A more comprehensive tally of such phrases would add information to studies that measure degrees of subjectivity or calculate strength of persuasiveness.

What I hope this book has demonstrated is that the act of evaluation plays an important role in a number of communicative areas, including persuasion and the construction of knowledge. It draws on the full range of resources in language; in English it is often associated with adjectives

and adverbs, but is by no means restricted to them. Many phrases as well as individual words are associated with evaluative meaning; this makes the approaches associated with corpus linguistics particularly suitable for studying evaluation. Once phraseology begins to be investigated, however, this opens up the avenue of investigation beyond the investigation of evaluative language itself.

Appendix to Chapter 5

FINITE VERB FORM (INCLUDING INDETERMINATE LINES)

democratic government privately	speculate whether social democra
melanin genes." What happened, they	speculate, when black and white m
me her dreams in great detail and	speculate as to what this stands for or
sit in my rocking chair and idly	speculate as to what the best thing abou
and Teheran. Many of the papers	speculate as to the effect the resumptio
snicker and order more scotch. We	speculate as to the exact nature of the
undertake difficult challenges and	persevere in them. Albert Einstein spent
over the course of several days	persevere in breaking an alpha-numeric c
warm and dry for lunch. <p> Few	persevere in field-based research withou
let me know your joy today as I	persevere in faithful service and as I
fighting to open an envelope, I	persevere in having sex." As in her
He said Egypt was insistent that it	persevere in its efforts to move the
of an important undertaking,	persevere in a fixed design, and work ou
member of the Billings Bighorns and	wonder about the team that practised jus
as much as anybody. <p> Simon: I	wonder about this. Warner Brothers didn'
phrase # killing instrument." I	wonder about the rage that has been his
pair of pants in the morning, so I	wonder about a nappy for his rest, or a
f only Mr Peck would talk enough. I	wonder about the orchids," said Jupe
by yet unanswered questions: I	wonder about my health, my financial
an be reunited. As the days go by I	wonder about what's going to happen in t
like Jimmy and go into business". I	wonder about their lives and whether the
slightly undercutting that. Erm I	wonder about the other two if you can se
Quite reserved in private. Some mps	wonder about his sexual orientation,
severe conditions. <p> But I still	wonder about this advice to ride out a
began to diminish. <p> JLG: I still	wonder about—and I guess I wish you
12 in the same way many of us today	wonder about the Holocaust of 60 years
ll time. <p> There may be those who	wonder about the viability of using what
last October. What odds now, you	wonder, about a multi-million pound expo
nds up with 7,000 redundancies, you	wonder about the process. These are
but managers and accountants	agonize over the choice anyway. The
care has gotten too good, and	agonize over whether it has become

(continued)

on a PC.'" <p> While many bankers agonize over the fork in the road, Wells
t, the journalism reviews endlessly agonize over it. You can't blame the
like Dorothy Carter and her husband agonize over what to tell their sons.
Every year at this time I agonize over what to give my wife for
uterus and fallopian tubes alone. I agonize over the number of post-
is the way of Tao. Tao people never agonize over problems but greet life wit
yes. <ZF1> We <ZF0> we sort of agonize over it. It you wouldn't believe
of slimness, more and more people agonize over whether or not they would b
to the wall. Many Americans still agonize over their own personal feelings
disease: the humane tacksmen agonize over distresses, that inability,
e struggle with these questions, we agonize over them, we suffer. As
nails over pathetic real ones; we agonize over every little nick a
that `the faculty as a whole agonize over their syllabi, keep
lives, peppers and bread, while you agonize over what to choose from
saying we are not democratic." I ponder on other `democratic" vie
and The organisation, and instead ponder on the remake currently m
night-time record to your journal - ponder on it. Note down what fee
of letters sipping coffee as they ponder on the day ahead. Sadly,
rivial incident, but the longer you ponder on it the more you wonder
go out there and play crap, you ponder on the match, you learn s
ass in the fairy tale, he says; you look into it and you see not yo
ill be those that recognise issues, grapple with them, and have the
S. government and business leaders grapple with ways to make the n
are being held up while managers grapple with the unfamiliar busi
work hard to avoid. While police grapple with strategies on how t
wen Bennett-Jones. As the Romanians grapple with the consequences of
in emphasis on both sides as they grapple with how to move forward
later the survivors of the tragedy grapple with health problems and
eminent historians, about how you grapple with public policy ques
step up its activities and turn its attention to Western—especially
their book, the authors turn their attention to the investigation of
summit in London today turn their attention to the difficult questi
people traditionally turn their attention to doing up their homes
later; for the moment we turn our attention to class size. Class si
labour movement politicking and consider the plan as a job-saving
to Croatia and the Bosnian Serbs consider Serbia as their natural
and it's important that I think we consider this as a future risk fo
Of course, there are many who consider themselves as scientist
will often differ for women. If you consider technology as culture, y
first and foremost or whether you consider yourselves as being

With operator *DO*

sleepy to fight well".31 He did not speculate whether the failure of
a Mateus Rosé bottle. We didn't speculate whether or not smoked c
a bucket load of goals but I did wonder about the ones he missed." Boyd

(continued)

olly, really, but one does slightly wonder about the kind of person who make
ow of behavior modification. Do you wonder about the people who are still
and laugh with them. They do not agonize over what others may or may not
Usually, if you do not agonize over it too much, it will sort
of writing fiction; she does not agonize over the ingrained deceitfulness
about it, don't reread them, don't agonize over them." As candor goes, it
was drunk that night. But she does ponder on the looks problem quit
people effusively? <p> Do you often ponder on previous misfortunes?
re still Travis. <p> I don't really ponder on what we've done, I'd a
about, um, something), but didn't ponder on them when there was Te
very much, and I think I don't look into it because I think th
too. <p> So why don't we turn our attention to something exotic -th
We know that many Palestinians do consider the PLO as their represe
nations. The Arab countries do not consider themselves as nations. T
And despite everything he did not consider himself as good a bet fo
find at our own fireside. I do not consider the newcomers as any ad
a psychosomatic illness. I do not consider her action as a suicide

MODAL MEANING: OBLIGATION/INCLINATION

Earhart's time, but would not speculate if it was part of Earha
39), but he won't boast about it or speculate whether his latest scor
<p> Lohr: Sloan wouldn't speculate how many more employees
that may be there. And he wouldn't speculate what effect Yeltsin's r
writing. Indeed, we do not have to speculate whether the worldly and
Beyond that, I'm not going to speculate what conclusions that m
yet. `It would be wrong for us to speculate when no decision has be headed
for Britain. I don't want to speculate where precisely it was
Pryderi's birth that one is led to speculate whether the two accoun
their trust." <p> It's tempting to speculate whether the changes in
feeling sad." It is tempting to speculate what her father might h
in their attitude. He wouldn't speculate as to the reason for this, but
representative in St. John's won't speculate as to what caused the sinking,
to be played and I don't want to speculate as to whether it is possible.
last night: `I wouldn't like to speculate as to what has prompted whoeve
such as these, he leaves us to speculate as to whether the area was, as
Mallory Ringess, and they liked to speculate as to the changes that a god
of this mirror? 7 would like to speculate as to the process and purpose
the incident, but I don't want to speculate as to motive or sponsorship.
tonight? <X02> I'm not going to speculate as to the time things might
knowledge, is not in a position to speculate as to the answer. On the same
it has made. We are left to speculate as to who ultimately controls
Humberside police are refusing to speculate as to whether the deaths are
they would assist in the war, and persevere in their friendship for the Ki

(continued)

ambitions, they had to accept and	persevere in the call to the priesthood
s. She had to forgo prestige and	persevere in her quieter way. All the
of what they produce. And we shall	persevere in our efforts to find the way
their story I encouraged them to	persevere in their life of devotion and
the cake-giving us the stimulus to	persevere in this magnificent hobby, whi
would probably encourage her to	persevere in prayer and Bible study desp
and misery but a determination to	persevere in well doing & to leave the r
my duty as UN Secretary General to	persevere in my efforts very strongly. A
admonished not to fall away but to	persevere in faithful endurance and foll
and all other parties involved to	persevere in negotiating a just end to t
and Peter Dean vowed yesterday to	persevere in the sport that killed their
outset of therapy, a willingness to	persevere in facing issues that you woul
the only principled course is to	persevere in the face of these twin
toil without pay drove me to	persevere in what promised to be a
deals from the AFL, but would	persevere in its attempts to gain a
War. Nevertheless, Graves would	persevere in the effort to maintain the
hear if the Irish Turf Club would	persevere in the protracted legal proces
than you think—and it makes me	wonder about you.' Rock said nothing; i
jobs are, I think you have to—to	wonder about the signals that the
person. He shouldn't have to	wonder about what your intentions are. A
do it in the big stuff. ONE has to	wonder about tennis and, in particular,
for some time. (You've got to	wonder about the other patrons!) He'd ha
annoying remarks which make you	wonder about your own judgement. He
underneath it that made you	wonder about the rumours of his past. <p
en and their achievements makes you	wonder about the old adage that countrie
the ready-to-wear, it does make you	wonder about the whole system. So it's
wouldn't have had to <pause>	agonize over this other <M01> Mm. <F01>
eficiencies, if one did not have to	agonize over the dilemma encountered by
the future, and by choosing not to	agonize over things that I can't change.
as any conscience at all, he should	ponder on the way he abused Alun
n the SAARC charter. Leaders should	ponder on this issue very seriou
float or sale. Shareholders should	ponder on whether fashion can be
the Chelsea manager had cause to	ponder on. Forest, whose own rec
Joan Smith would do well to	ponder on what it is that has ma
<p> John Smith would do well to	ponder on Laski's scathing Verdi
nationalism", they would do well to	ponder on the role of Soviet pol
it would be a good idea to	ponder on a general and integrat
position of the press may want to	ponder on the rights of the indi
he knock-about was over I decide to	ponder on this new-found knowle
my first company. You should also	look into real estate, though,
If you want to see the result-and	look into the future of commun
ll pop you on hold Miss FX and just	look into this for you. <F02> O
to have another drink, then I'll	look into things. What about yo
decide to do eventually, you should	look into all the pra
aid: `The secretary of state should	look into it and there should b

(continued)

asn't got her steaks. Could someone look into it, please?" <p> Morg
has asked that the Legislature to look into the possibility of bu
police review board be set up to look into misconduct charges.
the importance of being willing to look into his family's legacy o
want to exercise its discretion to look into those charges. <p> TO
feel. The two times I was asked to look into a murder case, I can
yesterday to set up a committee to look into the Norwegian bluepr
Republican had been appointed to look into the failure of the Ma
have set up a joint committee to look into the causes of Saturd
Exhibit 246, where he asked ACCO to look into the matter. <p> And M
nya, for a tribunal to be set up to look into the conduct of the tw
e could coax the downturned eyes to look into his own. <p> Who you
stablished a national commission to look into college-sports refor
ur situation Ian, so ask someone to look into it for you. There's
usband if David Blunkett refuses to look into the details she has s
mpetition authorities may decide to look into either offer. That co
levels of service and we will look into why it happened." How
new savings scheme? <p> Yes. I will look into it, if I get my head
1975, p. vii). All of us must grapple with the fact of our own
of us have to shop around. We must grapple with the difference betw
comes amid retailers' efforts to grapple with an increasingly gri
his ultimate intentions, who had to grapple with the problem of how
which the tabloid press have had to grapple with. In a sense she emb
for Racial Justice: `We have to grapple with endemic unemployme
2,000. The next synod will have to grapple with the ramifications o
urope. So NATO leaders will have to grapple with the problem of how
nalysts exploring influence have to grapple with the problem of meas
r Pfeiffer's successor will have to grapple with the challenge of in
onday. The new `City Plan" aims to grapple with the challenges of m
roblem? Scientists have long had to grapple with the fact that it is
product, are suddenly having to grapple with the unquantifiable
in France, Ross will now turn his attention to his new ITV career.
community should turn its attention to other regional probl
with the icing, he should turn his attention to the cake. <sect id=
Let me ask you to—to turn your attention to the Soviet Union for
Bush was expected to turn his attention to the matter this wee
I now urge him to turn his attention to Schweidler. This ex-
we would ask you to turn your attention to flip-over saws cost
I asked John Clarke to turn his attention to our Escort Cabriolet
Schrager who hopes to turn his attention to San Francisco and Lo
Last Rites, he wanted to turn his attention to something lighter in
because it may have to `turn its attention to batting down the dol
in asking people to turn their attention to higher things at wha
manager may have to turn his attention to players such as the
urged Schwarzkopf to turn his full attention to the Iraqi missiles.

(continued)

television will turn its potent attention to an unsigned group of
in Chapter 8, we will turn our attention to the validity of enti
dinosaur hunters will turn their attention to the rocky coastal ou
20 a month for a long period should consider unit trusts as represent
You would not have chosen to consider the French as a people o vagina.
It would be more correct toconsider such equations as involv
if he really would like lesbians to consider AIDS/HIV as a gay men's
viability for some time and had to consider its future as part of th
when a therapist is eager to consider his patients as healthy
the dove did not return. I would consider the prize as an olive br

MODAL MEANING: NECESSITY/IMPORTANCE/INTEREST

placers, but it is interesting to speculate whether just the uranium
and Ivanow; it is interesting to speculate whether the `English Am
It is therefore interesting to speculate how the fern spores can
judgment. It's fascinating to speculate what would happen if ou
those words in 1144. It's fun to speculate how he would have react
cases but {EDA}it's interesting to speculate if there is a correlat
influences. It is interesting to speculate where they will arrive
a hot pawn. It is interesting to speculate whether he would have
that <F01> But it is interesting to speculate how and why <M01> Mm. <
to lower incomes would lead me to speculate as to how much money it was
to Damascus. We need, therefore, to persevere in whatever method we choose,
the patient's family as a need to persevere in the face of inevitable loss
en to paper, parents need no longer agonize over every mistake or
most part, you don't even need to agonize over matters such as for
ordinary standards. Then again, why agonize over it? Perhaps there w
idn't answer the last question? Why agonize over how quickly your ne
such questions; but Labour needs to grapple with them intelligently,
states like Arkansas that need to grapple with the consequences of
When does the child need to grapple with his problems on his
unnecessary for the psychiatrist to grapple with the theoretical iss
Now Lord Irvine needs to turn his attention to the Court of Appeal.
insisted that it was necessary to consider the empire as a single

MODAL MEANING: POSSIBILITY/ABILITY/DIFFICULTY

Trevis), cynics will inevitably speculate whether Winslet, who re of
alliance is it anyway, they must speculate, when the ambassador of
<p> Naylor: I guess one can only speculate what the situation woul
by her arrival." One can only speculate how the mcfarlanes's da
on the government jet. One can only speculate what they might have be
and New Zealand. One can only speculate what the percentage inc

(continued)

Forest at home. One can only speculate whether Kidd will rue h
to do. It shouldn't be difficult to speculate what that means whenev
Police say it is too early to speculate whether it was the work
Funnell said it was too early to speculate whether troops would be
archives, but it is too early to speculate what that will include.
in the body. LUCK-BAKER: Can you speculate how long it might be b
if Synesius was right, then we can speculate as to how Hipparchus managed t
gave way. `At this stage we cannot speculate as to the cause," he said, aft
coherent plan. <p> One can easily speculate as to why this proposition mig
were indeed landmarks. One can only speculate as to the impact they have had
the results she wants. You can only speculate as to what strange household p
particular student. <p> We can only speculate as to why—possibly as a
left to answer them." One can only speculate as to what those questions wer
a gainst Scotland, but one can only speculate as to how much the past week's
audition, too, and we can only speculate as to how he might have `monke
as David Beckham. <p> We can only speculate as to what was going through
viewing figures. We can only speculate as to whether Tim Henman would
the public disclosure I can only speculate as to Nixon's and Haldeman's
09 July 1999 </dt> One can only speculate as to how the eternally
There is no real way to know or speculate as to what influences were on
officials said they couldn't speculate as to whether the agencies wou
A spokesman said he couldn't speculate as to when a new proposal woul
spokesman said: `I'm not able to speculate as to what has happened. We wo
broom of passion, it is easy to speculate as to the grounds for his two
the problem, it's very difficult to speculate as to how we might treat it.
It would take no great wit to speculate as to the origin of two mild h
Atlantic says it is too soon to speculate as to whether they will fly ov
to equity; and you cannot properly persevere in it, unless you prove that y
the most are those least likely to persevere in learning. If you want to
as I can see, all you can do is to persevere in seeking appropriate
could only encourage criminals to persevere in their wicked ways. The only
in the streets, has been able to persevere in great part because of the
Cod retreat. It is difficult to persevere in such environments, she tell
any great attention. And you could wonder about a few cancelled parking
ould Hofstadter have said, we might wonder, about the fact of NSC-68, which
start thinking it's less, you might wonder about it for a while, but if your
composers. Indeed a listener might wonder about the very nature of classica
the plane to Manchester. He must wonder about the expense, and why he is
you believe the study you might now agonize over the wasted years, looking f
uarter to six. There was no time to agonize over her outfit; she pul
at Cheltenham last March, may ponder on these words. He would
to Matron with the words: `You must ponder on the wisdom of sexual i
I quote: `San Siro fans are left to ponder on the transient nature o
Hart Lane will leave him much to ponder on, even though United's
1999 </dt> PETER FOX was left to ponder on a hatful of missed cha

(continued)

a short interval to enable you to ponder on what I've said." <p> O
sleep, and I had plenty of time to ponder on the foolishness of try
was his reply, leaving me to ponder on the thought that it's
but perhaps he had too long to ponder on it during the 12 overs
rtain things has happened, I got to look into it is all." But Harp
ransformation into a woman. Can you look into the future to see if
past history of Ireland that we can grapple with today's events. It
people, when fully informed, can grapple with difficult issues,
culture left a youth unprepared to grapple with this loss of his s
lesson in the tale, a better way to grapple with death in hollywood?
Thomas makes a valiant attempt to grapple with the various paradox
as possible. Instead it tried to grapple with the problem that "t
trouble is that when Labor tried to grapple with the problem, it fir
was a uniquely creative way to grapple with otherwise unconsci
or she is defined by being able to grapple with complex issues, ide
the Enlightenment's attempts to grapple with the dark secrets of
f we want a book which will help us grapple with the human condition
staff and he'll probably turn his attention to that matter beginni
he will feel free to turn his attention to other things. Soon h
Mr Milosevic was able to turn his attention to Bosnia; when the Day
ends. Now he could turn his full attention to his wife and her bro
sooner or later he would turn his attention to me. I was agile; I w
speculation that she would turn her attention to TV documentaries, an
despite all the madness, I try and consider it as a normal movie."
urged that Her Majesty could only consider the name as a compliment
adequate financial flexibility to consider new projects as opportun
and rupaul, that they seem to consider lesbians as butch, ungla
and too much dying for them to consider that as a real option. S
it is far too early for them to consider themselves as a force ju

OTHER USES

<p> Mr Lee said that rather than speculate what might happen after
began to plot future pleasures, to speculate how and on whom he woul
the case, then it is depressing to speculate how much those women wh
He was too self-effacing even to speculate what magic he might hav
censorship been a factor? Analysis Speculate as to why this particular grou
as an example to prove a point. Speculate as to the reasons that the aut
I can give them # P.W. <p> A Persevere in offering as varied a diet a
abut, which is unsettling. But persevere in building a better foundatio
were to think things out or to persevere in his various activities, he
when it comes along, than to persevere in misery. <p> In bed, though,
fundraising tactic in Gigha is to persevere in contacting the list of weal
induced the representee to persevere in a decision already reached.
US and Canadian governments will persevere in supporting Haiti as it

(continued)

o give. <p> Even then we will still wonder about the complex genius that is
Week, 1 May, p 20), I started to wonder about the environmental effects o
coming here. Inevitably I began to wonder about who could and couldn't play
on the way out and she started to wonder about his motives. Then she
scientists and theologians began to wonder about the propriety of "playing
as very appealing. But she began to wonder about him again. Why had he been
Sooner or later he would wonder about what the redhead was doing
often have I heard grown children agonize over what will happen to mother
Let Chateaus Haut-Brion and Latour agonize over the change to stainless st
utely stand by as she watched a man agonize over his Fisher King aspect. She
<p> While world leaders continue to agonize over how to stop the latest wav
realized how useless it had been to agonize over a past that could not be
Britain is only beginning to agonize over the apparent rise in female
Leave all that for some Scorpio to agonize over. <h> CANCER June 22-July 22
poorly)? Party members continue to agonize over this question (Mason, 1990)
<p> Spin and Marty. God, I used to agonize over that show!" <p> Jon
temporary—while I can leave you to agonize over your missing girlfr
in the past used to be that I would agonize over whether feeling <p>
reach of human beings to achieve, ponder on one awful statistic in
lue On Who Sell Out, the cover. And ponder on what stardom means Sat
count of western man. I began to ponder on what the world would b
THOUGHTS to ponder on while Michael Jackson
an hour. I was just beginning to ponder on the notion that the ex
that I will leave the reader to ponder on what I think is the ot
for the citizens of the Union to ponder on June 9th and 12th when
was on serious drugs. I started to ponder on the marvels of sexual
his face to the wall continued to ponder on the same question: Why
When she appears on television to ponder on what lies ahead, Ameri
a championship. It was a thought to ponder on the journey home but,
sponsored measure will also look into news leaks during the
understand that it's a joke to even look into it," Mr Ruxton said.
intimacy is IN-TO-ME-SEE—meaning look into me and see what's ins
but beware of high interest rates. Look into transferring your mor
back to that shortly. First let's look into the remarkable claim
by a team from a separate force to look into what are described as
a three-man fact-finding mission to look into conditions faced by
mission of inquiry to Jerusalem to look into the killings on Monda
have belonged to the commission to look into this and make a recom
women headed by Boni Robertson to look into the matter and find s
Criminal Justice Review Body would look into cases referred to it
o shut up, go out into the cold and grapple with the intractable: wh
hospitals are just beginning to grapple with the cultural revolu
the first since Germany began to grapple with the consequences of
ountry new bodies have sprung up to grapple with previously invisibl
destruction, let it also turn its attention to sanctions. By 1995,

(continued)

the case. <p> Let us now turn our attention to the show season and
Like them, you would then turn your attention to staging the show on
looking neat and tidy, turn your attention to weedy borders. Weeds
Cady Stanton, to turn their attention to the women's rights m
many Francoists to turn their attention to the succession. The
or 11%. <p> His failure to turn his attention to flagrant wastes of p
<p> Q: I'm going to turn your attention to a document at tab 12
profile. <p> Were you to turn your attention to politics on the othe
the merits of fish, let us turn our attention to an oil that we met b
JOURNAL <p> Bye, bye blue box. <p> Consider that as a possibility, if
lo ensnare my judgement. I shall consider myself as not having <p>
all, is not a career diplomat) to consider the work as being entire
common here in London for people to consider public transport as a ki
modern child-rearing continues to consider children as less fully h

Notes

NOTES TO CHAPTER 2

1. Wiebe et al. describe this as an objective speech event, but I have re-interpreted it as a judgement because Xirao-Nima is describing someone else's state of mind.
2. It should be noted, however, that Sinclair's concept of planes is not primarily about sets of linguistic features.

NOTES TO CHAPTER 3

1. http://news.bbc.co.uk/1/hi/entertainment/6714063.stm. Accessed 3 June 2007.
2. http://news.bbc.co.uk/1/hi/entertainment/6687091.stm. Accessed 3 June 2007.
3. http://www.mmrthefacts.nhs.uk/library/regressive.php. Accessed 3 June 2007.
4. http://www.fco.gov.uk. Accessed 26 February 2006.
5. For a discussion of the concept of 'default' interpretations, see Sinclair 1986.
6. http://www.fco.gov.uk. Accessed 26 February 2006.
7. 'What Doctors Don't Tell You,' issue 16, no. 9 (December 2005). http://www.wddty.co.uk/cms/files/WDDTY162qxd.pdf. Accessed 26 February 2006.
8. http://www.fco.gov.uk. Accessed 26 February 2006.
9. Speeding up 'unreal' segments is less common but not unheard of. In the documentary series *The Lost World of Mitchell and Kenyon* (BBC/BFI first shown starting 14 January 2005), the restored black-and-white film from the early 1900s is shown at normal speed, as it would have appeared when first shown. However, reconstructions of Sagar Mitchell and James Kenyon working in their studios appear slightly faster than normal, with jerky movements. This appears to be a humorous reference to the distortion of early film by the equipment used in the second half of the twentieth century, and has a time-distancing effect.
10. By analogy, the grammatical concept of projection (Halliday and Matthiessen 2004: 380) partly overlaps with the more general notion of attribution, but attribution covers instances which are not projection in grammatical terms.
11. From 'The Silent Castle' by Rosemary Lake: www.rosemarylake.com/ft_queenbee.html. Accessed 6 April 2006.
12. From a BBC report 'Sinn Fein British agent shot dead', 4 April 2006: news.bbc.co.uk/1/hi/northern_ireland/4877516.stm. Accessed 6 April 2006.

13. I would like to express my thanks to Geoff Thompson, whose remarks during a seminar at the University of Birmingham organised by Dr. Peter White directed my thinking along a path of which these paragraphs represent the outcome.

NOTES TO CHAPTER 4

1. Tim Johns' 'kibbitzers' website is not currently accessible but should soon be available at www.corpus.bham.ac.uk
2. In January 2003, Terry Pratchett wrote: 'Taking the piss out of our native dances and songs is part of the birthright of every Englishperson.' See http://www.terrypratchettbooks.com/ubb/ultimatebb.php?ubb=get_topic;f=11;t=000153;p=1. Accessed 5 March 2010.

NOTES TO CHAPTER 5

1. For example, modal auxiliaries do not inflect and form interrogatives and negatives without the use of *do*.
2. Each individual form of *BE* is less frequent than any of the forms shown in Table 5.3: *were* (3), *is* (2), *was, are*, *'re* (1 each).
3. Scanning 10,000 lines is not a preferred method of obtaining information of this kind.
4. In a similar way, Sinclair first assumed the existence of lemmas such as *YIELD* and *DECLINE* as objects of investigation in order to demonstrate that placing too much reliance upon them would be misleading.
5. The very low total percentage here is a result of considerable amounts of rounding off. In Leech et al. (2001) the figures given for the occurrence per million words of each wordform is not consistent with that given for the whole lemma, presumably because of rounding off in all cases.
6. In the original text, the phrase 'getting involved' is inside quotation marks.
7. I am grateful to Rosamund Moon of the University of Birmingham for pointing this out.

NOTES TO CHAPTER 6

1. This is a component of the Bank of English corpus. It comprises issues of *New Scientist* published between 1992 and 1999.
2. An advertisement shown on British television during the summer of 2007 plays on a similar distinction. The advertisement is for tourism in Ireland, and the script makes statements such as 'NAME OF GEOGRAHICAL FEATURE. First discovered by NAME OF PERSON'. It turns out that the time reference is very recent, that the person named is an 'ordinary individual' rather than an ancient explorer and that the discovery is a personal one.
3. Sinclair (2003: 31–40) gives examples of focusing adverbs and adjectives e.g. . . . *must have a foot or leg surgically removed* or *Tourism is as much ideology as physical movement*. In each case, the adverb or adjective concerned does not add new information (how else would a limb be removed? What other kind of movement is there?) but focuses attention on the salient aspect of the verb or noun following.
4. The examples given here involve textual studies—facts being transmitted through writing—but the project also involves studies of very different

kinds, such as building techniques being taken from one place to another by practitioners themselves.

5. Put very simply, Theme is the first element of a clause and Rheme the remainder of the clause (see Halliday 1994: 37ff.).

NOTES TO CHAPTER 7

1. It might be noted that *recover costs/damages* is a legal phrase where the costs or damages are not lost or stolen items that are got back in the way that *stolen birds* are got back.

NOTES TO CHAPTER 8

1. See chapter 3, note 8.
2. See chapter 3, note 7.

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Index

A

affect 20, 21, 54, 55, 95, 131, 134, 136–140, 143, 148–150
Aijmer 69, 86
appraisal vii, 3, 10–12, 14, 20, 22, 54, 130, 131, 136–138, 150
appraisal theory 19, 49, 54
appraisal, afforded 14
appraisal, inscribed 13, 14, 17, 58, 130
appreciation 17, 20, 21, 54, 55, 95, 130, 131, 134–137, 143, 144, 147–149
Atkins 121
attitude x, 1, 3, 6, 7, 11, 12, 20–24, 52, 54, 130, 140
attribution 17–19, 25, 28, 33–36, 38, 39, 43, 44, 49, 69, 139, 142, 166, 183
attribution, layering of 41, 54, 140
autonomous plane 24
averral 25, 33, 35, 37, 38, 41, 46, 62, 92, 117

B

Baker 4
Bank of English 8, 80, 82, 84, 135, 143, 154, 161, 164, 184
Barnbrook 120, 142, 143
Bednarek 119, 120, 149, 150
Biber 2, 3, 6, 7, 11, 22, 23, 51, 52, 66, 167, 168
Biber et al. 5, 7, 51, 69
Bloom et al. 54, 55
book review 1, 14, 16, 26

C

Chafe 33
Charles 2, 33, 49, 53
clause relations xi, 46, 52
Coates 66, 69
COBUILD 64, 120, 122
Coffin 64
cognitive linguistics 6
colligation 6, 56, 84, 109, 169
collocation 2, 4–8, 54, 56, 57, 70, 73, 109, 123, 144, 167, 169
concession and counter-assertion 2, 44, 46
concordance lines ix, 4, 5, 8, 15, 16, 56, 57, 59, 60, 66, 70–72, 74, 76, 79, 82, 83, 85, 89, 90, 94, 99, 107, 108, 123, 125, 126, 143, 168, 169
Conrad 3, 11, 22, 23, 51, 66
construction grammar 6, 123
conversation analysis 10, 23
corpus annotation 4, 54, 122, 144
corpus-driven 79, 85, 90
Coulthard 32, 55
Crismore 23

D

density vii, 151–153, 156, 157, 159, 161, 164, 165
dialogic 34, 36, 39, 45–47
dialogue 2, 18, 23, 33, 35, 39, 44, 49
directives 52, 53
documentary films 25, 26, 39, 40, 42, 183
Du Bois 1, 11, 12, 17, 23, 24

E

engagement xi, 2, 3, 20, 24, 25, 42, 44, 45, 47–49, 52
Englebretson 10, 23
epistemic status 2, 8, 21, 25
evaluation, contextual 14–16
evaluation, cumulative 3, 4, 14–16

evaluation, implicit 3, 4, 13, 16, 21, 50, 51, 55, 57, 62, 143, 151
evaluation, performing 139, 140, 142, 143
evaluation, reporting 139, 140, 142, 143
evaluative meaning, indicators of 13

F
Farnsworth 23
field 49
Fillmore 121, 126, 127
Fillmore et al. 119
Fletcher 54
FrameNet xii, 54, 119, 126, 127, 128, 142–147, 149, 150
Francis 30, 53, 108, 119, 120, 122, 143, 145–148
Francis et al. 69, 86, 88, 120, 122, 123, 125, 129, 131, 132, 134, 136, 143, 148
functional grammar 142

G
Gallie 12
general inquirer 162
Goldberg 119
graduation 20, 152
grammar patterns vii, 74, 75, 77, 119–122, 127, 128, 143, 144
Greene 54
Gries 6, 7, 167
Grieve 2, 54, 55
Groom 14, 114

H
Halliday 3, 6, 20, 33, 42, 43, 55, 68, 76, 80, 81, 83, 86, 87, 108, 129, 152, 183, 185
Hanks 81, 127
Hasan 3
Herbst et al. 122
Hoey 5–7, 11, 15, 46, 69, 81, 169
Holmes 5
How Well Do Facts Travel? 116
Hunston ix, 1, 2, 4, 6, 11–13, 16–19, 21, 22, 25–27, 32–34, 36, 39, 40, 44, 47, 57, 59, 60, 62, 65, 68, 70, 79, 80, 92, 94, 96, 108, 119–122, 128, 142–147, 149, 150, 157, 167
Hutchby 10
Hyland 2, 10–12, 23, 24, 33, 51–53, 128, 168

I
ideational 3, 19, 20
ideology 12, 13, 184
idiom principle 123
information extraction 12
intensification 3
intensifiers 1, 58, 129, 131, 164
intensifying phrases xii, 154–157, 160–164
intensity vii, 54, 55, 151–154, 156, 157, 159–161, 165
interactional 2, 23, 24
interactive 2, 3, 23, 24, 51
interactive plane 24
interpersonal 3, 12, 20, 23, 166
intersubjectivity 12, 23

J
Johns 58, 184
judgement 17, 20, 21, 54, 55, 130, 131, 133–137, 140, 143, 148, 149
Jullian 16

K
Kelly 162
Kennedy 68
Kress 39–41, 43

L
latent patterning 50, 55, 65
learner language 69
Leech 37, 89
Leech et al. 80, 81, 89, 184
lemma xi, 7, 9, 55, 59, 67, 79, 81–84, 89, 90, 99, 104, 123, 184
lexical items 13, 20, 55, 56, 60, 69, 70, 79–81, 88, 152, 169
lexical priming 5
lexical repetition 4
lexicography 55
lexis-grammar interface 5
Littman 54
local grammars vii, 8, 119–121, 127, 140, 142, 143, 147, 149, 150, 170
Louw 64, 65

M
Manning 64
Martin 2, 11, 12–14, 17, 19–22, 24, 39, 42, 44, 46, 47, 54, 55, 58, 95, 130, 131, 134–137, 144, 147, 152, 167
Matthiessen 5, 33, 42, 43, 80, 81, 83, 129, 170, 183

McCarthy 4, 5
metadiscourse 2, 10–12, 23, 24, 52
Mindt 68, 104
Mitchell 68, 76, 183
modal meaning xi, 8, 13, 66, 68–70, 73, 76, 77, 84, 85, 87–91, 169, 175, 178
modality 33, 40–44, 66, 68, 69, 76, 86, 88, 90, 116, 152, 170
modal-like vii, xi, 66–70, 72–74, 76–78, 89–91, 168, 169
Morgan 116–118
motifs 114–116
multi-modal texts vii, 25, 39, 49
multi-word units 5

N
Nesi 5
Nichols 33

O
O'Halloran 64
opinion mining 53

P
Palmer 68
Partington 57, 65
Patrick 54
pattern grammar 122, 123, 128, 138, 143, 144, 150, 170
patterns:
ADJ *about* n x, 120, 129, 137, 140, 149
ADJ *against* n 129, 137
ADJ *as* n 129, 137
ADJ *at* n 129, 137
ADJ *by* n 129, 137
ADJ *for* n 129, 137, 144
ADJ *for* n to-inf 144
ADJ *from* n 129, 137
ADJ *in* n 129, 137
ADJ n 128, 144
ADJ n to-inf 144
ADJ *of* n 129, 137
ADJ *on* n 129, 137
ADJ *over* n 129, 137
ADJ *to* n 129, 137
ADJ to-inf 129, 144, 147, 149
ADJ *towards* n 129, 137
ADJ *with* n 129, 137
it v-link ADJ *for* n to-inf 131, 132, 134, 144
it v-link ADJ *of* n to-inf 131
it v-link ADJ that 130, 139
it v-link ADJ to-inf 130, 144
it v-link ADJ-ing 129, 144
N that 93, 94
there v-link *something* ADJ *about* n/-ing 139
V *about* n 122, 123
V *from* n x, 124, 125
v *it* ADJ to-inf 130, 144
v n ADJ 144
v n ADJ to-inf 144
V n adj xii, 140–142
V n *as* adj xii, 140, 141
V n *as* n 111, 140, 142
V n *from* n x, xii, 121, 124, 125
V n ix, 122, 124, 128
V n n xii, 120, 121, 140–142
V pron-refl 122
V that 142
V to-inf 122
V wh 122
V x, 125, 128
verb + wh-clause 169
verb *about* noun 88
V-ing 122
v-link ADJ 144
v-link ADJ to-inf xii, 144, 148, 149
Perkins 68
persuasion 2, 3, 26, 49, 170
Pike 152
Plappert 91
preposition 70, 72, 73, 76, 77, 84, 90, 92, 93, 111, 113–115, 118, 122, 128, 129, 131, 134, 161, 162
prepositional phrase x, 22, 34, 37, 90, 122, 124–127, 130, 144, 147

Q
qualitative corpus research 50
quantitative corpus research 50
Quirk et al. 68

R
redundancy 152–157, 160, 161, 164, 165
register 4–6, 27, 50–52, 80, 109, 170
Reichardt 122
relevance 2, 21, 22, 25, 121, 139
Renouf 7, 119, 120
Resnik 54
Riloff 13, 54
Ruppenhofer et al. 54

S
Sampson 4

semantic preference 56
semantic prosody 10, 56–58, 60, 62, 64, 65
semantic reversal 119, 120, 127, 159, 165
semantic sequences 90, 91, 112, 170
Semino 33, 37
sentiment analysis 2, 53–55, 57, 119, 143, 162, 170
Short 33, 37
Sinclair 3, 5–7, 11–13, 24, 30, 32, 33, 55, 56, 60, 64, 69, 72, 79, 81, 119, 120, 121, 123, 142–144, 157, 159, 168, 170, 183, 184
Sinclair et al. 76, 80, 120, 122
Somasundaran et al. 54
stable meaning 13, 162
stance markers, directive 52
stance markers, epistemic 22, 52, 69
stance triangle 12, 17, 23
stance vii, 2, 3, 7, 10, 11, 13, 22, 23, 27, 50–53, 65, 123, 168
stance, markers/expressions of 51, 52
stance-taking 1, 10, 23, 53, 166
status markers 25
status modification 28, 33, 34
status vii, xi, 2, 8, 17, 21, 22, 25–40
Stewart 57
Stone 162
Stone et al. 162
Stubbs 2, 18, 19, 34, 55–59, 62, 63, 66, 69, 85, 153
subjectivity 1, 2, 11–13, 15, 18, 19, 21, 23, 25, 27, 29, 30, 33, 43, 54, 68, 77, 87, 128, 129, 142, 152, 159, 160, 170
Swales 33
systemic-functional linguistics 3, 11, 19, 20, 42

T

Taboada 2, 54, 55
tenor 49
Teubert 4, 14, 25, 119, 120, 122, 123, 134, 167, 169
that-clauses 28–31, 33, 37, 38, 52, 94, 106, 116, 122, 128, 136–138
The Real Da Vinci Code 26, 39, 41, 42
Thompson, G. 2, 11, 12, 18, 23, 33, 167, 184
Thompson, P. 33
to-infinitive clauses 52, 86, 122, 128, 134, 144, 145, 148, 149
Truth and Lies 26, 39, 40, 42
Tse 2, 10, 12, 23, 24, 51, 52, 128, 168
Turney 3, 54, 55

U

units of meaning 5, 7, 55, 170

V

valency 122, 127, 144
value 2, 3, 17, 19, 21–23, 25, 43, 51, 60, 87, 95, 111, 117
value system 1, 13
van Leeuwen 39–41, 43
Vande Kopple 23

W

Wallis 116, 117
White 2, 11–14, 17, 19, 20–22, 24, 39, 42, 44, 46, 47, 54, 55, 58, 95, 130, 137, 147, 152, 162, 167, 184
Whitelaw et al. 54, 55
Whitsitt 57, 58
Wiebe 13, 54
Wiebe et al. 12, 17, 18, 54
Willis 126
Wilson et al. 14, 54
Wooffitt 10
wordform xi, 7, 9, 50, 55, 66, 67, 73, 74, 76, 79, 81–84, 88, 91, 167, 168, 184
WordNet 54
words and phrases:
a dose of 163
a shred of ix, 60, 63
a wave of 164
afford 6, 81
agree 83, 123
alleviate 57, 58, 62
almost 157–160
as is humanly possible 154–156
assume ix, 99–105, 108
assumption ix, 52, 92–97, 99, 100, 102–104, 108
be greeted with 164
bordering on 157–159, 164
bring about ix, 57–59
carry 81–83
cause ix, 56–59, 94, 95, 97
claim 97, 105
clinical detachment 160
conclusion 24, 92, 97–99
consider 81, 82, 131, 174, 175, 178, 180, 182
decide ix, 70–73, 76, 84, 88, 169

difficult/difficulty 70, 74–78, 98, 129, 131–133, 143–147, 158, 168
disagree 83, 84
discover/discovery ix, 28, 92, 94–97, 104–109, 112
distinguish ix, xi, 70–77, 90, 168, 169
endure 162, 163
explain 80, 81, 84, 96, 113, 115
fact xii, 92, 108–118
feel 131
fill 80
for fear of xi, 78, 86, 87
forget 122, 123
gene 91
hypothesis xii, 108, 109
idea 92, 95–97
impression 95–97
in the face of 161, 164
involve 162, 163
months of xii, 162, 163
naked eye 56
notion 95–99
on the brink of 78, 164
on the verge of 78, 164
PAY . . . price ix, 58, 60, 61–64
perpetrate 58, 63
preserve 67
recover/recovery ix, x, 123–128, 144, 185
suffer 58, 125, 127, 128, 144, 162, 163
suggestion 95–97
terrible ix, 13, 57, 58, 60
the depths of 164, 165
the most . . . they can muster 154–156
theory 94–97
think 43, 69, 81–83
to the point of 157–161, 164
tragedy 57, 157, 161, 162
trust ix, 14, 15
view 94–97
walk 9, 81–83
whether, verbs followed by xi, 73, 84, 85

Y

Ye 18, 33